

Nonequilibrium Calibration and Faithful Records in Autonomous Effective Measurement Models

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Abstract

A calibrated final pointer distribution does not by itself establish that a record is faithful to an earlier event. We consolidate two finite autonomous Schrödinger–guidance constructions in which both questions have quantitative answers under explicitly restricted nonequilibrium initial laws. The first uses periodic focusing, a finite recoiling reference rotor and a delayed compensated oscillator. Its record error is at most 0.00443484008607784720002304 and its failure to copy and hold its actual earlier label is at most 0.00232725479707784720002304. The second starts with a small radial Gaussian, includes autonomous preparation and activation, and contains its writer from the original entrance. Its joint earlier-label and entire-record law has total variation distance below 0.006086770113 from a constant Born-labelled path; its actual-label copy-and-hold failure is below 0.000552421956. The proofs combine bounded-variation calibration, exact recoil reduction, conditional-rank currents, finite-clock derivative estimates and absolute holding flux. The two Hamiltonians and admissible law classes remain separate. Neither construction assumes full initial equilibrium, but both retain substantive readiness and statistical restrictions. Neither establishes a finite local source of every prescribed interaction or a realization in established matter. The appendices supply the specialized derivative, current and numerical estimates supporting these bounds. External specialist verification remains an open step.

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1 The problem and the contribution

An account of a measurement should explain not only the frequency of its final display but also what that display records. A pointer may have a suitable one-time distribution while subsequently changing sides, or while failing to copy the system's earlier position. We therefore specify an earlier configuration event and a whole holding interval. The statistical conclusion concerns that joint observable.

The two constructions below separate three mathematical requirements: calibration of the earlier event, physical copying under the same dynamics, and retention at every time of a finite interval. Their error estimates are uniform over stated input and initial-law classes. They are not claims of universal relaxation from an arbitrary configuration law.

Quantum equilibrium, subsystem statistics and Bohmian measurement theory have a substantial established literature [2, 3]. The method of arbitrary functions has also been applied to quantum probabilities: Bonds and collaborators study a random perturbation of a double-well model in long-time and small- $\hbar$ limits [1]. Our finite apparatus statements instead concern restricted configuration ensembles and explicit recorded histories. Inverse engineering of harmonic transport is established [12]; it is used here with an autonomous quantum clock and a separate historical-current estimate. Coarse-grained relaxation studies [13] address a different statistical question. We make no claim that any of these general ideas originates in this work.

The contribution being consolidated is the particular assumption-to-record chain, including its finite parameters and its current-sensitive error accounting. It consolidates the author's earlier research; the numerical constructions are not presented as new discoveries in this revision. Its relationship to the author's equilibrium and hybrid pilot-medium papers is comparative: those papers use different initial laws or dynamics and do not supply missing premises here [10, 8, 7].

1.1 What is and is not a physical claim

Throughout, the wave obeys a specified Schrödinger equation and actual configurations obey its guidance law. These are the constitutive assumptions. The potentials and coherent entrance stock are supplied mathematically. "Autonomous" means the complete displayed Hamiltonian has no externally switched laboratory-time parameter. It does not mean that its coefficients, supports and stock have been manufactured with established interactions. The intrinsic-circle model has an explicitly finite recoil coordinate. The radial model is an exact two-body s -wave reduction, but its clock and pointer remain effective scalar coordinates. Adding a field or a material support changes the model and requires a new same-model history proof.

2 Probability and complete record observables

For a spinor Ψ , put $\rho = \Psi^\dagger \Psi$ and $j_k = (\hbar/m_k)\Im(\Psi^\dagger \partial_k \Psi)$ in physical coordinates. Internal projectors label wave components; they are not independently sampled configuration bits. The wave measure and the actual initial law ν_0 are distinct. A bound $\nu_0 \leq C\rho_0 dq$ transports under the same conservative equivariant flow, and implies $\mathbb{E}_{\nu_0} F \leq C\mathbb{E}_{\rho_0} F$ for every nonnegative path functional F . It does not establish equilibrium or domination by an analytical comparison wave.

We use $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. Fix an earlier label $L \in \{0, 1, \dagger\}$, a compact holding interval I , and disjoint closed pointer regions R_0, R_1 . For a continuous pointer trajectory define $K_i = \{t \in I : Z_t \in R_i\}$, allowing the empty compact set, and

$$Y = (L, K_0, K_1), \quad \iota(0) = (0, I, \emptyset), \quad \iota(1) = (1, \emptyset, I).$$

The random compact sets give a measurable representation of the entire symbolic record; no unwarranted càdlàg assumption on a thresholded path is needed. For example, avoidance of a compact time set J is the condition $\min_{t \in J} \text{dist}(Z_t, R_i) > 0$, a measurable condition on continuous

paths. Countable rational interval approximations generate the corresponding compact-set sigma-field. A visit outside both record regions, however brief, counts as failure. Undefined L also counts as failure.

Lemma 2.1 (Direct ideal-record composition). *Let L_* be a binary reference on the same coupling. Suppose E, A, B are events such that outside $E \cup A$, $L = L_*$, and outside $E \cup B$ the entire actual record is the constant record of the defined label L . If $\text{TV}(\mathcal{L}(L_*), \text{Bernoulli}(p)) \leq e$, then*

$$\text{TV}(\mathcal{L}(Y), (\iota)_* \text{Bernoulli}(p)) \leq \mathbb{P}(E) + \mathbb{P}(A) + \mathbb{P}(B) + e.$$

Proof. On the complement of the three events, $Y = \iota(L_*)$. The coupling inequality bounds the distance to $\mathcal{L}(\iota(L_*))$ by the union probability; pushforward contracts the remaining distance. This proves the claim, including undefined outcomes. Shared events are paid once because their identity is established, not by subtracting unrelated upper bounds. $\square$

A comparison to an imperfect reference's entire path is a different statement: its own history failure generally must be paid as well. Likewise, conditioning on a rare successful event may amplify error. If $\text{TV}(P, Q) \leq \epsilon$ and $P(E) > \epsilon$, then the elementary normalization bound is $\text{TV}(P(\cdot | E), Q(\cdot | E)) \leq \min(1, 2\epsilon/P(E))$. No conditional conclusion is asserted for zero-probability branches.

2.1 Why absolute current is indispensable

For real even normalized functions g, h , the wave $\Psi(x, y) = g(x)h(y)e^{ikxy}$ has $j_x(0, y) = (\hbar k/m)yg(0)^2h(y)^2$. Its signed hidden-coordinate integral vanishes, whereas $\int |j_x(0, y)| dy > 0$ when $k \neq 0$. Thus averaging before taking the modulus can erase the quantity that bounds crossings. The record estimates below take the full-configuration absolute value first. This example diagnoses an invalid inference, not a no-go theorem for either apparatus.

Part I

A periodic detector with finite recoil

The first construction uses units $\hbar = 1$. The active rotor/pointer/clock coordinates and their law are specified afresh in this part. They are not identified with the physical radius and clock of Part II.

3 All moving coordinates and finite parameters

Let $x, R \in \mathbb{T}_2$ be a detector rotor and a new reference rotor, and set $X = x - R \pmod{2}$. Let $Z, S, C \in \mathbb{R}$, and retain the original 255 differential source coordinates $u \in \mathbf{1}^\perp \subset \mathbb{R}^{256}$. The unknown normalized qubit/reference vector is ψ , with exactly conserved projectors $P_0 + P_1 = I$. Set $p = \|P_1\psi\|^2$. The target projector P_1^{tar} is an orthogonal projector with $\|P_1 - P_1^{\text{tar}}\| \leq 10^{-4}$, and $p_{\text{tar}} = \|P_1^{\text{tar}}\psi\|^2$. Both act as the identity on any admitted inaccessible reference factor. That reference is an internal Hilbert-space factor and introduces no additional guidance coordinates. Thus $|p - p_{\text{tar}}| \leq 10^{-4}$. The periodic classifier uses the centred cell coordinate $u = ((X - o)/h) \bmod 1 \in [-1/2, 1/2)$: it is zero for $|u| \leq 1/4$ and one otherwise, with a fixed boundary convention.

Set

$$N = 256, \quad h = 1/128, \quad K = 2^{38}, \quad \eta_t = 2^{-16}, \quad \eta_p = 2^{-17}.$$

Let $m = m_x m_R / (m_x + m_R) = \alpha K / h^2$. Independently choosing

$$m_x, m_R \in [1.998, 2.002]K/h^2$$

ensures $\alpha \in [.999, 1.001]$. Nominally $m = 2^{52}$ and $m_x = m_R = 2^{53}$. The other masses are

$$M_s = 10^{18}, \quad M \in [.99, 1.01]10^{18}, \quad \mu \in [.00999, .01001], \quad M_C = 10^8.$$

The retained source has $\kappa_s = 2.5 \cdot 10^{43}$ and frequencies $\Omega_k = 10^{13} \sin(\pi k/256)$. It is a spectator Hamiltonian, not a field generator in this redesign.

Let $S_5(z) = 10z^3 - 15z^4 + 6z^5$. The even period-one U_{η_p} equals z^2 for $|z| \leq 1/2 - \eta_p$ and has positive-cap derivative

$$U'_{\eta_p}(z) = 2z [1 - S_5((z - 1/2 + \eta_p)/\eta_p)].$$

It is C^3 , nonnegative, and bounded by $1/4$. Define

$$V_i(X) = \frac{g_i K}{2} U_{\eta_p} \left(\frac{X - o}{h} - \frac{i}{2} - d_i \right), \quad g_i \in [.999, 1.001], \quad |d_i| \leq .01.$$

The common origin o is arbitrary. The relative sector offsets and gains can vary independently in these intervals; arbitrary nonperiodic site defects are not admitted.

For $B_9(a) = 126a^5 - 420a^6 + 540a^7 - 315a^8 + 70a^9$ on $[0, 1]$, with constant extensions, put

$$b(a) = LB_9((a - a_0)/w_b), \quad y = Z - z_0,$$

$$W_0(a, y) = \frac{y^2}{2\mu}, \quad W_1(a, y) = \frac{(y - b(a))^2}{2\mu} - \mu b''(a)(y - b(a)/2).$$

The finite writer family is

$$L \in [23.99, 24.01], \quad |z_0| \leq .001, \quad v \in [127, 129], \quad \lambda \in [.99999, 1.00001], \\ a_0 \in [1.2299, 1.2301], \quad w_b \in [.03999, .04001].$$

Its launch momentum is $Mv\lambda$. Spring coefficient $1/\mu$ and the scalar compensation are matched to the selected μ . General symmetry-breaking or independently mismatched compensation errors are outside this theorem.

4 The ordinary autonomous Hamiltonian

$$\boxed{\begin{aligned} H_{\text{per}} &= \frac{p_x^2}{2m_x} + \frac{p_R^2}{2m_R} + \frac{p_Z^2}{2\mu} + \frac{p_S^2}{2M} + H_s(u) + \frac{p_C^2}{2M_C} + 50000 \\ &\quad + \sum_{i=0}^1 P_i[V_i(x - R) + W_i(S/v, Z - z_0)], \\ H_s &= \frac{p_u^2}{2M_s} + \frac{\kappa_s}{2} \sum_j (u_{j+1} - u_j)^2. \end{aligned}} \tag{1}$$

No kinetic coefficient is switched off. No external time occurs in this expression. Every microscopic kinetic term shown is positive and quadratic; the relative reduced mass is a derived constant.

Proposition 4.1 (Operator realization). *The finite operator (1) has a self-adjoint semibounded form realization, with ordinary configuration currents $j_a = m_a^{-1} \text{Im}(\Psi^\dagger \partial_a \Psi)$. The shift makes it nonnegative. The added reference body is finite, not an infinitely massive frame.*

Proof. Use the free-rotor/clock and positive oscillator form. The angular potentials are bounded. Complete the pointer square:

$$W_1 = \frac{[y - b - \mu^2 b'']^2}{2\mu} - \frac{\mu b b''}{2} - \frac{\mu^3 (b'')^2}{2}.$$

The stated rectangle gives $|b| < 25$, $|b''| < 169000$, and a negative part below 43000. The affine terms are infinitesimally oscillator-form bounded. The form construction and ordinary Green identity give the assertion. Finite-horizon conservative currents are assumed on the usual differentiated domains; the initial active-law domination transfers their wave-null exceptions. The passive-coordinate issue is handled exactly in the comparison companion, not by an arbitrary-clock almost-sure argument. $\square$

5 Simple initial waves; independent actual-law data

Let $A_c(s) = 1$ on $|s| \leq .3$, $A_c(s) = 1 - B_9(5(|s| - .3))$ until $.5$, and zero outside. Set $Z_c = 176818/230945$ and $\varphi = A_c/\sqrt{Z_c}$. The supplied wave is

$$\Psi_0 = \frac{\psi}{2} \zeta(u) \gamma(Z - z_0) e^{iMv\lambda S} \varphi(S) e^{iMC^C} \varphi(C + 1), \quad \gamma(z) = \pi^{-1/4} e^{-z^2/2}.$$

The factor $1/2$ is the product of two flat rotor waves. No shaped rotor wave is supplied. ζ is the retained harmonic source ground wave.

In relative coordinates, the actual law is

$$f_0(X, Z, S) g(u \mid X, Z, S) \kappa_R(dR \mid X, Z, S, u) \kappa_C(dC \mid X, Z, S, u, R).$$

Here f_0 is a nonnegative density with integral one. Define $V_j = |D_j f_0|(\mathbb{R}^3)$ as the distributional total variation of its zero extension, for $j = X, Z, S$, including boundary jumps, and $B = V_X + V_Z + V_S$. The admitted same-support total-variation neighbourhood retains absolute continuity of the active configuration marginal, as required for the almost-sure flow; allowed singular passive conditionals retain their separately stated supports.

The original class is retained: support $[-1, 1]^2 \times [-1/4, 1/4]$, $V_X, V_Z \leq 2$, $V_S \leq 8$, $B \leq 12$, $f_0 \leq 2$, and $0 \leq g \leq 2|\zeta|^2$ normalized. κ_R is any normalized law on the reference circle; κ_C is any normalized law on $[-1.25, -.75]$. Neither must be Born, independent, or absolutely continuous. Taking $R_0 = 0$ embeds the original detector positions without even a common translation. An independent actual orientation law can also be appended to an original bare-position ensemble. Its translated mixture preserves the density cap, V_Z, V_S , circular rotor-marginal variation at most two, and rotor-marginal density at most one; its mixed source conditional preserves the source domination. Zero-extension rotor variation and the raw total B need not remain unchanged. The independent-orientation extension is a separate corollary because the active proof uses precisely the circular variation, marginal bound and density cap. This is not permission to correlate R arbitrarily with bare x while keeping the relative law unchanged. For example, $R = x$ would collapse the relative seed and is outside the declared relative-density class. The same-support 10^{-4} actual-law neighbourhood is charged once.

6 Exact relative-coordinate reduction with finite recoil

On the subspace $\Psi(x, R, \dots) = 2^{-1/2} \Phi(x - R, \dots)$,

$$\left(\frac{p_x^2}{2m_x} + \frac{p_R^2}{2m_R} \right) \Psi = -\frac{1}{2m} \partial_X^2 \Psi, \quad \frac{1}{m} = \frac{1}{m_x} + \frac{1}{m_R}.$$

Both the Hamiltonian and the product of flat initial rotor waves preserve this subspace. No physical mixed kinetic term has been introduced: these are the restrictions of two constant ordinary kinetic operators.

Theorem 6.1 (Finite recoil and spectator independence). *The complete wave has the exact form*

$$\Psi_t = \frac{\zeta(u) e^{-i(E_s + 50000)t} \chi_{C,t}(C)}{\sqrt{2}} \sum_i (P_i \psi) r_i(X, t) \Xi_i(S, Z, t).$$

Its active relative velocity, pointer velocity, and writer-clock velocity are independent of the instantaneous R, u, C . Moreover

$$\dot{x} = \frac{m}{m_x} v_X, \quad \dot{R} = -\frac{m}{m_R} v_X, \quad \dot{x} - \dot{R} = v_X.$$

The old differential source positions are stationary under their nodeless ground wave, with their Hamiltonian and phase retained. Their arbitrary normalized actual conditional law is not used as an equilibrium reference.

Proof. The differential identity gives an invariant wave subspace. Every u and C term commutes with the active Hamiltonian. Differentiating the exact factorization gives the displayed ordinary currents; division cancels spectator densities wherever defined. The source wave is positive and has a coordinate-independent phase. The free C tube proved below is positive at every allowed actual clock path. Active exceptions are wave-null and, by $f_0 \leq 16q_0$, actual-null. These exceptions depend only on active initial data, so singular conditional R, C laws cannot select them at otherwise good active positions. $\square$

In particular all active event probabilities equal those calculated from $f_0(X, Z, S)$ alone, after integrating the normalized g, κ_R, κ_C . This is stronger than assigning a Gaussian actual law, but weaker physically than generating the field from u : the latter coupling is absent.

7 The writer's exact driven comparison

Remove the common Galilean phase and set $\xi = S - v\lambda t$, $a = \lambda t + \xi/v$. For sector one let

$$q'' + \Omega^2 q = \Omega^2 b(a) + b''(a), \quad q(0) = q'(0) = 0, \quad \Omega = 1/\mu.$$

The normalized comparison is

$$G_1 = \varphi(\xi) e^{-it/(2\mu)} \exp\{i[\mu q'(y - q/2) + \vartheta]\} \gamma(y - q),$$

$$\vartheta' = \frac{\mu}{2} [\Omega^2 b + b''](q - b).$$

Sector zero has $q = \vartheta = 0$. Direct substitution proves the fibre Schrödinger equation when the clock's residual kinetic term is omitted. For $\lambda = 1$, $q = b$ and $\vartheta = 0$ exactly.

The difference $e = q - b(a)$ solves

$$e'' + \Omega^2 e = (1 - \lambda^2) b''(a).$$

Sine/cosine integration gives $|e| < .001$, $|e'| < .07$ on the complete finite rectangle. Derivatives obey

$$|q| < 25, \quad |q'| < 1630, \quad |q''| < 170000, \quad |q'''| < 36000000.$$

The clock-envelope derivative norms are below 6, 112, 4000, 190000. Appendix A gives the polynomial derivative and envelope integrals explicitly. Keeping the Gaussian polynomial before taking norms gives

$$\left\| \partial_\xi^2 G_i \right\| < 110000, \quad \left\| p_Z \partial_\xi^2 G_i \right\|, \quad \left\| (Z - z_0) \partial_\xi^2 G_i \right\| < 3 \times 10^6.$$

Appendix A derives these jet estimates from the two coherent-Gaussian derivative polynomials, with explicit coefficient bounds. Harmonic transport inverse engineering is established prior art [12].

8 Finite interacting clock, not an infinite-mass limit

The exact moving-frame Hamiltonian retains $-\partial_\xi^2/(2M)$. Consequently the comparison residual is $\partial_\xi^2 G_i/(2M)$. Its propagator is the full interacting S, Z propagator.

Lemma 8.1 (Full-time auxiliary errors). *For $0 \leq t \leq 3$ and $M_- = .99 \cdot 10^{18}$,*

$$\|\Xi_i - G_i\| \leq \frac{3(110000)}{2M_-} < 2 \times 10^{-13},$$

$$\|p_Z(\Xi_i - G_i)\| \leq \frac{3[6 \times 10^6 + 15000(110000)]}{2M_-} < 3 \times 10^{-9}.$$

Proof. The norm estimate is Duhamel with zero initial defect. For the derivative estimate solve the oscillator operator equations. The remaining Z force has norm below 5000, so

$$\|p_Z U(t, s)r\| \leq \|p_Z r\| + \|(Z - z_0)r\| + (t - s)5000 \|r\|.$$

Apply this to the residual and integrate. Clock recoil is in U , not prescribed away. No exponential in Ωt is paid, and no source phase enters these S, Z derivatives. $\square$

9 An earlier actual label is protected separately

For the full initial offset support,

$$\lambda_+ t_a + .5/127 < 1.2299, \quad \lambda_- t_b - .5/127 > 1.2301 + .04001.$$

The compact comparison is therefore still in the common potential before t_a and has completed writing at t_b . These guards concern a comparison support; they are not a claim that the exact clock has compact support at later times.

Before t_a , use

$$F_1(t, \xi, Z) = e^{-it/(2\mu)} \left[\varphi(\xi) + \frac{it}{2M} \varphi''(\xi) \right] \gamma(Z - z_0).$$

Its residual norm is at most $t \|\varphi''''\|/(4M^2)$. Both the exact common auxiliary evolution and each exact driven sector differ from F_1 by at most $t_a^2 \|\varphi''''\|/(8M^2)$. Therefore

$$\epsilon_{\text{pre}} \leq \frac{t_a^2(400000)}{4M_-^2}.$$

This expansion has not discarded any exact clock tails.

Let ρ_r, j_r be the isolated spinor rotor current. Its initial density is $1/2$, and circle Sobolev/energy estimates give

$$\|\rho_r\|_\infty \leq 3K/h, \quad \|j_r\|_\infty \leq 8K, \quad \|r'_i\|_2 \leq K/h.$$

Exact sector factorization bounds the pretrigger wave and its X derivative errors by ϵ_{pre} and $(K/h)\epsilon_{\text{pre}}$. Thus

$$\|\delta j_X\|_1 \leq 4h\epsilon_{\text{pre}}, \quad \|\delta \rho\|_1 \leq 3\epsilon_{\text{pre}}.$$

The isolated lifted cumulative rank along the coupled ordinary path obeys

$$\rho \dot{\mathcal{K}} = \rho_r j_X - j_r \rho,$$

including its circulation. Hence

$$\boxed{\mathbb{E}_Q \text{Var}_{[0, t_a]} \mathcal{K} \leq 120K\epsilon_{\text{pre}} < 6 \times 10^{-18}.}$$

This is the physical clock-to-label error used in the periodic-current section. A global wave norm alone is not used as its replacement.

10 Every admitted preparation-clock position is harmless for a reason

The old preparation clock C is retained but no longer controls the active programme. It is an ordinary free particle with $M_C = 10^8$. In its co-moving coordinate, the free propagator and one-dimensional Sobolev bound give, through time three,

$$e_0 \leq \frac{3(112 + 4000)}{2M_C}, \quad e_1 \leq \frac{3(4000 + 190000)}{2M_C}, \quad |\dot{C} - 1| \leq \frac{e_1}{M_C(1 - e_0)} < 3 \times 10^{-11}.$$

The displacement error is below 9×10^{-11} , less than the .05 plateau clearance. A first-exit argument proves the tube for every $C_0 \in [-1.25, -.75]$. No averaging over $|\chi_C|^2$ is used. Its correlations are irrelevant to active histories by Theorem 6.1.

The active writer clock is S . Its actual distribution is controlled by the original joint cap $f_0 \leq 2$, not by a new Born-clock premise. A pointwise trajectory tube for every S_0 is neither claimed nor needed by the active-law current proof.

11 Reduced active law and periodic averaging

Use the finite Hamiltonian and initial data of the periodic-model section. The recoil and auxiliary calculation above proves the exact relative dynamics and spectator reduction. The active initial reference density is

$$q_0(X, Z, S) = \frac{1}{2} |\gamma(Z - z_0)|^2 |\varphi(S)|^2.$$

On the readiness box, $q_0 > 1/8$. Indeed $\exp((1.001)^2) < 11/4$, $\sqrt{\pi} < 9/5$, $Z_c < 4/5$, and $2(11/4)(9/5)(4/5) < 8$. Thus every active actual law is dominated by $16q_0$. This reference is only a proof measure.

Lemma 11.1 (Periodic event averaging). *Let A be any measurable h -periodic subset of $\mathbb{T}_2$, with volume fraction θ . For a normalized actual marginal f_X of zero-extension variation at most V_X ,*

$$|P_{f_X}(A) - \theta| \leq \frac{hV_X}{8} = \frac{V_X}{4N}.$$

Proof. The mean-zero function $\mathbf{1}_A - \theta$ has an h -periodic primitive of oscillation at most $h\theta(1 - \theta)$. Subtract its midrange value; its sup norm is at most $h\theta(1 - \theta)/2 \leq h/8$. Integration by parts against the circular BV derivative proves the estimate. Its variation is bounded by the given zero-extension variation, including boundary jumps. No smoothness of the event boundary and no equilibrium actual law is needed. $\square$

12 Rotor wave and velocity-weighted error

In one sector write $z = (X - o)/h - i/2 - d_i$, $\omega_i = \sqrt{g_i/\alpha}$, and $b_i = \cos(\omega_i t)$. Let a_{η_t} be the unnormalized quintic taper: one on $|z| \leq 1/2 - 2\eta_t$, zero on $|z| \geq 1/2 - \eta_t$, and a S_5 transition between. The harmonic compact comparison carries amplitude $a_{\eta_t}(z/b_i)/\sqrt{2b_i}$ and phase $-\alpha K \omega_i \tan(\omega_i t) z^2/2$ in each cell.

The physical initial wave is $1/\sqrt{2}$, so its initial error is not zero. Exact integrals give

$$d_0^2 = \frac{643\eta_t}{231}, \quad A_1^2 = \frac{20}{7\eta_t}, \quad A_2^2 = \frac{240}{7\eta_t^3}, \quad A_3^2 = \frac{1440}{\eta_t^5}.$$

The comparison lies in the quadratic potential core whenever it is used. Its residual norm is $A_2/(2\alpha K b_i^2)$.

Lemma 12.1 (Finite rotor comparison). *Through capture time $t_b = 1.28$, let r_i be the exact rigid-sector wave. Then*

$$\|r_i - \Phi_i\| \leq d_0 + \epsilon_R =: d, \quad \epsilon_R = \frac{2A_2}{.999K},$$

$$\frac{\|p_X(r_i - \Phi_i)\|}{mh} \leq \frac{A_1}{.999K} + .501d_0 + \frac{4.02A_3}{K^2} + 1.002\epsilon_R =: q.$$

For the selected row, $d < D = .007233$ and $q < Q = .003983$.

Proof. The residual norm integral uses $\omega_i t < 1.3$ and $\tan 1.3 < 4$. The static nonnegative rotor Hamiltonian propagates its own square-root norm. Multiplying that estimate by $\sqrt{2/(\alpha K)}$ gives the initial derivative contribution $A_1/(\alpha K) + \omega_i d_0/2$. Differentiating the residual introduces $A_3/(2\alpha^2 K^2 b_i^3)$ and its harmonic phase derivative; the potential part adds the same $\omega_i/2$ residual bound. Integrate with $\int_0^{1.3} \sec^3 \theta d\theta < 8$, $\omega_i < 1.002$, and $4/(\alpha^2 \omega_i) < 4.02$. Weak H^3 taper derivatives suffice; no fourth derivative of the sharp taper is invoked. The outward rational square-root calculation in Appendix A proves the final decimals. $\square$

At $t_a = 1.22$, each Φ_i is supported in its own outcome region even with $|d_i| \leq .01$. The wrong-side wave probability is at most D^2 . For the isolated spinor rotor, the guidance map commutes with $X \mapsto X + h$. The initial preimage of the classifier is therefore periodic, including its circle circulation. Lemma 11.1 and the wrong-side estimate yield the actual isolated-label error

$$|P(\ell(X_{t_a}^{\text{iso}}) = 1) - p| \leq \frac{1}{2N} + D^2.$$

13 Pretrigger disturbance belongs to the same ordinary model

The recoil and auxiliary calculation above supplies the exact ordinary auxiliary estimates, including all clock tails and backreaction. In particular,

$$\epsilon_{\text{pre}} \leq \frac{t_a^2(400000)}{4M_-^2}, \quad I_{\text{pre}} \leq 120K\epsilon_{\text{pre}} < 6 \times 10^{-18}.$$

Here I_{pre} is expected variation of the isolated rotor's lifted rank along the exact active trajectory. It is not the old source-inclusive logical-current quantity with a substituted constant.

The initial rotor marginal has density at most one because its variation is at most two. Its initial cumulative-rank density is at most two. The $2N$ pulled-back classifier boundaries therefore have actual strip mass at most $8Nr$. With $r = 10^{-9}$,

$$\epsilon_{\text{prelabel}} \leq 8Nr + \frac{16(6 \times 10^{-18})}{r} = .000002144.$$

The first term uses the actual marginal directly; the second uses the derived active domination factor16.

14 Copying the actual earlier label

Let B_X be the union of the intervals at relative distances $h/5$ to $3h/10$ between adjacent zero and one wells. Let $H_X \in [0, 1]$ be the quintic soft classifier, equal to the binary label outside these bands, with $|H'_X| \leq 20/h$.

During $[t_a, t_b]$, both Φ_i and $\partial_X \Phi_i$ vanish in these bands. Exact sector factorization gives

$$Q(X_{t_a} \in B_X) \leq D^2, \quad \mathbb{E}_Q \text{Var}_{[t_a, t_b]} H_X(X_t) \leq \frac{6}{5} DQ.$$

For clarity Q in the last product denotes the fixed number .003983; $\mathbb{E}_Q$ denotes wave-reference expectation.

Use $I_0 = [-8, 8]$, $I_1 = [16, 32]$ and define

$$G_{\text{mis}}(x, z) = 1 - (1 - H_X(x))\mathbf{1}_{I_0}(z) - H_X(x)\mathbf{1}_{I_1}(z).$$

For any continuous paths,

$$\mathbf{1}_{Z_{t_b} \notin I_{\ell(X_{t_a})}} \leq \mathbf{1}_{B_X}(X_{t_a}) + \text{Var } H_X(X_t) + G_{\text{mis}}(X_{t_b}, Z_{t_b}).$$

The comparison writer has completed its drive for every retained offset at t_b and its wrong-inner-region amplitude is below 10^{-12} . With the exact auxiliary error $e_A < 2 \times 10^{-13}$,

$$\mathbb{E}_Q G_{\text{mis}} \leq (D + e_A + 10^{-12})^2.$$

Theorem 14.1 (Actual historical copy). *Uniformly over the original actual class and all qubit inputs,*

$$\epsilon_{\text{copy}} = 16 \left[D^2 + \frac{6}{5} DQ + (D + 2 \times 10^{-13} + 10^{-12})^2 \right] = .00222725479707774720002304.$$

This bounds failure to copy the actual coupled relative rotor label at 1.22 into the pointer at 1.28.

The internal sector in this proof is an orthogonal wave component, not a sampled actual spin. On the full wave the pointer velocity depends on the actual relative rotor coordinate through its branch weights.

15 Whole-interval retention, including later rotor evolution

The outer regions are $\mathcal{R}_0 = [-10, 10]$ and $\mathcal{R}_1 = [14, 34]$. Four width-two bands separate them from the corresponding inner regions. After capture, the auxiliary comparison center stays within .012 of its intended value and has speed below .07. Its band wave and momentum norms are below 10^{-14} and 10^{-13} .

The exact rotor waves are used during retention; no harmonic comparison is continued through a caustic. Their unit norms suffice. With $e_A = 2 \times 10^{-13}$, $p_A = 3 \times 10^{-9}$, $\mu_- = .00999$, the total actual probability of leaving after starting in an inner region is bounded by

$$16 \cdot 4 \cdot \frac{15}{16} \cdot \frac{43}{25} \left[.07 10^{-28} + \frac{e_A 10^{-13} + 10^{-14} p_A + e_A p_A}{\mu_-} \right] < 10^{-16}.$$

This uses band-integrated absolute current and is simultaneous over [1.28, 3]; Appendix A supplies the cutoff and Gaussian-tail calculation. Later rotor revival or source phase evolution is not frozen. The same-support actual-law neighbourhood adds 10^{-4} once to the complete history event.

16 Finite complete theorem

Let $X = x - R$. Define $t_a = 1.22$, $t_b = 1.28$, $T_H = 3$ and

$$I_0 = [-8, 8], \quad I_1 = [16, 32], \quad \mathcal{R}_0 = [-10, 10], \quad \mathcal{R}_1 = [14, 34].$$

The record symbol is 0 or 1 in the corresponding outer region and $\perp$ elsewhere. The relative rotor classifier is halfway between the nominal sector centers. No observation is performed at t_a ; it identifies the earlier actual outcome which the pointer records.

16.1 A genuinely nonequilibrium admissible stock

On the readiness box take $f_0 = (1 + 0.8X)\mathbf{1}_{[-1,1]^2 \times [0,1/4]}$. Its mass is one, maximum is 1.8, directional variations are (1.8, 1, 8) and rotor half-probability is 0.7. Its clock is confined to one half of a symmetric wave packet, so the initial full-law TV distance from the wave measure is at least 1/2. An admitted source conditional is $(1 + \frac{1}{2} \operatorname{sgn} u_1)|\zeta|^2$. These are examples, not new assumptions imposed on the whole class. Invertibility of the full flow preserves fine-grained TV. Calibration of a coarse record therefore does not prove that the full ensemble equilibrates or that previously retained information is destroyed.

Part II

A radial apparatus with preparation and a writer from entrance

17 The fixed radial preparation operator

We now use physical units. Put $\ell_0 = 10^{-4}$ m, $T_0 = 0.03$ s, $m = 2^{52}\hbar T_0/\ell_0^2$, $M = 10$ kg and $v = 1$ m/s. The relative radial Hamiltonian has Dirichlet boundary at zero. The two ordinary Cartesian bodies each have mass $2m$. Their supplied normalized free centre Gaussian and constant angular factor give an exact s -wave reduction; the centre and constant angular factor separate from the active dynamics.

All smoothing choices are fixed. Let

$$\kappa_\epsilon(x) = Z_\epsilon^{-1} e^{-1/(1-(x/\epsilon)^2)} \mathbf{1}_{|x| < \epsilon}, \quad \Theta(x) = \begin{cases} 1 & x \leq 0, \\ [1 + e^{-1/x+1/(1-x)}]^{-1} & 0 < x < 1, \\ 0 & x \geq 1. \end{cases}$$

Here Z_ϵ normalizes the mollifier. Let B_9 be the clamped polynomial already defined, and let b be the convolution of $1 + 99B_9(s/0.1)$ with $\kappa_{10^{-5}}$. Its constant extensions are one and one hundred. Set $a = b'/b$; derivatives of these preparation functions are in clock time $s = S/v$. The distinct pointer displacement introduced below will always carry its own argument and derivative convention.

Take $L_i = 10^{-4}$ m, $L_f = 100L_i$. The radial comparison amplitude is

$$\chi(s, r) = \frac{2r}{\pi^{1/4}[L_i b(s)]^{3/2}} e^{-r^2/(2L_i^2 b(s)^2)}.$$

The initial wave uses χ on the $b = 1$ plateau, the specified centre Gaussian, an arbitrary normalized qubit, and a clock envelope centred at $S_c = -0.002$ m with carrier Mv . The envelope is the normalized convolution of $\sqrt{64/(35\sigma)} \cos^4(\pi\xi/(2\sigma)) \mathbf{1}_{|\xi| \leq \sigma}$, $\sigma = 10^{-3}$ m, with $\kappa_{10^{-10}}$. Its actual support halfwidth is 0.0010000001 m; it is not replaced by an unsmoothed packet.

Define an odd tent by $f_0(r) = r$ on $[0, R]$, $f_0(r) = 2R - r$ on $[R, 2R]$ and zero beyond, with $R = 0.201$ m. Set $f = f_0 * \kappa_{0.001}$ and $F(r) = 2 \int_0^r f(u) du$. Thus $f = r$, $F = r^2$ through 0.2 m, $f = 0$ from 0.403 m, $\|f\|_\infty \leq 0.201$ m, $\|f'\|_\infty \leq 1$ and $F_\infty \leq 0.080802$ m². Put

$$\omega_i = \hbar/(mL_i^2), \quad E_0 = 3\hbar\omega_i/2, \quad (2)$$

$$Q(s, r) = m\omega_i^2 r^2/(2b(s)^4) - E_0/b(s)^2, \quad (3)$$

$$Q_c(S, r) = \Gamma(S)\Theta((r - 0.2 \text{ m})/(0.001 \text{ m}))Q(S/v, r), \quad (4)$$

$$V_c(S, r) = -\frac{m}{2}a'(S/v)F(r) - \frac{m}{2}a(S/v)^2 f(r)^2 - \frac{m^2 a'(S/v)^2 F(r)^2}{8Mv^2}. \quad (5)$$

The clock cutoff Γ is one on $[-0.004, 0.105]$ m, zero outside $[-0.005, 0.106]$ m, with left factor $1 - \Theta((S + 0.005 \text{ m})/(0.001 \text{ m}))$ and right factor $\Theta((S - 0.105 \text{ m})/(0.001 \text{ m}))$. In particular, the clock-only far-radial part of V_c is retained. It is not silently subtracted as a constant.

In dimensionless radius $x = r/\ell_0$, let $h = 1/128$, $K = 2^{38}$, $\eta = 2^{-17}$ and let U_η be the capped quadratic of Part I. The physical measured potentials are

$$W_i^{\text{phys}}(r) = \frac{\hbar}{T_0} \Theta(x - 1000) \frac{K}{2} U_\eta(x/h - i/2).$$

The outer cutoff ends at 0.1001 m. Its cap transition width is 5.9604644775390625 picometres. The finite radial classifier uses the centred representative $(r/\ell_0) \bmod h \in [-h/2, h/2)$. It is zero when the absolute value of that representative is at most $h/4$, one otherwise, and undefined unless $0 \leq r/\ell_0 < 1000$. The boundary convention is the frozen one; ordinary nonnegative modulo is not substituted. There are 256000 internal cuts plus an exterior ray. Define $\gamma(s) = B_9((s - 0.104 \text{ s})/(0.0001 \text{ s}))$. The preparation-plus-activation operator is exactly

$$H_{\text{rad}} = \frac{p_S^2}{2M} + \frac{p_r^2}{2m} + V_c + (1 - \gamma(S/v))Q_c + \gamma(S/v) \sum_i \Pi_i W_i^{\text{phys}}. \quad (6)$$

It is this operator, not a subsequently fitted effective wave, that is enlarged by the pointer. Laboratory entrance is -0.106 s , handoff is -0.002 s , witness is 0.0366 s , and holding is $[0.0384, 0.09] \text{ s}$.

17.1 The full initial-law contract

The new radius is coupled to a periodic-reference initial coordinate by $X = (100r_0/\ell_0) \bmod 2$. Retain one complete reference law $\nu_{\text{ref}}(\text{d}X, \text{d}\zeta_{\text{ref}})$, where $\zeta_{\text{ref}} = (Z_{\text{ref}}, S_{\text{ref}}, u_{\text{ref}}, R_{\text{ref}}, C_{\text{ref}})$. The comparison rotor is fixed to the nominal Part I member: $o = d_0 = d_1 = 0$, $g_0 = g_1 = 1$, and reduced mass 2^{52} in its reference units. The radial theorem does not inherit the entire independent gain, offset and mass rectangle of Part I. It satisfies the periodic readiness and BV restrictions and, for this radial construction, the additional full reference cap $\nu_{\text{ref}} \leq 32\mu_{\text{ref}}$, where μ_{ref} is the complete original reference entrance wave measure. Its marginal obeys $f_X \leq 1$. The earlier admitted arbitrary singular passive laws are not all in this narrowed class.

Let $w_{\text{new}}(\text{daux} \mid X)$ be the normalized disintegration of the specified new entrance *wave measure*. The winding weights are $\chi_{100}(X + 2k)^2/W_{100}(X)$ for positive radius, together with the centre, angular, clock and pointer wave factors. The dimensionless amplitude is $\chi_{100}(y) = 2ye^{-y^2/(2 \cdot 100^2)}/(\pi^{1/4} 100^{3/2})$ for $y > 0$, and zero otherwise. Here

$$W_{100}(X) = \sum_{k: X+2k>0} \chi_{100}(X + 2k)^2 \geq \frac{67499}{135000} = W_{\min}.$$

Choose one normalized input-independent conditional measure

$$0 \leq g_{\text{new}}(\text{daux} \mid X, \zeta_{\text{ref}}) \leq 2w_{\text{new}}(\text{daux} \mid X).$$

The complete actual coupling is $\nu_{\text{ref}}g_{\text{new}}$. Correlations among new auxiliaries and with every retained reference variable are allowed. The new physical clock and pointer are not identified with the reference clock and pointer. Integration gives

$$C_{\text{micro}} = 2/W_{\min} = 270000/67499, \quad C_{\text{joint}} = 32/W_{\min} = 4320000/67499.$$

For the joint comparison measure use

$$\mu_{\text{joint}} = 2W_{100}(X) \mu_{\text{ref}}(\text{d}X, \text{d}\zeta_{\text{ref}}) w_{\text{new}}(\text{daux} \mid X).$$

The reference wave has uniform X marginal $\text{d}X/2$, so this is normalized. The two likelihood bounds then give $\nu_{\text{joint}} \leq (32/W_{\min})\mu_{\text{joint}}$. These are consequences of the complete coupling,

not substitutes for it. All original reference neighbourhood and projector-calibration allowances remain charged in the final comparison.

For a concrete nonequilibrium example, take the biased reference law of Part I and reference passives whose combined full likelihood ratio is below $23.04 < 32$. Choose $g_{\text{new}} = 2\mathbf{1}_{S>S_c}w_{\text{new}}$. Evenness of the entrance clock makes this conditional normalized for each X . It is input independent, satisfies the single cap and has actual-wave TV at least $1/2$. Thus the permitted class is genuinely nonequilibrium while remaining strongly restricted.

17.2 The preparation gauge and why it is legitimate

After the common Galilean carrier is removed, use the analytical unitary $e^{-i\theta}$ with $\theta = ma(S/v)F(r)/(2\hbar)$. Expanding the kinetic squares and (5) gives exactly

$$H_g = \frac{p_r^2}{2m} + \frac{p_S^2}{2M} + vp_S + \frac{1}{2}\{af, p_r\} + \frac{1}{2}\{\delta, p_S\} + Q_c + V_A, \quad \delta = \frac{ma'F}{2Mv}, \quad V_A = \gamma(W - Q_c).$$

The clock drift δ remains. During preparation write $\tau = t_{\text{lab}} + 0.106\text{ s}$ for elapsed time, so $\tau = 0$ at the entrance. The real helper $G_g = \phi(S - S_c - v\tau)\chi(S/v, r)$ has residual

$$\begin{aligned} R_{\text{core}} &= \frac{\hbar^2}{2M}[(\phi\chi)_{SS} + 2i\theta_S(\phi\chi)_S + i\theta_{SS}\phi\chi], \\ R_{\text{tail}} &= i\hbar a\phi[(f - r)\chi_r + (f' - 1)\chi/2] + (Q - Q_c)\phi\chi. \end{aligned}$$

The tails begin at twenty maximum packet widths and are bounded, never deleted. Since the helper has no remote activation support during preparation, activation acts on the error and is controlled by moving clock collars. Those collars are proof multipliers, not restrictions of the physical initial law.

For error momenta P_r, P_S and a right-collar amplitude n , the mass-weighted norm $Z_E = (P_r^2/m + P_S^2/M)^{1/2}$ satisfies a positive system $Z'_E \leq \gamma_0 Z_E + e_0 + c_0 dt + L_A n$, $n' \leq g_0 + b_0 Z_E$. The fixed weight $10^{-6}\sqrt{J}$ makes both row growth coefficients below 141 s^{-1} . Gronwall and Gaussian residual integrals then bound the exact gauged error. The stronger anisotropic and localized Hessian estimates used by the writer are derived in Appendix B. A scalar norm alone is never used to infer those derivatives.

18 Model, stock, and scope

The active configuration is $(S, r, Z) \in \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}$, with a two-component spinor. The relative radius r is the exact s -wave reduction of the frozen two-body scalar model. The free centre coordinate and constant spherical harmonic factor from the active Hamiltonian. The clock S and pointer Z are effective scalar coordinates; no three-dimensional lifting error for those two coordinates is asserted.

Write Π_0, Π_1 for the conserved qubit projectors. Retain the exact radial preparation baseline bounded-phase preparation, smooth activation, finite radial potentials, and all clock self terms. Denote that operator by H_{rad} . Its physical clock mass is $M = 10\text{ kg}$, its carrier speed is $v = 1\text{ m/s}$, and its reduced radial mass is

$$m_r = 2^{52}\hbar T_0/\ell_0^2, \quad T_0 = 0.03\text{ s}, \quad \ell_0 = 10^{-4}\text{ m}.$$

The scalar preparation operator is defined in Section 17. Its fixed cutoffs and admitted laws are part of the theorem. The appendices derive the specialized coefficient and current estimates used below.

The enlarged autonomous Hamiltonian is

$$H = H_{\text{rad}} + \frac{\hbar}{T_0} \left[\frac{-\partial_Z^2 + Z^2 - 1}{2\mu_Z} + \Pi_1\{-F(s)Z + C_b(s)\} \right], \quad (7)$$

where

$$\mu_Z = 0.01, \quad s = \frac{S - S_{\text{on}}}{vT_0}, \quad S_{\text{on}} = 0.14145 \text{ m}, \quad w = 0.003,$$

$$b(s) = 24B_9(s/w), \quad F = b/\mu_Z + \mu_Z b'', \quad C_b = b^2/(2\mu_Z) + \mu_Z b b''/2.$$

Here $B_9(x) = 126x^5 - 420x^6 + 540x^7 - 315x^8 + 70x^9$ on $[0, 1]$, extended by 0 and 1. Primes denote s derivatives. Thus the pulse width is $90 \mu\text{s}$ in the characteristic clock, and its last spatial point is 0.14154 m . The physical pointer mass is 10^{-19} kg ; its length unit is $\sqrt{\hbar\mu_Z T_0}/10^{-19} \text{ kg} \simeq 0.562469 \text{ nm}$. Every displayed compensation term remains in (7).

At laboratory time $t_0 = -0.106 \text{ s}$, the state is precisely the original full small-Gaussian radial and smooth compact clock stock, with its Galilean carrier and arbitrary normalized qubit, tensored with

$$g_0(Z) = \pi^{-1/4} \exp(-Z^2/2).$$

There is no wave truncation or later preparation reset. The original radial width is 10^{-4} m ; the clock is the positive 10^{-10} m mollification of its normalized $\cos^4$ packet of half-width 10^{-3} m . Its initial centre is $S = -0.002 \text{ m}$. The analytical handoff is $t_h = -0.002 \text{ s}$, with clock centre 0.102 m .

The narrowed nominal reference-law class of Section 17.1 is retained exactly. The pointer is included in the same normalized, input-independent joint auxiliary conditional rule $g_{\text{new}} \leq 2w_{\text{new}}$. This is one rule for the whole auxiliary configuration, allowing the declared correlations; it is not a separate cap or independent Born assignment for each added coordinate. Its disintegration gives

$$\nu_{\text{micro},0} \leq C|\Psi_0|^2 dq, \quad C = \frac{270000}{67499}. \quad (8)$$

The retained periodic reference marginal, its allowed neighbourhood, and its target-calibration allowance remain part of the hypotheses. The cap alone is not substituted for that complete reference-coupling contract.

19 Observable and theorem

Let L be the frozen radial detector classifier evaluated at the *actual* radius at $t_a = 0.0366 \text{ s}$, with the inherited cemetery outcome for a failed earlier event. Set

$$R_0 = [-10, 10], \quad R_1 = [14, 34], \quad R(Z) = \begin{cases} 0 & Z \in R_0, \\ 1 & Z \in R_1, \\ \dagger & \text{otherwise.} \end{cases}$$

The complete holding interval is $I = [0.0384, 0.09] \text{ s}$ and the observable is

$$Y = (L, (R(Z_t))_{t \in I}).$$

An incorrect or undefined record at any holding time is a failure. An undefined earlier label is a failure even if a cemetery record could be assigned the same symbol. No time grid replaces the interval.

Theorem 19.1 (Effective joint-path theorem). *For Hamiltonian (7), the full entrance stock above, every normalized qubit $\alpha_0|0\rangle + \alpha_1|1\rangle$, and every admitted original reference/joint auxiliary law, there is the unchanged compatible reference coupling for which*

$$\mathbb{P}(L \neq L_{\text{per}}) < 0.003331233001, \quad (9)$$

$$\mathbb{P}(L \text{ fails or } \exists t \in I : R(Z_t) \neq L) < 0.000552421956. \quad (10)$$

Writing $p = |\alpha_1|^2$ and $B \sim \text{Bernoulli}(p)$ only as an ideal comparison variable,

$$\text{TV}(\mathcal{L}(Y), \mathcal{L}(B, (B)_{t \in I})) < 0.006086770113 < 0.01. \quad (11)$$

The microscopic dynamics contains no sampled sector variable B .

Theorem 19.1 is about one effective scalar Hamiltonian. It is not a claim that the finite source candidate realizes this Hamiltonian. The difference to 0.01 is an unachieved sufficient allocation for a future compatible microscopic comparison, not a measured or proved error of added hardware.

20 A conservative flow for the effective model

For current-based existence background see [11]. The three active kinetic operators have positive coefficients. The pointer-linear term has bounded clock coefficients and is infinitesimally bounded relative to the free kinetic plus oscillator operator. Square completion gives a finite lower bound on the potential. Consequently the full operator is self-adjoint on that reference operator's domain.

After odd extension in r , the original stock belongs to $D(H^2)$. The writer coefficients vanish on its clock support. Functional calculus therefore gives $\Psi \in C_t D(H^2)$ and $\partial_t \Psi \in C_t D(H)$. The writer potential has the locally bounded weak second derivatives needed for local elliptic regularity: B_9 is C^4 , while its occurrence through b'' gives only the required C^2 statement. Thus $D(H^2) \subset H_{\text{loc}}^4$ and $D(H) \subset H_{\text{loc}}^2$. In three active dimensions these imply the local continuity and local Lipschitz velocity needed for a unique nonnodal guiding flow. No unjustified H^6 claim for the writer is used.

Energy conservation and $\|H\Psi\| < \infty$ give, on any finite interval,

$$\int dt \int |J_j| dq \leq \frac{\hbar}{m_j} \int dt \|\Psi\| \|\partial_j \Psi\| < \infty,$$

$$\int dt \int \rho |D_t \log \rho| dq \leq \int dt \left(2\|\Psi\| \|\partial_t \Psi\| + 2 \sum_j \frac{\hbar}{m_j} \|\partial_j \Psi\|^2 \right) < \infty.$$

Stopped partial equivariance and exhaustion exclude finite-time escape and node hitting almost surely. The origin is an odd-extension node. The free centre flow and constant angular factors then restore the two-body Cartesian description. Domination (8) transfers the null exceptional set and transports the same cap along this one flow. The much higher-dimensional interacting finite-field model requires its own existence proof.

21 Preparation and prewitness comparison

Appendix B proves the preparation estimates used here. The writer's presence during preparation is paid by a remote-support Duhamel estimate and weighted oscillator hierarchy, from the original stock. It gives the handoff norm change below 2.768×10^{-30} , first moving-annihilator norm below 5.050×10^{-24} , and second annihilator norm below 1.229×10^{-17} . The strengthened handoff clock-error derivative is

$$\|\partial_S^2(\Psi_{\text{new}} - \Psi_{\text{old}}g_0)\| < 1.829 \times 10^{19} \text{ m}^{-2}.$$

The proof uses ordinary fourth spatial derivatives of the smooth preparation error, with nonzero Gaussian tails and cutoff derivatives retained. It never assumes the capped reference entrance belongs to a fourth radial Hamiltonian graph domain.

The preparation history itself is rederived with the original explicit Gaussian material rank $K(S, r)$. Its Z derivative is zero, but its clock and radial currents are those of the new exact wave. Both phase restoration corrections are included. For $N = 256001$ rank cuts, including the exterior ray, the preparation event and handoff conditional-CDF bridge are respectively below

$$3.084950 \times 10^{-6}, \quad 6.1578014 \times 10^{-5}.$$

For the later prefix, use the constant Galilean frame for both waves: $\Psi_b = e^{-iMvS/\hbar + iMv^2t/(2\hbar)}\Psi$ and $H_b = H - i\hbar v\partial_S$, with the identical scalar terms. Let ψ_{old} denote the exact old wave in that frame, with $\psi_{\text{old},i} = \Pi_i\psi_{\text{old}}$ including its qubit coefficient. Define the normalized Weyl comparison $G_* = V(S)(\psi_{\text{old}}g_0)$, with sector-one factor

$$\gamma_1(s, Z) = \pi^{-1/4} \exp\left[-(Z - b)^2/2 + i\mu_Z b'(Z - b/2)\right]$$

and sector-zero factor g_0 . Direct differentiation gives the exact residual

$$(H_b - i\hbar\partial_t)G_* = -\frac{\hbar^2}{2M} \sum_i (2\psi_{\text{old},i,S}\gamma_{i,S} + \psi_{\text{old},i}\gamma_{i,SS}).$$

It is supported only on the pulse. The displaced stationary plateau has no residual source. Keeping the entire old clock tail and the leading edge of its characteristic stock gives

$$\sup_{t_h \leq t \leq t_a} \|\Psi_{\text{new},b} - G_*\| < 5.056 \times 10^{-22}.$$

This norm estimate is not used as a trajectory-agreement assertion.

For $\delta = \Psi_{\text{new},b} - G_*$, two fixed left localizations and the oscillator moment hierarchy give

$$\sup \|\delta_S\| \leq 0.1 \text{ m}^{-1}, \quad \sup \|\delta_{SS}\| < 1.118 \times 10^{21} \text{ m}^{-2}.$$

The differentiated residual includes the coefficients $1, 5/2, 2, 1/2$ on $\gamma_S\psi_{SSS}$, $\gamma_{SS}\psi_{SS}$, $\gamma_{SSS}\psi_S$, and $\gamma_{SSSS}\psi$. Its old-wave input is

$$\int_{t_h}^{t_a} \|\psi_{\text{old},SSS}\| dt < 9.747 \times 10^{46} \text{ s/m}^3.$$

On the remote preparation-force support the characteristic helper is identically zero, so propagation of this third clock derivative uses the already bounded old *error* derivatives there. All activation terms elsewhere remain.

22 Rank current and actual earlier label

Appendices C and D provide the quantitative inputs for this step. Use the new wave's own radial conditional CDF given the live pair (S, Z) . Its material derivative has clock and pointer contributions; the radial current cancels pointwise in the conditional-rank identity. The hidden-coordinate absolute values precede all integrations. The hybrid comparison yields

$$I_{\text{post}} = I_{\text{old}} + \Delta I_S + I_Z < 3.695139 \times 10^{-15}, \quad \Delta I_S < 4.547787 \times 10^{-16}, \quad I_Z < 2.574815 \times 10^{-15}.$$

The rank collar is optimized once on this sum, giving $2C\sqrt{2NI_{\text{post}}}$. Independently optimized component costs are not silently substituted for that expression.

The common good-clock event requires $|S_h - 0.102| \leq 0.000839 \text{ m}$ and whole-path clock deviation at most 10^{-5} m through 0.09 s . On that event $S_t < S_{\text{on}}$ for every prewitness time, with a strict $1 \mu\text{m}$ margin at t_a . Hence $G_* = \psi_{\text{old}}g_0$ there. The same event places the witness conditional age in $[355/300, 377/300]$, completes the writer by 0.038389 s , and keeps the clock beyond the pulse throughout the hold.

The earlier-label composition retains the new preparation event, both distinct exact-CDF bridges, the original conditional age-union and outer-ray events, the common good-clock complement, the actual radial exit event, and the unchanged conditional-reference endpoint comparison. The new absolute barrier estimate gives radial-exit probability below 9.127218×10^{-8} through readout. These terms prove (9) on the original reference coupling. Conditional rank uniformity is used only under its own wave law, then transferred with the same cap C ; no conditional Born law is assigned to actual states.

23 Copy, whole hold, and the direct path bound

Appendix E proves the complete copy and holding inequalities. The exact moving annihilator is

$$A = \partial_Z + Z - \Pi_1(b + i\mu_Z b').$$

With $c = b + i\mu_Z b'$ and the characteristic variables, its forced equation has the sole finite-clock source

$$(i\partial_\tau - \tilde{H} - 1/\mu_Z)A\Psi = -\epsilon\Pi_1(c'\partial_s + c''/2)\Psi, \quad \epsilon = \frac{\hbar}{Mv^2T_0}.$$

In the completed-clock region the pointer current is controlled by $\text{Im}(\Psi^\dagger A\Psi)/\mu_Z$. Positive inner pointer regions $[-8, 8]$ and $[16, 32]$, followed by soft barriers to R_0, R_1 , give an entire-interval estimate; recrossings are charged by absolute variation. A soft radial classifier on the same exact trajectories connects the conserved-sector proxy to the actual earlier position label. It does not replace that label by a sampled logical bit.

The copy proof retains its baseline radial band current, finite-clock radial correction, positive pointer-inner event, entire hold current, and the shared clock-core, clock-deviation and radial-exit exceptions. The hold-current term is below 3.272×10^{-9} ; the full sum is (10). Every term concerns the new Hamiltonian and original entrance law.

In the union of failed earlier-label transfer and failed record history, the same three exceptional events need be charged only once. Their identity is explicit in both proofs; this is a union of events, not subtraction of unrelated numerical upper bounds. The outward arithmetic in Section 26 gives

$$p_{\text{union}} < 0.003879184824.$$

The preserved reference earlier-label estimate, including both original 0.0001 allowances, is $e_{\text{label},26} = 0.002207585289$. Coupling first to the correct constant record of L_{per} and then to its target Bernoulli label proves

$$\text{TV}(\mathcal{L}(Y), \mathcal{L}(B, (B)_{t \in I})) \leq p_{\text{union}} + e_{\text{label},26} < 0.006086770113.$$

This establishes the whole observable of Theorem 19.1.

The former comparison to the entire imperfect periodic reference apparatus has the different sufficient upper expression

$$p_{\text{union}} + h_{\text{per}} < 0.006206439621, \quad h_{\text{per}} = 0.00232725479707784720002304.$$

That expression does not meet the old target 0.00556515991392215279997696. Its failure is not an actual error lower bound. The theorem uses the expressly permitted direct ideal joint-path alternative.

24 Conditional-rank stability with zero fibres included

This section supplies the nonlinear step behind the radial proof. It is useful precisely because the actual clock/pointer marginal need not have a positive lower bound. Let $q = \psi(S, \cdot)$ be a radial spinor slice, $u = \psi_S$, $z = \psi_{SS}$, $a = \|q\|$, $\eta = q/a$, and let P_r multiply by $\mathbf{1}_{[0,r]}$. Set $K = \langle \eta, P_r \eta \rangle$ and $Z_\eta = (P_r - K)\eta$. The constant clock carrier is removed from all derivatives, but its actual drift is retained in continuity.

Lemma 24.1 (Exact weighted rank current). *On nonzero fibres, with $\kappa = \hbar/M$,*

$$\rho \frac{dK}{dt} = \mathcal{F}[\psi] = \kappa \left\{ 2\Im(\eta(r)^\dagger u(r))\Re\langle Z_\eta, u \rangle - a|\eta(r)|^2\Im\langle Z_\eta, z \rangle \right\}. \quad (12)$$

For two live conditioning coordinates S, Z , the right side is the sum of the corresponding two functionals, with coefficients $\hbar/M$ and $1/(\mu_Z T_0)$ respectively.

Proof. Write $w = \int \rho dr$, $A_r = \int_0^r \rho dy$, and $K = A_r/w$. Let δB and δj be the cumulative and full clock-current integrals after subtracting the constant carrier. Continuity and zero origin flux give

$$\frac{dK}{dt} = \frac{\delta J}{\rho} K_S - \frac{\partial_S \delta B - K \partial_S \delta j}{w}.$$

The radial current has cancelled pointwise against $K_r \dot{r}$. Now $K_S = 2\Re\langle q, (P_r - K)u \rangle / a^2$ and $\partial_S \delta B - K \partial_S \delta j = \kappa \Im\langle q, (P_r - K)z \rangle$; the $u^\dagger u$ term is real. Substitution proves (12). Applying continuity with both transverse currents gives the two summands, not an additional mixed-derivative term. Each local product sums the original spinor components. No component has been sampled as an actual sector. $\square$

Lemma 24.2 (Stability without an exact-marginal denominator). *For normalized ψ, G , write $E = \psi - G$, and suppose the quantities*

$$D = \|E\|, \quad e_1 = \|E_S\|, \quad e_2 = \|E_{SS}\|, \quad g_j = \|\partial_S^j G\|, \quad Q_G^2 = \int \frac{\|G_S(S, \cdot)\|^4}{\|G(S, \cdot)\|^2} dS$$

are finite. The ratio is zero where G and its required derivatives vanish. Then

$$\|\mathcal{F}[\psi] - \mathcal{F}[G]\|_1 \leq \frac{\hbar}{M} (4g_1 e_1 + 2e_1^2 + 14DQ_G + 9Dg_2 + e_2). \quad (13)$$

Here $\mathcal{F}[G]$ is algebraic evaluation, including whatever residual the comparison has under the actual equation.

Proof. At a fixed slice put $g = G(S, \cdot)$, $b = \|g\|$, $\xi = g/b$, $U = G_S$, $V = G_{SS}$, and $d = \|q - g\|$, $d_1 = \|u - U\|$, $d_2 = \|z - V\|$. The elementary estimates

$$\|\eta - \xi\| \leq 2d/b, \quad \|Z_\eta\| \leq 1/2, \quad \|Z_\eta - Z_\xi\| \leq 3\|\eta - \xi\|, \quad \left\| |\eta|^2 - |\xi|^2 \right\|_1 \leq 2\|\eta - \xi\|$$

follow by adding and subtracting q/b , by projection variance, and by $|K_\eta - K_\xi| \leq 2\|\eta - \xi\|$. Split the first term of (12) first in $u - U$ and then in normalized directions. Its derivative part is at most $4\|U\|d_1 + 2d_1^2$; its direction part is at most $7\|\eta - \xi\|\|U\|^2 \leq 14d\|U\|^2/b$. For the second term, the $z - V$ part is at most ad_2 . Splitting the remaining amplitude, density and projection yields $(d/2 + 4b\|\eta - \xi\|)\|V\| \leq 9d\|V\|$. Integrate these inequalities and use Cauchy–Schwarz and $\int a^2 dS = 1$. If $a = 0$, direct bounding of $\mathcal{F}[G]$ works with $d = b$; if $b = 0$ and its derivatives vanish, the bound $2d_1^2 + ad_2$ suffices. Thus zero fibres are included, not discarded. $\square$

For $G = \phi(S - S_c - vt)u(S, r)$ with real ϕ and normalized u ,

$$Q_G \leq \left\| \phi'^2 / \phi \right\|_2 + \sup_S \|u_S\|^2, \quad \left\| \phi_0'^2 / \phi_0 \right\|_2 = \frac{4\pi^2}{\sigma^2} \sqrt{\frac{3}{35}}.$$

The last equality is a direct trigonometric integral for the unsmoothed cosine packet. Positive convolution and weighted Cauchy–Schwarz give $\phi'^2 / \phi \leq Z^{-1} \kappa * (\phi_0'^2 / \phi_0)$; Young’s inequality then adds only the normalization multiplier at most 1.000001. No actual conditional law is asserted uniform in this argument.

Lemma 24.3 (Deterministic-time CDF interface). *For every measurable radial cut $b(S)$,*

$$\int w(S) |K_\psi(S, b(S)) - K_G(S, b(S))| dS \leq 2\|\psi - G\|.$$

Under the already transported actual-law cap C , the corresponding binary-test mismatch is at most $2C\|\psi - G\|$ per cut.

Proof. Multiply the CDF difference by $w = \int |\psi|^2$. The exact identity is

$$w(K_\psi - K_G) = \int (\mathbf{1}_{r \leq b} - K_G)(|\psi|^2 - |G|^2) dr.$$

The coefficient has modulus at most one and the density difference has L^1 norm at most $2 \|\psi - G\|$. Uniformity of the exact rank under its own conditional *wave* measure identifies this integral with the test mismatch. Domination then supplies the actual-law bound. Zero comparison fibres are charged by their exact mass. The same proof works with live conditioning pair (S, Z) . It is a fixed-time result; it is not substituted at a random clock hitting time. $\square$

25 How the exact writer controls the new rank terms

Let $G_* = V(S)(\psi_{\text{old}}g_0)$ and $\delta = \Psi - G_*$ as above. Throughout this prefix section $T = t_a - t_h = 0.0386$ s, with elapsed time measured from the handoff $t_h = -0.002$ s. On $S < S_{\text{on}}$ the displacement and its derivatives are the identity. Every hybrid functional integral in this section is restricted to $Q_{\text{pre}} = \{S < S_{\text{on}}\}$ before integration. The separately charged good-clock event keeps actual prefix trajectories in that set. Global error and moment norms may upper-bound these restricted integrals, but $G_* = \psi_{\text{old}}g_0$ and the corresponding small pointer-jet substitution are asserted only there. No derivative of the region's indicator is taken, and no full-space equality of the two references is being asserted.

For the clock component use the original characteristic helper and its already proved positive current majorant. Inserting the sum of its old error and δ into (13) gives the increment

$$\Delta I_S \leq \frac{\hbar}{M} T [4(g_{1,o} + e_{1,o})d_S + 2d_S^2 + 14dQ_{G,o} + 9dg_{2,o} + d_{SS}].$$

For the pointer component use the *different* reference $\psi_{\text{old}}g_0$, whose radial direction is independent of Z . Its pointer rank functional is exactly zero. Gaussian integration gives $g_{1,Z} = 1/\sqrt{2}$ and $g_{2,Z} = Q_Z = \sqrt{3}/2$. Therefore

$$I_Z \leq \frac{T}{\mu_Z T_0} [2\sqrt{2}d_Z + 2d_Z^2 + (23\sqrt{3}/2)d + d_{ZZ}].$$

The references are chosen separately because each has the appropriate proved coefficients; no wave or current is transferred between them without the displayed identity.

Write $A = \partial_Z + Z - \Pi_1 c(s)$, $c = b + i\mu_Z b'$. On the quiet prefix $A = B = \partial_Z + Z$, and $B\delta = A\Psi$, $B^2\delta = A^2\Psi$. Oscillator integration by parts gives

$$d_Z \leq \sqrt{A_1^2 + d^2}, \quad d_{ZZ} \leq \sqrt{A_2^2 + 4A_1^2 + 3d^2}, \quad A_j = \|A^j \Psi\|.$$

Indeed $\|(p_Z^2 + Z^2)f\|^2 = \|B^2 f\|^2 + 4\|Bf\|^2 + \|f\|^2$, while $\Re\langle p_Z^2 f, Z^2 f \rangle = \|Zf'\|^2 - \|f\|^2$. All identities hold on the complete vector-valued fibres before integration. A clock indicator is not differentiated.

25.1 Derivative closure and its proof

Appendix D supplies the Gaussian polynomial recursion, the four simultaneous positive moment inequalities, and the strict supersolution tests for the same new wave. It retains the additional $-\kappa\Pi_1(c')^2\Psi/(vT_0)^2$ source in the $A^2\Psi$ equation. The localized calculation then improves the pointer-weighted first clock-error estimate to $Q(\delta_S) < 616735 \text{ m}^{-1}$ and proves

$$\sup \|\delta_S\| < 0.1 \text{ m}^{-1}, \quad \sup \|\delta_{SS}\| < 1.117215 \times 10^{21} \text{ m}^{-2}.$$

The old third-clock input and all differentiated residual coefficients are included there. Appendix B derives the original-stock handoff data, including the fourth-spatial estimate and its Gaussian-tail terms. Thus these derivative bounds are established for the declared entrance wave, rather than imposed on a replacement handoff wave.

26 Outward arithmetic and the complete event sum

The preceding arguments identify each error as an event or a nonnegative current functional under the same initial coupling. The preparation, old-wave, new-prefix and copy/hold appendices specify their coefficient formulae and evaluation bounds. No numerical certificate is used in place of an analytic inequality.

For reproducible square-root enclosures, given a nonnegative rational x and $n = 10^{60}$, choose the least integer k with $k^2 \geq n^2 x$; then k/n is an upper bound. Polynomial integrations are exact rational operations, and positive Taylor remainders or explicit derivative bounds control the nonpolynomial terms as detailed in the appendices. The following current ceilings have been rounded outward before combining:

$$\begin{aligned} I_{\text{old}} &< 6.655453611235769 \times 10^{-16}, \\ \Delta I_S &< 4.547786946512146880 \times 10^{-16}, \\ I_Z &< 2.574814383253091344 \times 10^{-15}. \end{aligned}$$

The rank row below is evaluated as $2C\sqrt{2N(I_{\text{old}} + \Delta I_S + I_Z)}$, with $C = 270000/67499$ and $N = 256001$. Every displayed table entry is itself a rational upper bound; unlike a shortened diagnostic display, these entries may be added directly. The harmless rounding of the pointer-inner event is deliberately included in the sum.

Event or contribution	Outward upper bound
Conditional reference endpoint	0.002588532236512050
Preparation rank	0.000003084949592861
Initial exact CDF	0.000061578013525596
Conditional age union	0.000263432327209778
Conditional exterior union	0.000000015476669581
Terminal exact CDF	0.000062145214303409
Aggregate own-rank history	0.000347974650032331
Copy baseline	0.000537032962395582
Finite-clock copy correction	0.000010915589215099
Positive pointer-inner event	0.000000000000000001
Holding current	0.000000003271569844
Shared clock core	0.000004378860028174
Shared whole-interval clock deviation	0.000000000000106690
Shared actual radial exit	0.000000091272176629

The first seven rows belong to earlier-label transfer, the next four to copying and retention, and the last three to both arguments. In both proofs the last three are the same clock-core, entire-clock-deviation and actual-radial-exit events of the new wave. Their sum is therefore charged once in the union. Direct addition of the displayed rational ceilings gives

$$\begin{aligned} p_{\text{label}} &\leq 0.003331233000157099 < .003331233001, \\ p_{\text{copy/hold}} &\leq 0.000552421955492019 < .000552421956, \\ p_{\text{union}} &\leq 0.003879184823337625, \\ p_{\text{union}} + .002207585289 &\leq 0.006086770112337625 < .006086770113. \end{aligned}$$

The reference-label ceiling .002207585289 already includes both original 10^{-4} allowances. Applying Lemma 2.1 proves the advertised joint-record bound without counting either allowance again. The explicit outward sums also show that the last displayed headline digits do not depend on substituting rounded intermediate values into an unreported computation.

27 Comparison and physical boundary

The two theorems share a statistical and current-based proof strategy, but their constants cannot be interchanged. The periodic theorem allows its explicitly admitted inaccessible quantum reference and singular passive-coordinate laws. The radial theorem is stated for a normalized two-component qubit with no arbitrary inaccessible quantum-reference extension, and narrows the reference law to a full-cap class before adding its single joint auxiliary conditional. Its small entrance, preparation, finite radial cutoff and added writer require new proofs.

The periodic apparatus does not generate its prescribed potential from the retained elastic modes: they are decoupled spectators. The radial apparatus does not acquire a finite material source merely because a separate field candidate exists. Source preparation, actual-position loading or proxy agreement after that enlargement, finite propagation, clock/pointer support and whole-interval retention must be shown for one combined Hamiltonian. The unspent distance below one percent is only a possible allocation in such a future proof.

28 Conclusion

The achieved statements concern calibrated, historically faithful finite records under explicit deterministic dynamics and restricted nonequilibrium laws. Their mathematical content can be evaluated without adopting a broader philosophical framework. What is not derived is the physical provenance of every initial-law restriction, the selection of guidance among alternative dynamics, or an established-matter realization of every interaction. Independent mathematical review should first test the precise current/domain and event-composition claims under the stated assumptions. A physical programme would additionally need a material model and its own quantitative reduction to the retained-record observable.

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A Explicit estimates for the periodic apparatus

This appendix supplies the polynomial and Gaussian calculations used in Part I. All parameter bounds refer to that part's finite rectangle. They do not enlarge the law class of Part II.

A.1 Polynomial derivatives and clock normalization

Writing $z = 1 - 2a$, direct differentiation gives

$$B'_9(a) = 630a^4(1-a)^4, \quad B''_9(a) = \frac{315}{8}z(1-z^2)^3, \quad B'''_9(a) = -\frac{315}{4}(1-z^2)^2(1-7z^2).$$

The first maximum is $315/128$; the second occurs at $|z| = 1/\sqrt{7}$ and is below 10. For the third, the only interior stationary squared arguments are 0 and $3/7$, giving a bound $315/4$. Consequently the writer displacement has derivative bounds

$$\|b'\|_\infty < 1478, \quad \|b''\|_\infty < 150200, \quad \|b'''\|_\infty < 2.958 \times 10^7.$$

For the four envelope derivatives the exact integrals $J_k = \int_0^1 (B_9^{(k)}(a))^2 da$ are

$$(J_1, J_2, J_3, J_4) = \left(\frac{4410}{2431}, \frac{5040}{143}, \frac{272160}{143}, \frac{1814400}{11} \right).$$

Also $\int A_c^2 = 176818/230945 = Z_c$. The change of variable on both transition intervals gives $\|\varphi^{(k)}\|^2 = 2 \cdot 5^{2k-1} J_k / Z_c$. Squaring the four claimed upper bounds 6, 112, 4000, 190000 verifies all four inequalities using rational arithmetic alone.

A.2 Driven Gaussian jets

Let $d = v\lambda$, $r = y - q$, and let u denote the normalized coherent Gaussian in $G = \varphi(\xi)e^{-it/(2\mu)}u$. The start of every characteristic in the envelope support precedes the drive. Thus the characteristic solution depends on $t + \xi/d$, and differentiation in ξ is time differentiation divided by d on q, q', ϑ . The forcing $\mathcal{F} = \Omega^2 b + b''$ and the error $e = q - b$ obey

$$|e| \leq \frac{|1 - \lambda^2| 2\|b'\|_\infty}{\lambda\Omega} < .001, \quad |e'| \leq \frac{|1 - \lambda^2| 2\|b'\|_\infty}{\lambda} < .07.$$

Indeed the total variation of the single positive hump b' is $2\|b'\|_\infty$; integrate the sine and cosine representations of e, e' against $b''(\lambda t + \xi/v)$. It follows that $|q| < 25$, $|q'| < 1630$, $|q''| < 170000$, and $|q'''| < 36000000$, using $q'' = b'' - \Omega^2 e$ and $q''' = \lambda b''' - \Omega^2 e'$. Moreover $|\mathcal{F}| < 421000$ and $|\dot{\mathcal{F}}| < 5.24 \times 10^7$.

Define the complex coefficient a_1 and real coefficient β by

$$a_1 = \frac{q' + i\mu q''}{d}, \quad \beta = \frac{\mu}{2d}(qq'' - q'^2 + \mathcal{F}e).$$

Direct differentiation, including the scalar compensation phase, gives

$$\begin{aligned} u_\xi &= (a_1 r + i\beta)u, \\ u_{\xi\xi} &= \left[a_1^2 r^2 + ((a_1)_\xi + 2i\beta a_1)r + i\beta_\xi - a_1 q_\xi - \beta^2 \right] u, \\ (a_1)_\xi &= (q'' + i\mu q''')/d^2, \\ \beta_\xi &= \frac{\mu}{2d^2}(qq''' - q'q'' + \dot{\mathcal{F}}e + \mathcal{F}e'). \end{aligned}$$

Since $d \geq 126.99873$, substitution of the preceding bounds yields

$$|a_1| < 19, \quad |(a_1)_\xi| < 25, \quad |\beta| < 273, \quad |\beta_\xi| < 367, \quad |q_\xi| < 13, \quad \mu|q'| < 17.$$

For example the numerator bounding β is $\mu_+(25 \cdot 170000 + 1630^2 + 421000 \cdot .001)$; for β_ξ it is $\mu_+(25 \cdot 36000000 + 1630 \cdot 170000 + 5.24 \cdot 10^7 \cdot .001 + 421000 \cdot .07)$. The denominators are respectively $2d_-$ and $2d_-^2$.

For the Gaussian measure $|u|^2 dr$, $\|r^k u\|^2 = (2k-1)!!/2^k$. The quadratic polynomial in $u_{\xi\xi}$ therefore has coefficient moduli bounded by 361, 10399, 75143. Applying the triangle inequality to these three monomials and to their derivatives proves

	u_ξ	$u_{\xi\xi}$
$\ \cdot\ $	287	83000
$\ r \cdot\ $	211	63000
$\ p_y \cdot\ $	5110	1485000

as strict upper bounds. Here $p_y(P(r)u) = [-iP'(r) + (ir + \mu q')P(r)]u$, which explains the last row without omitting the phase momentum. Leibniz' rule now gives

$$\begin{aligned} \|G_{\xi\xi}\| &< 112 + 12(287) + 83000 = 86556 < 110000, \\ \|p_y G_{\xi\xi}\| &< 112(18) + 12(5110) + 1485000 < 1.55 \cdot 10^6, \\ \|y G_{\xi\xi}\| &< 25(86556) + 112/\sqrt{2} + 12(211) + 63000 < 2.23 \cdot 10^6. \end{aligned}$$

These explicit calculations prove both weighted bounds used in Lemma 8.1.

A.3 Rotor enclosures and the quiet-prefix current

For completeness, an outward square-root operation on a nonnegative rational x can be defined using integers alone. Set $n = 10^{32}$, $k = \lfloor \sqrt{n^2 x} \rfloor$, and return k/n if $k^2 \geq n^2 x$, otherwise $(k+1)/n$. This is an upper enclosure whose error is at most 10^{-32} . Applying it to the four rational squares in Lemma 12.1 gives

$$\begin{aligned} d_0 &< .006517177225775, & \epsilon_R &< .000715486047289, \\ d &< .007232663273064, & q &< .003982026605165. \end{aligned}$$

The strict comparisons with $D = .007233$ and $Q = .003983$ follow. The trigonometric estimates used there also have elementary rational checks: alternating Taylor sums through degrees 14, 16 bound $\cos(1.3)$, and degrees 15, 17 bound $\sin(1.3)$. They give $\cos(1.3) > 1/4$, $\tan(1.3) < 4$, and $\sec(1.3) + \tan(1.3) < 8$. Since the positive exponential Taylor sum through degree 14 at 2.1 exceeds 8,

$$\int_0^{1.3} \sec^3 t \, dt = \frac{1}{2}[\sec(1.3) \tan(1.3) + \log(\sec(1.3) + \tan(1.3))] < 8.$$

The same cosine upper sum at .998(1.22) gives $\cos(.998(1.22))/2 + .01 < 1/5$, placing both compact waves strictly inside their respective soft-classifier plateaux.

The Sobolev constants in the quiet-prefix estimate require no private auxiliary result. For a normalized sector rotor r_i with $0 \leq V_i \leq g_i K/8$, energy and graph-norm conservation from the initial flat wave give

$$\|r'_i\|^2 \leq m g_i K/4, \quad \|r''_i\| \leq m g_i K/2.$$

Thus $\|r'_i\| \leq K/h$ and $\|r''_i\| \leq K^2/h^2$. On a circle of length two, $\|f\|_\infty^2 \leq \|f\|^2/2 + 2\|f\|\|f'\|$. Apply this first to r_i and then to r'_i to obtain $\|\rho_r\|_\infty \leq 3K/h$ and $\|j_r\|_\infty \leq 8K$. The factorized pretrigger wave errors then give

$$\|\delta j_X\|_1 \leq 4h\epsilon_{\text{pre}}, \quad \|\delta \rho\|_1 \leq 3\epsilon_{\text{pre}},$$

and hence the lifted-rank integrand is at most $36K\epsilon_{\text{pre}}$. Integrating up to 1.22 is bounded by $120K\epsilon_{\text{pre}} < 6 \cdot 10^{-18}$ as used in the main text.

A.4 Gaussian tails and the whole holding interval

After capture, every comparison centre is within .012 of its intended centre, including the offset z_0 . For $a > 0$, integration by parts gives

$$\|\mathbf{1}_{|r|\geq a}\gamma\|^2 \leq \frac{e^{-a^2}}{a\sqrt{\pi}}, \quad \|\mathbf{1}_{|r|\geq a}r\gamma\|^2 \leq \frac{e^{-a^2}}{\sqrt{\pi}} \left(a + \frac{1}{2a}\right).$$

Take $a = 7.98$. The positive Taylor sum of e^{a^2} through degree 200 exceeds $4 \cdot 10^{27}$. With $\mu|q'| < .000701$, these inequalities give comparison tail norms below 10^{-14} and momentum norms below 10^{-13} on all four holding bands. They also imply the weaker wrong-inner-region amplitude bound 10^{-12} at capture.

For each inner/outer pair choose a smooth cutoff changing from zero to one across the intervening band of width two, with $|\chi'| \leq 15/16$. Any path that starts in the inner region and leaves its outer region has cutoff variation at least one. Equivariance therefore bounds its wave probability by $\int_{t_b}^3 \int |\chi'(Z)j_Z| dq dt$. This is a band integral of absolute current; no pointwise boundary trace estimate is needed. Expanding the exact auxiliary wave as $G + E$ bounds its integrand by the comparison current plus $(e_A 10^{-13} + 10^{-14} p_A + e_A p_A)/\mu_-$. Summing the four cutoffs, multiplying by the actual-law cap 16, and using $3 - 1.28 = 43/25$ proves the displayed holding bound in Part I.

B Preparation estimates and transfer to the enlarged Hamiltonian

All momenta in this appendix have the constant Galilean carrier removed. A superscript g additionally removes the bounded preparation phase θ . Thus an ordinary clock momentum here means $p_S - Mv$ in the laboratory frame. We give the finite coefficient construction as well as its numerical enclosures; the latter are consequences of the construction, not extra hypotheses.

B.1 Coefficient conventions and the Gaussian residual

Write $T = 0.104$, $T_0 = 0.03$, $L_i = 10^{-4}$, $L_f = 10^{-2}$, $M = 10$, $v = 1$, $K = 100$, and $\lambda = 2^{52}T_0/L_i^2$, so that $m = \hbar\lambda$. Lengths and times in this appendix are in metres and seconds. Let b_p denote the preparation dilation, to distinguish it from the writer displacement. Set $a = b'_p/b_p$, $f_* = 0.201$, $F_* = 0.080802$, $g_1 = 0.0005$, $g_2 = 0.00025$ and $\gamma_* = 141$. Here $F(r) = 2 \int_0^r f$ is the bounded preparation profile; the subscript on F_* in this subsection is unrelated to the writer force. Here $b_0(s) = 1 + 99B_9(s/0.1)$ has its constant extensions, and b_p is its convolution with the fixed 10^{-5} -scale mollifier. The explicit clamped ninth-degree polynomial and its fixed positive mollifier give the derivative bounds

$$B_1 = 99(315/128)/0.1, \quad B_2 = 99(2520/64)/0.1^2, \quad (14)$$

$$B_3 = 99(7560/16 + 5040/64)/0.1^3, \quad B_4 = 99(1360800)/0.1^4. \quad (15)$$

The sharper relative bounds $b'_p \leq 140b_p$ and $|b''_p| \leq 24000b_p$ follow from polynomial positivity before convolution. One finite verification is to express each of $140b_0 - b'_0$, $24000b_0 - b''_0$, $24000b_0 + b''_0$ in Bernstein form on dyadic subintervals of $[0, 1]$; bisect a subinterval if any coefficient is negative. All leaves are nonnegative by depth five. For reproducibility, if $P(x) = \sum_{j=0}^n p_j x^j$, its Bernstein coefficient of index k on $[u, u+w]$ is

$$\sum_{j=0}^k \frac{\binom{k}{j}}{\binom{n}{j}} \sum_{l=j}^n p_l \binom{l}{j} u^{l-j} w^j.$$

Positive convolution preserves these inequalities. Put

$$a_0 = 140, \quad (16)$$

$$a_1 = 24000 + a_0^2 = 43600, \quad (17)$$

$$a_2 = B_3 + 3a_0(24000) + 2a_0^3 = 70141750, \quad (18)$$

$$a_3 = B_4 + 4a_0B_3 + 3(24000)^2 + 12a_0^2(24000) + 6a_0^4 = 1387431060000. \quad (19)$$

These bound $|a^{(j)}|$. Let B_5 be the largest absolute Bernstein coefficient on $[0, 1]$ of d^5b_0/ds^5 and set

$$a_4 = B_5 + 5a_0B_4 + 10(24000)B_3 + 20a_0^2B_3 + 30a_0(24000)^2 + 60a_0^3(24000) + 24a_0^5.$$

The normalized clock envelope has derivative norms bounded by

$$\begin{aligned} p_0 &= 1, & p_1 &= q\sqrt{16/7}\pi/(2\sigma), & p_2 &= q\sqrt{512/35}[\pi/(2\sigma)]^2, \\ p_3 &= q\sqrt{1024/7}[\pi/(2\sigma)]^3, & p_4 &= q(2\pi/\sigma)^4, \end{aligned}$$

where $\sigma = 0.001$, $q = 1.000001$. The convolution scale is 10^{-10} and $q(1 - 10^{-10}p_1/q) > 1$, which proves the normalization allowance. These are weak-derivative estimates for the compact envelope, not a band-limit assertion.

Let $A = 2r \exp[-r^2/(2L^2)]/(\pi^{1/4}L^{3/2})$, $L = L_i b_p(S)$, and $G = \phi(S - S_c - vt)A$. Set $z = (r/L)^2$, $M_j = (2j+1)!!/2^j$ and $\nu_j = \sqrt{M_{2j}}$. Thus $\|z^j A\| = \nu_j$. For a polynomial $P = \sum c_j z^j$ define $\mathcal{N}(P) = \sum |c_j| \nu_j$. This coefficient norm is deliberately subadditive; signs of independently bounded derivatives cannot cancel. Use

$$H_1 = z - \frac{3}{2}, \quad H_2 = z^2 - 5z + \frac{9}{4}, \quad H_3 = z^3 - \frac{21}{2}z^2 + \frac{79}{4}z - \frac{27}{8}.$$

Put $k = B_1$, $k_1 = B_2 + B_1^2$, $k_2 = B_3 + 3B_1B_2 + 2B_1^3$, and

$$\begin{aligned} C &= \lambda L_i^2(100B_2 + B_1^2)/2, \\ C_k &= \lambda L_i^2(B_1B_2 + B_1^3)/2, \\ C_2 &= \lambda L_i^2(100B_3 + 3B_1B_2 + 2B_1^3)/2, \\ C_3 &= \lambda L_i^2(100B_4 + 4B_1B_3 + 3B_2^2 + 12B_1^2B_2 + 6B_1^4)/2. \end{aligned}$$

The three nonnegative coefficient bounds

$$\begin{aligned} r_0 &= p_2 + 3kp_1 + \frac{3}{2}k_1 + \frac{9}{4}k^2, \\ r_1 &= 2kp_1 + k_1 + 5k^2 + 2Cp_1 + 3C_k + C_2, \\ r_2 &= k^2 + 2C_k \end{aligned}$$

give the residual norm coefficient $R_0 = \sum_{j=0}^2 r_j \nu_j$. For its radial derivative set

$$R_r = \sum_{j=0}^2 r_j [2j\sqrt{M_{2j-1}} + \sqrt{M_{2j+1}}],$$

where the first summand is zero at $j = 0$. This follows equally by the unitary radial reduction of the three-dimensional Gaussian gradient; its integration-by-parts identity includes the reduced amplitude's factor r . For the clock derivative set

$$\begin{aligned} U_1 &= k\mathcal{N}(H_1), \\ U_2 &= k_1\mathcal{N}(H_1) + k^2\mathcal{N}(H_2), \\ U_3 &= k_2\mathcal{N}(H_1) + 3kk_1\mathcal{N}(H_2) + k^3\mathcal{N}(H_3), \\ R_S &= p_3 + 3p_2U_1 + 3p_1U_2 + U_3 + 2p_2C\nu_1 + 4p_1kC\mathcal{N}(zH_1) + 3p_1C_2\nu_1 \\ &\quad + 2C[k_1\mathcal{N}(zH_1) + k^2\mathcal{N}(zH_2)] + 3kC_2\mathcal{N}(zH_1) + C_3\nu_1. \end{aligned}$$

These are obtained by differentiating the complete gauged residual

$$\frac{R^g}{\hbar} = \frac{\hbar}{2M} [G_{SS} + 2i\theta_S G_S + i\theta_{SS} G] + ia\phi[(f-r)A_r + (f'-1)A/2] + (Q - Q_c)G/\hbar. \quad (20)$$

In particular, no derivative of a restored phase is added to this *gauged* residual.

Here are all coefficients needed to include its tails and the weak trap. Write $\omega_i = (\lambda L_i^2)^{-1}$, $q_2 = m\omega_i^2/2$, $q_0 = 3\hbar\omega_i/2$, $q_{2S} = 4a_0q_2$, $q_{0S} = 2a_0q_0$, $r_c = 0.201$, $c_1 = 5000$, and $\epsilon_{20} = 10^{12}/7^{100}$. Set

$$\begin{aligned} Q_0 &= q_2 r_c^2 + q_0, & Q_r &= 2q_2 r_c + c_1 Q_0, \\ Q_S &= q_{2S} r_c^2 + q_{0S} + c_1 Q_0, & Q_t &= q_2 L_f^2 + q_0, & Q_{tS} &= q_{2S} L_f^2 + q_{0S}. \end{aligned}$$

The complete residual rates are

$$d = \hbar R_0/(2M) + (\lambda\omega_i^2 L_f^2/2 + 3\omega_i/2 + 3a_0)\epsilon_{20}, \quad (21)$$

$$e_r = \hbar^2 R_r/(2ML_i) + [c_1 Q_t + 2q_2 L_f + Q_t/L_i + \hbar a_0(3/0.2 + 7/L_i + 2000)]\epsilon_{20}, \quad (22)$$

$$e_S = \hbar^2 R_S/(2M) + [Q_{tS} + Q_t(p_1 + 5a_0/2) + \hbar\{3(a_1 + a_0 p_1) + 19a_0^2/2\}]\epsilon_{20}. \quad (23)$$

The Gaussian tail recurrence $\int_{20}^{\infty} y^n e^{-y^2} dy = 20^{n-1} e^{-400}/2 + (n-1) \int_{20}^{\infty} y^{n-2} e^{-y^2} dy/2$ proves the common tail norm used here; $e^2 > 7$ makes its stated bound rational.

B.2 First moments and preparation current

In the gauge the velocity coefficients are $u_r = af$ and $u_S = v + \delta$, $\delta = ma'F/(2M)$. Put

$$\begin{aligned} d_r &= ma_1 f_*/M, & d_S &= ma_2 F_*/(2M), & u_{rS} &= a_1 f_*, \\ z_r &= Q_r + \frac{\hbar}{2}(a_0 4000 + ma_2 f_*/M), \\ z_S &= Q_S + \frac{\hbar}{2}(a_1 + ma_3 F_*/(2M)). \end{aligned}$$

If $W_0 = \hbar 2^{35}/T_0$ and $W_1 = \hbar(2^{52}/256 + 5 \cdot 2^{35})/(T_0 L_i)$, the remote activation forces are bounded by $F_r = W_1 + Q_r$, $F_S = 25000(W_0 + Q_0) + Q_S$. Let $P_j = \|p_j(\Psi^g - G)\|$ and $D = \|\Psi^g - G\|$. Differentiating the symmetrized drift gives

$$D(t) \leq dt, \quad (24)$$

$$P'_r \leq a_0 P_r + d_r P_S + F_r n + e_r + z_r dt, \quad (25)$$

$$P'_S \leq u_{rS} P_r + d_S P_S + F_S n + e_S + z_S dt, \quad (26)$$

$$n' \leq g_0 + P_S/(Mg_1), \quad (27)$$

$$g_0 = \hbar[p_1 + a_0\sqrt{3/2}]/(Mg_1). \quad (28)$$

The collar defining n translates at $v + \|\delta\|_{\infty}$ on the right and $v - \|\delta\|_{\infty}$ on the left; its convective contribution is nonpositive. Both collars vanish on the initial stock. Their final full-one edges are respectively below 0.1035000001262 and above 0.1004999998738, so that they cover the remote activation and the left phase support respectively. Thus the force term $F_j n$ is valid throughout preparation, including the exact wave's tails.

For the sharper clock estimate take $\alpha = 10^{-12}$ and

$$s = e_r + e_S/K + \alpha g_0, \quad c = z_r + z_S/K, \quad E_* = (11/4)^{15}.$$

The three scaled row sums $a_0 + Kd_r + F_r/\alpha$, $u_{rS}/K + d_S + F_S/(K\alpha)$ and $\alpha K/(Mg_1)$ are less than 141. A positive supersolution gives

$$B(t) = e^{141t}(st + cdt^2/2), \quad P_r(t) \leq B(t), \quad P_S(t) \leq KB(t), \quad (29)$$

$$B_T = E_*(sT + cdT^2/2), \quad B_I = E_*(sT^2/2 + cdT^3/6), \quad (30)$$

$$n_{1T} = g_0 T + KB_I/(Mg_1). \quad (31)$$

Using $e^{141T} < E_*$ is legitimate since $141T < 15$ and $e < 11/4$. These terminal majorants are increasing and apply at every earlier time.

For the smaller radial estimate and the preparation current retain also the mass-weighted system. Put $\ell = 10^{-6}$,

$$\begin{aligned} e_0 &= e_r/\sqrt{m} + e_S/\sqrt{M}, & c_0 &= z_r/\sqrt{m} + z_S/\sqrt{M}, \\ s_0 &= e_0 + \ell g_0, & Z_T &= E_*(s_0T + c_0dT^2/2), \\ Z_I &= E_*(s_0T^2/2 + c_0dT^3/6), & n_w &= g_0T + Z_I/(\sqrt{M}g_1). \end{aligned}$$

Indeed the weighted momentum plus ℓn has growth at most 141: the drift row bound is $a_0 + \sqrt{m/M}a_1f_* + d_S$, increased by $\ell/(\sqrt{M}g_1)$, and $(F_r/\sqrt{m} + F_S/\sqrt{M})/\ell < 141$. Consequently the ordinary radial error at the handoff is bounded by

$$P_{r,o} = \sqrt{m}Z_T + ma_0f_*n_w, \quad P_{S,o} = KB_T + ma_1F_*n_{1T}/2, \quad D_o = dT. \quad (32)$$

The phase terms here are localized to the reflected collar; no phase is set to zero on an exact wave tail.

For the Gaussian rank let $k_r = 6/(7L_i)$, $k_S = 2a_0$, $c_r = k_r\sqrt{3/2}/L_i$, $c_S = k_S(p_1 + a_0\sqrt{3/2})$. The exact material-rank identity and the bilinear current expansion give

$$I_o = (c_r/\lambda + \hbar c_S/M)dT^2/2 + (k_r/\sqrt{m} + k_S/\sqrt{M})Z_I, \quad (33)$$

$$\beta = mL_i^2(B_1B_2 + a_0B_1^2)T/M + Ta_032000/7^{200}. \quad (34)$$

The first term of β bounds $K_S\delta$ pointwise; the second bounds the nonquadratic radial tail. The remaining current has absolute value before integration over any live coordinate. This proves $I_o < 1.974548968 \cdot 10^{-19}$ and $\beta < 2.638660 \cdot 10^{-13}$.

B.3 Second spatial moments and the second collar

Here is an explicit construction of every second-order residual coefficient. Continue H_j by $H_0 = 1$, $H_{j+1} = (z - 3/2)H_j - 2zH'_j$, and take absolute coefficients before adding independently bounded terms. The positive Gaussian clock polynomials are

$$\begin{aligned} A_0 &= 1, & A_1 &= a_0|H_1|, \\ A_2 &= a_1|H_1| + a_0^2|H_2|, & A_3 &= a_2|H_1| + 3a_0a_1|H_2| + a_0^3|H_3|, \\ A_4 &= a_3|H_1| + (4a_0a_2 + 3a_1^2)|H_2| + 6a_0^2a_1|H_3| + a_0^4|H_4|. \end{aligned}$$

Here $|P|$ means coefficientwise absolute value. To evaluate radial norms, form $P_{j,r}(y) = (\partial_y - y)^r y^{2j+1}$ and set

$$\mathcal{R}_r(j; s) = L_i^{-r} \sum_e |[y^e]P_{j,r}| \sqrt{M_{e-1+s}}, \quad M_{-1} = 2.$$

Only indices at least -1 occur. For a factor F/L^2 and $r \leq 2$ replace this by

$$\mathcal{R}_r^F(j) = \sum_{l=0}^r \binom{r}{l} c_l L_i^{-l} \mathcal{R}_{r-l}(j; 2-l), \quad (c_0, c_1, c_2) = (1, 2, 2).$$

This is the product rule using $F \leq r^2$, $|F'| \leq 2r$, $|F''| \leq 2$. Define

$$J_{n,r} = \sum_{j=0}^n \binom{n}{j} p_{n-j} \sum_l [z^l] A_j \mathcal{R}_r(l; 0),$$

and $J_{n,r}^F$ by replacing $\mathcal{R}_r$ with $\mathcal{R}_r^F$. With $C_j^* = \lambda L_f^2 a_j / 2$ for $1 \leq j \leq 4$, the sources are

$$\begin{aligned} e_{rr} &= \frac{\hbar^3}{2M} (J_{2,2} + 2C_1^* J_{1,2}^F + C_2^* J_{0,2}^F) + 10^{-100}, \\ e_{rS} &= \frac{\hbar^3}{2M} (J_{3,1} + 2C_1^* J_{2,1}^F + 3C_2^* J_{1,1}^F + C_3^* J_{0,1}^F) + 10^{-100}, \\ e_{SS} &= \frac{\hbar^3}{2M} (J_{4,0} + 2C_1^* J_{3,0}^F + 5C_2^* J_{2,0}^F + 4C_3^* J_{1,0}^F + C_4^* J_{0,0}^F) + 10^{-100}. \end{aligned}$$

For example the last formula differentiates $G_{SS} + 2i\theta_S G_S + i\theta_{SS} G$ twice and retains coefficients 2, 5, 4, 1. The nonquadratic and trap tails after two derivatives have Gaussian degree at most seven and total frequency coefficient below 10^{30} ; their momentum contribution is less than $\hbar^2 10^{30} \epsilon_{20} < 10^{-100}$ in each row. This can also be checked by the general product-rule construction below.

For clarity we spell out the second-force coefficients. Set $c_2 = 1.28 \cdot 10^8$, $q_{t1} = q_{2S} r_c^2 + q_{0S}$, $q_{t2} = (4a_1 + 16a_0^2) q_{2r}^2 + (2a_1 + 4a_0^2) q_0$. Then

$$\begin{aligned} Q_{rr} &= 2q_2 + 4c_1 q_{2r} r_c + c_2 Q_0, \\ Q_{rS} &= c_1 Q_r + 2q_{2S} r_c + c_1 q_{t1}, \\ Q_{SS} &= c_2 Q_0 + 2c_1 q_{t1} + q_{t2}. \end{aligned}$$

Put $\gamma_1 = 25000$, $\gamma_2 = (2520/64)/(0.0001)^2$, and

$$\begin{aligned} W_2 &= \frac{\hbar}{T_0 L_i^2} [2^{52} (1 + (15/16) 2^{17}) + 5 \cdot 2^{52}/128 + 10^5 \cdot 2^{35}], \\ F_{rr} &= W_2 + Q_{rr}, \\ F_{rS} &= \gamma_1 (W_1 + Q_r) + Q_{rS}, \\ F_{SS} &= \gamma_2 (W_0 + Q_0) + 2\gamma_1 Q_S + Q_{SS}. \end{aligned}$$

Let v_{jr} and v_{jS} denote the component derivatives of $(af, ma'F/(2M))$. The second-derivative ceilings used below are

	rr	rS	SS
u_r	$a_0 4000$	a_1	$a_2 f_*$
δ	ma_1/M	$ma_2 f_*/M$	$ma_3 F_*/(2M)$
$\text{div } u$	$a_0 10^7 + ma_2/M$	$a_1 4000 + ma_3 f_*/M$	$a_2 + ma_4 F_*/(2M)$

Write $d_r^{\text{div}} = a_0 4000 + ma_2 f_*/M$ and $d_S^{\text{div}} = a_1 + ma_3 F_*/(2M)$ for the first divergence ceilings. If U_{jk}, V_{jk}, D_{jk} denote the three rows of this table, all scalar coefficients of the Hessian comparison are

$$\begin{aligned} e_2 &= e_{rr} + e_{rS}/K + e_{SS}/K^2, \\ c_B &= \hbar(U_{rr} + KV_{rr} + d_r^{\text{div}}) + 2Q_r \\ &\quad + \{\hbar[U_{rS} + KV_{rS} + (Kd_r^{\text{div}} + d_S^{\text{div}})/2] + KQ_r + Q_S\}/K \\ &\quad + \{\hbar[U_{SS} + KV_{SS} + Kd_S^{\text{div}}] + 2KQ_S\}/K^2, \\ c_D &= \frac{\hbar^2}{2}(D_{rr} + D_{rS}/K + D_{SS}/K^2) + \hbar(Q_{rr} + Q_{rS}/K + Q_{SS}/K^2), \\ c_n &= \hbar(F_{rr} + F_{rS}/K + F_{SS}/K^2) + 2F_r \hbar/(g_2 K) + 4F_S \hbar/(g_2 K^2), \\ c_A &= 2F_r + 2F_S/K. \end{aligned}$$

These follow directly by commuting two momenta through $\{u_j, p_j\}/2 + Q_c + V_A$. For example the mixed commutator contributes $\hbar \sum_l \|u_{l,jk}\| P_l$ and $\hbar(\|\partial_j \text{div } u\| P_k + \|\partial_k \text{div } u\| P_j)/2$; both occur above. The three homogeneous scaled row sums are below 282.

With $Z_2 = \max(\|p_r^2 E^g\|, \|p_r p_S E^g\|/K, \|p_S^2 E^g\|/K^2)$, nested clock cutoffs give

$$Z'_2 \leq 282Z_2 + c_A \sqrt{n_1} \sqrt{Z_2} + e_2 + c_B B + c_D dt + c_n n_1.$$

Indeed a cutoff supported where the first collar is one satisfies $\|\chi p_r E^g\| \leq \sqrt{n_1} \|p_r^2 E^g\|$ and $\|\chi p_S E^g\| \leq \sqrt{n_1} \|p_S^2 E^g\| + 2\hbar n_1/g_2$. These inequalities are integration by parts, not finite propagation. Let $b_n = K/(Mg_1)$, $\gamma = 141$, and define

$$\begin{aligned} J_n &= \frac{\sqrt{g_0 \pi}}{2\gamma^{3/2}} + \frac{\sqrt{b_n s/2}}{(\gamma/2)^2} + \frac{3\sqrt{\pi b_n c d/6}}{4(\gamma/2)^{5/2}}, \\ J_f &= \frac{e_2}{2\gamma} + c_B(s/\gamma^2 + cd/\gamma^3) + \frac{c_D d}{(2\gamma)^2} \\ &\quad + c_n\{g_0/(2\gamma)^2 + b_n(s/\gamma^3 + cd/\gamma^4)\}, \\ Z_{2T} &= e^{2\gamma T}(c_A J_n/2 + \sqrt{J_f})^2, \quad H_{SS} = K^2 Z_{2T}. \end{aligned}$$

For the last display, extend positive integrals to infinity after using $n_1(t) \leq g_0 t + b_n e^{\gamma t}(st^2/2 + cdt^3/6)$. The comparison follows by setting $y = e^{-\gamma t} \sqrt{Z_2}$: the sum of the separate positive supersolutions for $y' \leq A + B/(2y)$ is a supersolution, with zero handled by regularization. The factor $e^{\gamma T}$ is enclosed by its 90-term Taylor sum and the remaining positive geometric tail with ratio $\gamma T/91$; its upper enclosure is squared when evaluating $e^{2\gamma T}$.

The second collar amplitude and the physical phase restoration are

$$n_{2T} = \frac{T}{Mg_2} [\sqrt{n_{1T} H_{SS}} + 2\hbar D_o/g_1], \quad (35)$$

$$L_S = \sqrt{n_{2T} H_{SS}} + 2\hbar n_{1T}/g_2, \quad L_r = \sqrt{n_{2T} H_{SS}/K^2}, \quad (36)$$

$$H_{SS}^o = H_{SS} + 2q_1 L_S + (q_1^2 + \hbar q_2^*) n_{2T}, \quad q_1 = ma_1 F_*/2, \quad q_2^* = ma_2 F_*/2. \quad (37)$$

Both reflected second collars fit before the relevant remote support: their final edges are below 0.103750000127 and above 0.100249999873. These same increasing majorants hold for all preceding times. In particular the ordinary derivatives on the writer support are bounded by L_S, L_r , because both the helper and the phase vanish there.

B.4 Fourth spatial closure and its complete coefficient ledger

Only ordinary spatial derivatives are used here. There is no assumption that the unmollified radial reference lies in the domain of a fourth power of its Hamiltonian. Use the commuting scaled momenta $D_r = p_r$, $D_S = p_S/K$, and $Z_j = \max_{a+b=j} \|D_r^a D_S^b E^g\|$.

The higher coefficient construction is finite. Define $q_j = b_p^{(j)}/b_p$. Starting with $P_0(q) = q_1$, form

$$P_{j+1} = \sum_{l=1}^6 (q_{l+1} - q_1 q_l) \partial_{q_l} P_j, \quad 0 \leq j < 6.$$

Evaluate the absolute coefficients at

$$(\bar{q}_1, \dots, \bar{q}_7) = (140, 24000, B_3, B_4, B_5, 4B_5/10^{-5}, 300B_5/10^{-10})$$

to obtain $\bar{a}_j \geq |a^{(j)}|$. For $j \leq 4$ these agree with the bounds above. The last two entries use convolution of $b_0^{(5)}$ with the first and second derivative of the mollifier. Similarly set $p_5 = 4p_4/10^{-10}$, $p_6 = 300p_4/10^{-20}$. The normalized unit bump has derivative polynomials generated by

$$U_0(x, u) = 1, \quad U_{j+1} = \partial_x U_j + 2xu^2 \partial_u U_j - 2xu^2 U_j, \quad u = (1 - x^2)^{-1}. \quad (38)$$

Its normalization is greater than $1/4$; hence its j th derivative supremum is bounded by $4 \sum_{l,k} |[x^l u^k] U_j| k^k / (8/3)^k$, with the $k = 0$ factor defined as one. This proves the quoted mollifier bounds (the first derivative also uses unimodality), as well as all tent jets below.

Let the positive partial Bell coefficients be

$$\mathcal{B}_{0,0} = 1, \quad \mathcal{B}_{n,k} = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} \bar{a}_{j-1} \mathcal{B}_{n-j,k-1}.$$

All unspecified entries are zero. For $n = 0$ put $\mathcal{A}_0 = 1$; otherwise $\mathcal{A}_n = \sum_{k=1}^n \mathcal{B}_{n,k} |H_k|$. For $\varepsilon = 0, 1$, define the computable Gaussian norm majorant

$$\mathcal{J}_{n,r}^{(\varepsilon)} = \sum_{j=0}^n \binom{n}{j} p_{n-j} \sum_l [z^l] \mathcal{A}_j L_i^{-r} \sum_e |[y^e] (\partial_y - y)^r y^{2l+1+2\varepsilon}| \sqrt{M_{e-1}}.$$

For the higher-core calculation one may use $\pi \leq 22/7$ in p_j and $\hbar \leq \bar{\hbar} = 1.055 \cdot 10^{-34}$. The fourth physical source coefficient is bounded by

$$s_4 = \frac{\bar{h}^5}{2M} \sum_{n=0}^4 K^{-n} \left[\mathcal{J}_{n+2,4-n}^{(0)} + \sum_{j=1}^{n+2} c_{n,j} \frac{\lambda L_f^2 \bar{a}_j}{2} \mathcal{J}_{n+2-j,4-n}^{(1)} \right], \quad (39)$$

$$c_{n,j} = 2 \binom{n}{j-1} + \binom{n}{j-2}, \quad (40)$$

where a binomial coefficient outside its range is zero. Direct rational arithmetic and outward square roots give $s_4 < 7.141 \cdot 10^{-127}$. The physical phase-square cancellation in the gauge is essential to this small coefficient.

We next give a complete way to check the nonquadratic tail bound, including higher derivatives of $F - r^2$. Let b_j denote the bump derivative suprema just constructed and use

$$(f_0, f_1, f_2, f_3, f_4, f_5) = (0.201, 1, 4000, 10^7, 2b_2/0.001^3, 2b_3/0.001^4), \\ F_0 = 0.080802, \quad F_j = 2f_{j-1} \quad (1 \leq j \leq 6).$$

The disjoint jumps of the unsmoothed tent give these bounds. The scaled velocity derivative sum at order j is explicitly

$$V_j = \max_{0 \leq k \leq j} \frac{\bar{a}_k f_{j-k}}{K^k} + \max_{0 \leq k \leq j} \frac{K \bar{h} \lambda \bar{a}_{k+1} F_{j-k}}{2M K^k}. \quad (41)$$

It gives $V_2 < 560001$, $V_3 < 1.400001 \cdot 10^9$, $V_4 < 3.669751 \cdot 10^{13}$, $V_5 < 2.786850 \cdot 10^{18}$.

For an entirely algebraic tail check replace every Gaussian norm in $\mathcal{J}_{n,r}^{(0)}$ by the absolute sum of its polynomial coefficients; call the result $\mathcal{C}_{n,r}$. Keep powers of y as formal factors. The derivatives of $F - r^2$ have coefficient ceilings $\Delta F = (2, 2, 4, 2f_2, 2f_3)$, and those of $f - r$ have $\Delta f = (2, 2, f_2, f_3, f_4, f_5)$. The unit logistic cutoff has order- j derivative bounded by

$$t_j = 8j! \sum_{k=1}^j \binom{j-1}{k-1} 2^k \frac{(j+k)^{j+k}}{(8/3)^{j+k}} \quad (1 \leq j \leq 4).$$

To prove it on $x \leq 1/2$, put $u = 1/x$: the exponent is at most $-u + 2$, its j th derivative at most $2j! u^{j+1}$, and apply the Bell product rule and $\max u^n e^{-u} \leq n^n / (8/3)^n$. Symmetry handles the other half. Set $t_0 = 1$ and $\tau_j = t_j / 0.001^j$. For $p = 2, 4$ define $J_p(0) = 1$, $J_p(n) = \sum_{k=1}^n \mathcal{B}_{n,k} p^k$. With $A_2 = 4 \cdot 10^{-41}$ and $A_0 = 1.2 \cdot 10^{-48}$, put

$$q(n, r) = A_2 J_4(n) (L_f^2, 2L_f, 2)_r + \mathbf{1}_{r=0} A_0 J_2(n), \\ \tilde{q}(n, r) = q(n, r) + \sum_{c=0}^n \sum_{e=0}^r \binom{n}{c} \binom{r}{e} \tau_c \tau_e q(n-c, r-e).$$

The three-entry sequence is zero outside $r = 0, 1, 2$. A bound for the sum of all fourth-derivative tail frequency coefficients is the following finite positive sum, with $r = 4 - n$:

$$\mathcal{T} = \sum_{n=0}^4 \sum_{a=0}^n \sum_{b=0}^r \binom{n}{a} \binom{r}{b} \left[\frac{1.425 \cdot 10^{-12}}{2M} \bar{a}_{a+1} \Delta F_b \mathcal{C}_{n-a+1, r-b} \right. \quad (42)$$

$$\left. + \frac{1.425 \cdot 10^{-12}}{4M} \bar{a}_{a+2} \Delta F_b \mathcal{C}_{n-a, r-b} + \bar{a}_a \Delta f_b \mathcal{C}_{n-a, r-b+1} \right. \quad (43)$$

$$\left. + \frac{1}{2} \bar{a}_a \Delta f_{b+1} \mathcal{C}_{n-a, r-b} + 10^{34} \hat{q}(a, b) \mathcal{C}_{n-a, r-b} \right] < 10^{70}. \quad (44)$$

This is simply the product rule applied respectively to the two phase terms, transport tail, and $Q - Q_c$ in (20). Taking absolute coefficients before forming $\mathcal{C}$ prevents cancellation between independently bounded clock jets. As a rough independent size check, $p_j \leq 10^{9j}$, $\bar{a}_j < 10^3 10^{9j}$, and Gaussian mixed coefficients through order six are below $10^{12} 10^{9j}$. The transport term has at most five derivative slots and fewer than 10^5 product summands, giving $10^{3+3+12+45+5} = 10^{68}$. In the phase term $\hbar\theta$ replaces the large phase coefficient by the physical mass: six derivative slots give at most $10^{-12+3+3+12+54+5} = 10^{65}$. The pure G_{SS} term is smaller. The explicit positive sum above also checks the weak potential term. No nonzero tail is discarded.

The degree is at most twelve. Integration by parts gives

$$\|\mathbf{1}_{y \geq 20y^{12}} A\|^2 \leq \frac{2 \cdot 20^{25}}{1 - 25/800} e^{-400} < 10^{40} / 7^{200}.$$

Thus the additional fourth physical source is $s_{\text{tail}} = \bar{h}^4 10^{90} / 7^{100} < 10^{-130}$. The same finite product rule for the quiet potential gives, for total order $j \leq 4$,

$$Q_j \leq 10^{-40} 10^{9j}. \quad (45)$$

For an explicit verification use the preceding $J_p(n)$ and cutoffs, replace $(L_f^2, 2L_f, 2)$ by $(1, 1, 2)$ on the cutoff support, and evaluate $\sum_{a=0}^n \sum_{b=0}^r \binom{n}{a} \binom{r}{b} \tau_a \tau_b [A_2 J_4(n-a)(1, 1, 2)_{r-b} + \mathbf{1}_{r=b} A_0 J_2(n-a)]$ for $n+r=j$.

For the remote potential define $W_j = \bar{h} w_j / (T_0 L_i^j)$, where

$$\begin{aligned} w_0 &= 2^{35}, & w_1 &= 2^{52}/256 + 5 \cdot 2^{35}, \\ w_2 &= 2^{52}(1 + (15/16)2^{17}) + 5 \cdot 2^{52}/128 + 10^5 \cdot 2^{35}, \\ w_3 &= 2^{112}, & w_4 &= 2^{160}. \end{aligned}$$

Set

$$(\gamma_0, \dots, \gamma_4) = \left(1, 25000, \frac{2520/64}{10^{-8}}, \frac{7560/16 + 5040/64}{10^{-12}}, \frac{1360800}{10^{-16}} \right),$$

$$R_j = \max_{0 \leq k \leq j} K^{-k} \left[\gamma_k W_{j-k} + \sum_{l=0}^k \binom{k}{l} \gamma_l Q_{j-l} \right].$$

These bounds follow by differentiating the displayed capped radial potential and activation polynomial; the last two deliberately loose bounds require only its bounded fourth weak derivative. All remote supports lie at $S \geq 0.104$.

Add two smooth analytic cutoffs of width $g = 10^{-4}$, with final transition intervals $(0.103751, 0.103851)$ and $(0.103851, 0.103951)$. They fit between the second old collar and activation. Their first two derivative bounds are $5/g$ and $128/g^2$. Put $a_c = \bar{h}/(Kg)$. With n the second-collar amplitude, integration by parts and interpolation give local derivative bounds

$$N_1 \leq \sqrt{nZ_2} + 10a_c n, \quad (46)$$

$$N_2 \leq \sqrt{n}(\sqrt{Z_4} + 18a_c \sqrt{Z_2}), \quad (47)$$

$$N_3 \leq \sqrt{N_2 Z_4} + 10a_c N_2. \quad (48)$$

For example the squared second-derivative norm is at most $nZ_4 + 20a_c n \sqrt{Z_2 Z_4} + 306a_c^2 n Z_2$; its square root is bounded as displayed since $306 < 18^2$. The global interpolation $Z_3 \leq \sqrt{Z_2 Z_4}$ moves one commuting derivative between the two factors in its squared norm. Odd extension removes the radial boundary term.

The useful time envelopes, with $Z_{1T} = B_T$, are

$$\begin{aligned} Z_1(t) &\leq Z_{1T} e^{-\gamma(T-t)}, & Z_2(t) &\leq Z_{2T} e^{-2\gamma(T-t)}, \\ n(t) &\leq A_n t^{3/2} e^{\gamma t} + B_n t^2 e^{3\gamma t/2} + C_n t^2, \\ A_n &= \sqrt{K^2 Z_{2T} g_0 / (M g_2 e^{\gamma T})}, \\ B_n &= \sqrt{K^2 Z_{2T} n_{1T}} / (M g_2 T e^{3\gamma T/2}), \\ C_n &= 2\bar{h} D_T / (M g_2 g_1 T), \quad D_T = 3.006684 \cdot 10^{-11}. \end{aligned}$$

Here D_T is used only in the fourth-order majorant, not to replace the sharper handoff norm in the final probability accounting. Writing $Z = Z_4$, four commutations yield

$$Z' \leq 4\gamma Z + \alpha(t) Z^{3/4} + \beta(t) Z^{1/2} + f(t), \quad (49)$$

$$\alpha = 4R_1 n^{1/4}, \quad (50)$$

$$\beta = 4R_1 [\sqrt{18a_c} n^{1/4} Z_2^{1/4} + 10a_c n^{1/2}] + 6\bar{h} R_2 n^{1/2} + (8\bar{h} V_2 + 4Q_1) \sqrt{Z_2}, \quad (51)$$

$$f = (720R_1 a_c^2 + 108\bar{h} R_2 a_c) \sqrt{n Z_2} + 4\bar{h}^2 R_3 (\sqrt{n Z_2} + 10a_c n) + \bar{h}^3 R_4 n \quad (52)$$

$$+ (7\bar{h}^2 V_3 + 6\bar{h} Q_2) Z_2 + (3\bar{h}^3 V_4 + 4\bar{h}^2 Q_3) Z_1 + (\bar{h}^4 V_5/2 + \bar{h}^3 Q_4) dt + s_4 + s_{\text{tail}}. \quad (53)$$

The factors $8, 7, 3, 1/2$ in the drift terms are the sum of the first-order drift and its divergence product-rule coefficients. Putting $y = e^{-\gamma t} Z^{1/4}$ and adding the separate positive supersolutions gives

$$Z_4(T)^{1/4} \leq e^{\gamma T} \left[\frac{1}{4} \int_0^T \alpha e^{-\gamma t} dt + \sqrt{\frac{1}{2} \int_0^T \beta e^{-2\gamma t} dt} + \left(\int_0^T f e^{-4\gamma t} dt \right)^{1/4} \right]. \quad (54)$$

This includes the nonlinear third-derivative coupling; it is not an exponential multiplying a terminal forcing value.

For completeness all integrals in this expression can be evaluated by one finite rule. For $0 < p \leq 1$ use subadditivity on the three terms of $n(t)$; combine with $Z_2(t)^q$ and the integrating factor, and extend each positive integral to infinity. If its time power is $u \geq 0$ and its decay is $\rho > 0$, with $j = \lfloor u \rfloor$, use

$$\int_0^\infty t^u e^{-\rho t} dt \leq j!(j+1)^{u-j} / \rho^{u+1}.$$

This is Hölder interpolation of adjacent integer moments. All decay rates in (54) are positive. Integer roots are enclosed rationally; no numerical gamma function is needed. For a final far cutoff of width 0.01 before $S = 0.14145$, put

$$P_4 = K^4 Z_4(T), \quad L_{SS} = K^2 \sqrt{n_{2T}} [\sqrt{Z_4(T)} + 18\bar{h} \sqrt{Z_{2T}} / (K \cdot 0.01)]. \quad (55)$$

The unshortened positive constructions define the values used in later interfaces; their shortened displays are $P_4 < 1.075567 \cdot 10^{-76}$ and $L_{SS} < 2.534075 \cdot 10^{-57}$ uniformly through preparation.

B.5 Original-stock transfer to the writer

Let Ψ_o be the exact preparation without the writer, and Ψ_n the enlarged exact wave, with their common initial state $\Psi_n(0) = \Psi_o(0)g_0$. Here $g_0(Z) = \pi^{-1/4} e^{-Z^2/2}$. Write $\Delta = \Psi_n - \Psi_o g_0$. The ground oscillator energy is subtracted and the additional potential is $J = (\hbar/T_0) \Pi_1 [C_w(s) - F_w(s)Z]$. Thus

$$i\hbar \partial_t \Delta = H_n \Delta + J \Psi_o g_0, \quad \Delta(0) = 0.$$

No comparison flow or initial law is substituted in this identity. The writer is supported on $S \geq 0.14145$. Throughout preparation, the old helper and its bounded phase both vanish on this support. The source amplitudes and momentum norms are therefore bounded by n_{2T}, L_S, L_r, L_{SS} established above.

The following constants are exact rational majorants. Put $\mu = 0.01$, $w = 0.003$ and

$$(b_0, b_1, b_2, b_3, b_4) = 24 \left(1, \frac{315}{128w}, \frac{2520}{64w^2}, \frac{7560/16 + 5040/64}{w^3}, \frac{1360800}{w^4} \right).$$

In this subsection b_j denotes a writer derivative bound, not the preparation dilation or bump derivative. Define

$$\begin{aligned} F_j &= b_j/\mu + \mu b_{j+2} \quad (0 \leq j \leq 2), \\ C_0 &= b_0^2/(2\mu) + \mu b_0 b_2/2, \\ C_1 &= b_0 b_1/\mu + \mu(b_1 b_2 + b_0 b_3)/2, \\ C_2 &= (b_1^2 + b_0 b_2)/\mu + \mu(b_2^2 + 2b_1 b_3 + b_0 b_4)/2, \\ c_0 &= b_0 + \mu b_1, & c_1 &= b_1 + \mu b_2. \end{aligned}$$

These bound F_w, C_w and their dimensionless derivatives. Hereafter $F_{\text{old}} = 8 \cdot 10^{-6} + 10^{-17}$ and $F_{\text{old},r} = 10^{-6}$. These force ceilings follow by differentiating the displayed potential: the preparation clock force is bounded by

$$ma_2(0.080802)/2 + ma_0 a_1(0.201)^2 + m^2 a_1 a_2(0.080802)^2/(4M) < 8 \cdot 10^{-6},$$

and its radial force by $ma_1(0.201) + ma_0^2(0.201) + m^2 a_1^2(0.080802)(0.201)/(2M)$. Adding the quiet and activation derivatives gives the stated ceilings; their clock contribution is below 10^{-17} . A second differentiation gives the conservative bound $F_{\text{old},SS} \leq 1$: its preparation part is

$$ma_3(0.080802)/2 + m(a_1^2 + a_0 a_2)(0.201)^2 + m^2(a_2^2 + a_1 a_3)(0.080802)^2/(4M),$$

with $Q_{SS} + \gamma_2(W_0 + Q_0) + 2\gamma_1 q_{t1}$ added.

For concise exact formulas set $n_o = n_{2T}$ and

$$d_\Delta = (C_0 + F_0/\sqrt{2})n_o/T_0, \quad q_\Delta = (C_0 + \sqrt{3/2}F_0)n_o/T_0, \quad (56)$$

$$r_S = (C_0 + F_0/\sqrt{2})L_S/T_0 + \hbar(C_1 + F_1/\sqrt{2})n_o/T_0^2, \quad (57)$$

$$r_r = (C_0 + F_0/\sqrt{2})L_r/T_0. \quad (58)$$

The exact oscillator identities $\|Zg_0\| = \|p_Z g_0\| = 1/\sqrt{2}$ and $\|Z^2 g_0\|^2 = \|p_Z Zg_0\|^2 = 3/4$ prove these coefficients. Unitary Duhamel and oscillator rotation in the direct sum of its two quadratures then give increasing bounds

$$D_\Delta(t) = d_\Delta t, \quad (59)$$

$$Q_\Delta(t) = q_\Delta t + F_0 d_\Delta t^2/(2T_0), \quad (60)$$

$$P_{\Delta,S}(t) = r_S t + F_{\text{old}} d_\Delta t^2/2 \quad (61)$$

$$+ \frac{\hbar}{T_0^2} [F_1(q_\Delta t^2/2 + F_0 d_\Delta t^3/(6T_0)) + C_1 d_\Delta t^2/2], \quad (62)$$

$$P_{\Delta,r}(t) = r_r t + F_{\text{old},r} d_\Delta t^2/2. \quad (63)$$

Here Q_Δ bounds $(\|Z\Delta\|^2 + \|p_Z \Delta\|^2)^{1/2}$, not an unbounded coordinate times a bare norm estimate. In the clock inequality the term $F_1 Q_\Delta$ is retained explicitly.

Let $B = \partial_Z + Z$. Its exact equations have sources $-\Pi_1 F_w \Psi_n$ and $-2\Pi_1 F_w B \Psi_n$ for $B \Psi_n$ and $B^2 \Psi_n$, respectively. Because B annihilates the old product wave, integration gives

$$B_1(t) = F_0(n_o t + d_\Delta t^2/2)/T_0, \quad (64)$$

$$B_2(t) = F_0^2(n_o t^2 + d_\Delta t^3/3)/T_0^2, \quad (65)$$

$$A_h = B_1(T) + c_0[n_o + D_\Delta(T)], \quad (66)$$

$$A_{2h} = B_2(T) + 2c_0 B_1(T) + c_0^2[n_o + D_\Delta(T)]. \quad (67)$$

The last quantity bounds the norm of the squared moving annihilator; it is not the square of its norm. These are operator identities on the full oscillator Gaussian domain, with no truncation of Z . For $f = B \Psi_n$, oscillator algebra gives $\|Zf\|^2 + \|p_Z f\|^2 = \|Bf\|^2 + \|f\|^2$. Consequently, with integrals over $[0, T]$,

$$\begin{aligned} P_B &= F_{\text{old}} \int B_1 + \frac{\hbar}{T_0^2} \left[F_1 \int (B_2 + B_1) + C_1 \int B_1 \right] \\ &\quad + \frac{F_0}{T_0} [L_S T + T P_{\Delta, S}(T)] + \frac{\hbar F_1}{T_0^2} (n_o T + d_\Delta T^2/2), \\ U_h &= [P_B + c_0(L_S + P_{\Delta, S}(T)) + \hbar c_1(n_o + D_\Delta(T))/T_0]/\hbar. \end{aligned}$$

This proves $\|\partial_S(A \Psi_n)(T)\| \leq U_h$. All displayed integrals are polynomials; for example $\int B_1 = F_0(n_o T^2/2 + d_\Delta T^3/6)/T_0$ and $\int B_2 = F_0^2(n_o T^3/3 + d_\Delta T^4/12)/T_0^2$.

The exact symbolic entrance values used subsequently are

$$D_h = 3.006683 \cdot 10^{-11} + D_\Delta(T), \quad (68)$$

$$n_h = n_{1T} + D_\Delta(T), \quad (69)$$

$$P_h = \hbar p_1 + P_{S, o} + P_{\Delta, S}(T), \quad (70)$$

$$P_{r, h} = P_{r, o} + P_{\Delta, r}(T) + \epsilon_q, \quad 0 \leq \epsilon_q < 10^{-100}, \quad (71)$$

$$R_h = 1 + Q_\Delta(T) < 2. \quad (72)$$

Here ϵ_q is the quiet-reference comparison error in the radial momentum, as bounded in the activation appendix. In particular the larger norm $3.006684 \cdot 10^{-11}$ used in the fourth-moment estimate is *not* substituted for D_h .

B.6 Localized repair of the transfer Hessian

A coarse second-momentum bound is useful only to generate source-free left collars. We give it explicitly to avoid circular reuse of the improved result. Let $P = P_{\Delta, S}(T)$, $D = D_\Delta(T)$, $Q = Q_\Delta(T)$, and

$$\begin{aligned} r_{SQ} &= (C_0 + \sqrt{3/2} F_0) L_S / T_0 + \hbar (C_1 + \sqrt{3/2} F_1) n_o / T_0^2, \\ Q_S &= F_0 P T / T_0 + F_{\text{old}} \int Q_\Delta + \frac{\hbar}{T_0^2} \left[C_1 \int Q_\Delta + F_1 \int (B_2 + 2B_1 + 2D_\Delta) \right] + r_{SQ} T, \\ r_{SS}(L) &= (C_0 + F_0/\sqrt{2}) L / T_0 + 2\hbar (C_1 + F_1/\sqrt{2}) L_S / T_0^2 + \hbar^2 (C_2 + F_2/\sqrt{2}) n_o / T_0^3, \\ W_\Delta &= 2\hbar T (C_1 P + F_1 Q_S) / T_0^2 + \hbar^2 T (C_2 D + F_2 Q) / T_0^3, \\ H_c &= r_{SS}(H_{SS}) T + 2F_{\text{old}} P T + \hbar D T + W_\Delta. \end{aligned}$$

The quadrature estimate $\|Z^2 \Delta\|$ and its companion are controlled by $B_2 + 2B_1 + 2D_\Delta$, so that Q_S bounds the mixed weighted momentum without replacing Z by a bounded operator. Differentiating the inhomogeneous equation twice proves $\|p_S^2 \Delta\| \leq H_c$; the coarse value is below $\hbar^2(2.486519 \cdot 10^{27})$.

Choose decreasing left cutoffs with transitions $[0.126, 0.136]$ and $[0.116, 0.126]$, width $g = 0.01$. The writer forcing vanishes on both supports and every old clock force is contained in their full-one regions, because it vanishes for $S \geq 0.106$. Positive drift favours exit from each left region. Continuity and one integration by parts give

$$n_{\Delta 1} = TP/(Mg), \quad n_{\Delta 2} = \frac{T}{Mg}[\sqrt{n_{\Delta 1}H_c} + 2\hbar D/g]. \quad (73)$$

For the new Hessian $H(t) = \|p_S^2 \Delta(t)\|$ the old-force momentum is instead localized by $\|\mathbf{1}_{\text{oldforce}} p_S \Delta\| \leq \sqrt{n_{\Delta 2}H(t)} + 2\hbar n_{\Delta 1}/g$. Use the proved far old-wave Hessian L_{SS} in the source. The differential inequality has the form $H' \leq 2\alpha\sqrt{H} + f$, where $\alpha = F_{\text{old}}\sqrt{n_{\Delta 2}}$. The function $[\alpha t + \sqrt{\int_0^t f}]^2$ is a supersolution, because its derivative is at least $2\alpha\sqrt{H} + f$ when evaluated at that function. Bounding the nonnegative source integral gives

$$H_{\Delta} = \left[F_{\text{old}}T\sqrt{n_{\Delta 2}} + \sqrt{r_{SS}(L_{SS})T + 4F_{\text{old}}\hbar n_{\Delta 1}T/g + \hbar n_{\Delta 2}T + W_{\Delta}} \right]^2. \quad (74)$$

It proves $\|p_S^2 \Delta(T)\| \leq H_{\Delta}$ and $H_{\Delta}/\hbar^2 < 1.828510 \cdot 10^{19}$. The coarse Hessian only generated the amplitude collar; it was not relabelled as the improved source or inserted into this nonlinear old-force term.

B.7 Handoff enclosures and the preparation event

The constructions above give the following outward decimal enclosures. Every endpoint in the table is a rational number. The unrounded expressions, not shortened displays, define the quantities when used in the final event sum.

Quantity	Strict upper bound
D_o	$3.00668265254473406785 \cdot 10^{-11}$
$P_{S,o}$	$6.98319079275307899984 \cdot 10^{-32}$
n_{1T}	$6.24924697825783043607 \cdot 10^{-30}$
H_{SS}	$3.23800849281954087759 \cdot 10^{-49}$
H_{SS}^o	$3.24499145042629721024 \cdot 10^{-49}$
n_{2T}	$5.97036914776425754723 \cdot 10^{-38}$
P_4	$1.07556686771573853653 \cdot 10^{-76}$
L_{SS}	$2.53407404062459981525 \cdot 10^{-57}$
$D_{\Delta}(T)$	$2.767839 \cdot 10^{-30}$
$P_{\Delta,S}(T)$	$7.597256 \cdot 10^{-36}$
$P_{\Delta,r}(T)$	$2.08385966319223143790 \cdot 10^{-37}$
A_h	$5.04959232864420059458 \cdot 10^{-24}$
A_{2h}	$1.22824296520291159974 \cdot 10^{-17}$
U_h	276122.969821611569892
P_h	$3.20281666862106903351 \cdot 10^{-31}$
$H_{\Delta}/\hbar^2$	$1.82850947100693947781 \cdot 10^{19}$

B.8 Preparation history under the enlarged exact flow

The preparation rank $K(S, r)$ is independent of Z ; hence its material derivative under the enlarged flow has no pointer-current term. Its explicit drift β in (34) is unchanged. The error in its current must nevertheless be recomputed for that flow. In the bounded gauge the additional momentum ceilings are

$$d_r^g = P_{\Delta,r}(T) + ma_0(0.201)D_{\Delta}(T), \quad d_S^g = P_{\Delta,S}(T) + ma_1(0.080802)D_{\Delta}(T)/2.$$

The exact normalized wave stays in the derivative-error factor of the bilinear expansion, retaining the quadratic error terms. Consequently

$$\Delta I = T[(\hbar c_r/m + \hbar c_S/M)D_{\Delta}(T) + (k_r/m)d_r^g + (k_S/M)d_S^g] < 1.304454833 \cdot 10^{-22}. \quad (75)$$

Under the single original joint domination constant $C = 270000/67499$, N fixed radial cuts therefore have preparation failure probability bounded by

$$E_p(N) = 2CN\beta + 2C\sqrt{2N(I_o + \Delta I)} + C[D_o + D_\Delta(T)]^2. \quad (76)$$

This follows by taking initial rank collars of width $\beta + \epsilon$, using Markov only on the unknown absolute current, and minimizing in ϵ . The last term covers the terminal clock suffix where the helper vanishes. It is an estimate on the enlarged flow with its original initial law; closeness of two waves alone has not been used to assert history agreement. The initial conditional-CDF interface costs, separately,

$$E_{\text{cdf}}(N) = 2CN[D_o + D_\Delta(T)]. \quad (77)$$

For the final cut family, including the exterior ray, use $N = 256001$ in both formulas. Evaluating the finite expressions gives

$$\begin{aligned} E_p(256001) &< 0.000003084949592860634224432805, \\ E_{\text{cdf}}(256001) &< 0.0000615780135255954031515710. \end{aligned}$$

The pointer remains part of the one jointly dominated auxiliary stock; no independent pointer cap, sector sample, equilibrium reset, or new law at handoff enters these estimates.

C Radial conditional comparison and the old clock current

This appendix supplies the radial comparison estimates used before the writer is active. Detector coordinates and ages are used until physical clock units are explicitly restored. Put

$$\begin{aligned} m &= 2^{52}, \quad h = 1/128, \quad L = 100, \quad V = mh^2/8 = 2^{35}, \\ \eta &= 2^{-17}, \quad g = 1/300, \quad s_- = 355/300, \quad s_+ = 377/300. \end{aligned}$$

Write $W_i = \Theta(r - 1000)V_i(r)$, $F = \chi_q U_f$, $\chi_q(r) = \Theta((r - 2000)/10)$, $U_f = (r^2/L^2 - 3)/(2mL^2)$, and

$$H_i(s) = -(2m)^{-1}\partial_r^2 + \gamma(s)W_i + (1 - \gamma(s))F, \quad \gamma(s) = B_9(s/g).$$

The decreasing cutoffs have their specified constant continuations. The exact conditional solution starts from $\chi(r) = 2re^{-r^2/(2L^2)}/(\pi^{1/4}L^{3/2})$ on $r > 0$. All estimates hold separately in each conserved sector and therefore for their coherent spinor, without sampling a sector. Norms are half-line norms or, equivalently, norms of the normalized odd extension.

C.1 Gaussian folds and transported cell comparisons

Gaussian integration gives

$$d_j^2 = \|\chi^{(j)}\|^2 = \frac{(2j+1)!!}{2^j L^{2j}}, \quad B_j = (d_j^2 + 2hd_j d_{j+1})^{1/2}.$$

Indeed $h \sum_k f(y + kh) \leq \int f + h \text{TV}(f)$ for nonnegative integrable f of bounded variation: compare the value in each cell to its cell average and sum the cell variations. Apply this to the square of the j th derivative of the odd extension, whose variation is at most $2d_j d_{j+1}$. Multiplication by an h -periodic taper of normalized cell norm A_k consequently costs at most $B_j A_k h^{-k}$. For a taper polynomial P rising from zero to one on a unit transition, with two transitions of width η and two omitted edge strips of width η , define

$$\begin{aligned} e_0^2 &= 2\eta + 2\eta \int_0^1 (1 - P)^2, & A_0 &= 1, \\ A_k^2 &= 2\eta^{1-2k} \int_0^1 (P^{(k)})^2, & C_k &= \sum_{j=0}^k \binom{k}{j} B_j A_{k-j} h^{-(k-j)}, \\ D_0 &= B_0 e_0, & c &= C_2/(2m). \end{aligned}$$

All taper integrals here are polynomial integrals with rational coefficients. Two different analytical tapers will be used for the same exact conditional wave. Their estimates are not interchanged.

The lens $b'' + \gamma b = 0$, $b(0) = 1$, $b'(0) = 0$ satisfies, for $0 \leq s < \pi/2$,

$$\cos s \leq b \leq 1, \quad b' \leq 0, \quad |b'| \leq \sin s, \quad b'^2 + \gamma b^2 \leq 1.$$

For the first inequality, $b - \cos s$ is the sine convolution of $(1 - \gamma)b$ until a hypothetical first zero; it prevents that zero. Also $(b'^2 + b^2)' = 2b'b(1 - \gamma) \leq 0$, and differentiating $b - \cos s$ or integrating the equation yields the derivative bound. In each cell centered at c_i set $x = r - c_i$, $y = x/b$, and

$$\Phi_i(s, r) = b^{-1/2} e^{imb'x^2/(2b)} \chi(c_i + y) a(y/h).$$

The core residual has norm $\leq cb^{-2}$. Its radial derivative, divided by mh , has norm at most $c|b'|/(2b^2) + C_3/(2m^2hb^3)$, by differentiating its amplitude and quadratic phase. At an origin well the helper is odd; at an origin barrier it vanishes in a collar. Thus the even extension of each potential and the odd extension of the helper justify integration by parts without an origin force or a self-adjoint half-line momentum assertion.

Here and below use the explicit weak-potential bounds

$$\begin{aligned} F_0 &= \frac{2010^2}{2mL^4} + \frac{3}{2mL^2}, \\ F_1 &= F_0/2 + 2010/(mL^4), \\ F_2 &= 1000F_0 + 2010/(mL^4) + 1/(mL^4). \end{aligned}$$

For the outer mismatch the harmonic backflow of $r \geq 1000$ is larger than 990. The recurrence

$$I_k(z) = \frac{1}{2} z^{k-1} e^{-z^2} + \frac{k-1}{2} I_{k-2}(z), \quad I_0(z) \leq e^{-z^2}/(2z)$$

for Gaussian tails, together with $e^2 > 7$, gives

$$T_0^2 = 21/7^{49}, \quad T_1^2 = 2040/(L^2 7^{49}), \quad T_2^2 = 205100/(L^4 7^{49})$$

as strict upper bounds for the first three Gaussian tail norms. Put $\epsilon = VT_0 + F_0$.

For the quintic taper $P = 10z^3 - 15z^4 + 6z^5$, direct integration gives

$$e_0^2 = 2\eta + 181\eta/231, \quad A_1^2 = 20/(7\eta), \quad A_2^2 = 240/(7\eta^3), \quad A_3^2 = 1440/\eta^5.$$

Define $M_1 = 15/(8\eta h)$, $f = 1/2 + 5h/8 + F_1/(mh)$, $q_0 = (B_1 e_0 + B_0 A_1/h)/(mh)$, and $J_3 = (\sec s \tan s + \log(\sec s + \tan s))/2$. Duhamel's inequality and its differentiated equation give the explicit increasing majorants

$$D(s) = D_0 + c \tan s + \epsilon s, \tag{78}$$

$$\begin{aligned} q(s) &= q_0 + f\{D_0 s + c \log \sec s + \epsilon s^2/2\} + c(\sec s - 1)/2 + C_3 J_3/(2m^2 h) \\ &\quad + (1/2 + 5h/8)T_0 s + VT_0(1 - \cos s)/2 + \frac{V(T_1 + M_1 T_0)}{mh} \log(\sec s + \tan s) \\ &\quad + \frac{F_1 s}{mh} + F_0 \left\{ (1 - \cos s)/2 + \frac{C_1}{mh} \log(\sec s + \tan s) \right\}. \end{aligned} \tag{79}$$

They bound $\|u_i - \Phi_i\|$ and $\|\partial_r(u_i - \Phi_i)\|/(mh)$. In detail, the force term in the differentiated error equation is $\leq f\|u_i - \Phi_i\|$; the outer-mismatch derivative contributes $(1/2 + 5h/8)T_0 + V|b'|T_0/2 + V(T_1 + M_1 T_0)/(mh)$; and the weak-trap derivative contributes $F_1/(mh) + F_0\{|b'|/2 + C_1/(mh)\}$. Integrating these terms and the core residual proves (78)–(79), rather than assuming an abrupt switch.

C.2 The fixed-age sorting bound and the complete age union

Let $G_1(s) = \int_0^s \gamma(x) dx$ and $G_2(s) = \int_0^s (s-x)\gamma(x) dx$ during the gate. For $s \leq g$, $b \leq 1 - \cos(g)G_2(s)$ and $|b'| \leq G_1(s)$. The identities $G_1(g) = g/2$, $G_2(g) = 3g^2/22$ imply, for $s \geq g$,

$$\begin{aligned} b_U(s) &= (1 - 3g^2 \cos g/22) \cos(s-g) - (g \cos g/2) \sin(s-g), \\ p_U(s) &= \sin(s-g) + (g/2) \cos(s-g). \end{aligned}$$

Use the preceding bounds as b_U, p_U before g and put $G_-(s) = 1/2 - (1/2 - \eta)b_U(s)$, $G'_+(s) = (1/2 - \eta)p_U(s)$. These functions are nonnegative and increasing on the needed interval; G_- is a lower gap and G'_+ an upper gap speed, not derivatives of a common artificial lens.

For completeness the moving-gap estimate is obtained with linear CDF cutoffs centered on each positive barrier, with half-width $w(s)$. The helper vanishes in these gaps, so the mass and integrated absolute current there are at most $D(s)^2$ and $hD(s)q(s)$ after disjoint gaps are summed. Differentiating the expectation of a moving linear cutoff costs at most $w'D^2/(2w) + hDq/(2w)$. Initial symmetric smoothing costs ϵ_0 below, and replacing a soft terminal cut by each of the two associated sharp classifier cuts costs their intervening error mass. Summing over both cuts at every positive barrier gives

$$S \leq 2\epsilon_0 + 2D(s_-)^2 + \int_0^{s_-} \frac{G'_+(s)D(s)^2 + D(s)q(s)}{G_-(s)} ds. \quad (80)$$

There is no collar across zero: its cumulative mass is exactly zero, and the single first-cell classifier is paid by its share of D^2 . At s_- the helper gap contains the classifiers because $G_-(s_-) > 1/4$. The initial Gaussian density $\rho_0 = \chi^2$ obeys

$$\text{TV}(\rho_0) < 12/(7L), \quad \text{TV}(|\rho'_0|) \leq 20/L^2.$$

The first follows from its single maximum $4/(e\sqrt{\pi}L) < 6/(7L)$; the second from integrating the Gaussian derivative polynomial. Symmetric linear smoothing and linear interpolation in half-cells of length $\ell = h/2$ give

$$\begin{aligned} \epsilon_0 &= \frac{(h\eta)^2}{6} \left\{ \frac{12}{7Lh} + \frac{20}{L^2} \right\}, \\ J &= \frac{\ell}{8} \frac{12}{7L} + \frac{\ell^2}{8} \frac{20}{L^2}, \quad R = 0.000052316289 \left(1 + \ell \frac{12}{7L} \right). \end{aligned}$$

For the interpolation estimate, the integrated error of a linear interpolant on a cell is at most one eighth of the cell length times the variation of the function there; apply this first to ρ_0 and then to its derivative. The reference preimage in each half-cell is shifted from the ideal sector cut by at most 0.000052316289ℓ , as follows from the periodic rotor error $(0.007233)^2$. Summing the corresponding initial masses with the Gaussian fold inequality gives R . Thus R refers to the original reference age 1.22, not s_- .

Here is also an explicit finite-range bound used in this comparison. For the periodic conditional solution z_i starting flat on a circle of length two, energy differentiation gives $\|z'_i\| \leq mh/2$, since $\frac{d}{ds}\langle T + \gamma V_i \rangle = \gamma'\langle V_i \rangle \leq \gamma'V$. Each periodic cell has mass $h/2$. The auxiliary wave $\sqrt{2}\chi z_i$ has residual norm at most

$$\beta = B_2/(2m) + (h/2)B_1 + VT_G + F_0B_0, \quad T_G = (22/7^{49})^{1/2}. \quad (81)$$

The two kinetic products use the Gaussian folds; the last two terms pay the finite outer and weak cutoffs. Its outer norm is at most T_G , because $h \sum_{k \geq 0} \rho_0(1000 + kh) \leq \int_{1000}^{\infty} \rho_0 + h\rho_0(1000) < 22/7^{49}$. Consequently the fixed-age mismatch with the original periodic label is

$$0.000002144 + C\{S + J + R + (s_- \beta + T_G)^2 + 201/(10 \cdot 7^{50})\}, \quad C = 270000/67499. \quad (82)$$

This is less than 0.002588532236512050. The quiet-prefix allowance is included once.

For the whole age interval use the fixed transition bands between relative distances $h/5$ and $3h/10$ from neighboring wells. The quintic helper and its derivative vanish on these bands because $b(s) \leq \cos(s - g)$ and $\cos(s_- - g) < 0.4$. A Lipschitz soft binary label with slope at most $10/h$ must vary by at least $1/2$ along any trajectory that starts outside the bands and subsequently changes its label. Hence, directly under the reference wave law,

$$q_{\text{age}} = D(s_-)^2 + 20 \int_{s_-}^{s_+} D(s)q(s) ds, \quad Cq_{\text{age}} < 0.000263432327209778. \quad (83)$$

This is a pathwise union estimate and permits the final age to depend on all other coordinates. One does not replace an age union by its largest single-age probability.

For an outer threshold the situation is different: reference ranks that escape at age s form the upper ray $[F_s(1000), 1]$. Their union is one ray. Therefore, on $[-1/8, 13/10]$, its mass is at most $q_{\text{out}} = ((13/10)\beta + T_G)^2$, and

$$Cq_{\text{out}} < 1.547666958047712 \times 10^{-8}. \quad (84)$$

The negative-age extension is the exact finite-trap evolution treated below and has a still smaller tail. If the actual rank differs from the original Gaussian rank by at most δ , expanding this ray by δ also covers the actual terminal escape. Adding it to the binary moving cuts uses $N = 256001$ cuts in the conservative $2N\delta$ allowance. With known drift β_* and remaining wave-law mean absolute drift I , Markov's inequality and optimization in $\delta - \beta_*$ give

$$C(q_{\text{age}} + q_{\text{out}}) + 2CN\beta_* + 2C\sqrt{2NI} + \mathbb{P}(\text{clock-age failure}). \quad (85)$$

Only domination of the original law is used. The same expanded outer ray pays whole-history escape if clock confinement and the absolute rank integral hold throughout the whole interval.

C.3 Energy graphs and the reference clock functional

The following coefficients also supply the energy inputs for holding. Put $A = W - F$, $A_0 = V + 1$, and use

$$A_1 = 2^{46}, \quad A_2 = 2^{80}, \quad A_3 = 2^{112}, \quad A_4 = 2^{160}$$

for derivative ceilings of A , with $\|U'\| \leq A_1$. They are deliberately loose bounds, not additional hypotheses. For the cap its first derivative is $-S_5(z)(1 - 2\eta z)$; the sum of absolute coefficients is at most 32, and the next three derivatives are bounded by $326^{k-1}\eta^{-(k-1)}$ with the cell powers of h restored. Leibniz's rule gives the displayed bounds after the outer cutoff. The smooth logistic cutoff has derivatives through order four bounded by 10^{10} : on its central half the logit derivatives are bounded by 32, 256, 3072, 49152; at an edge put $x = 1/t \geq 4$, use sigmoid derivatives $\leq 75e^{-A}$, $e^{-A} \leq 4e^{-x}$, and the Bell-polynomial bound $50x^8$. Its maximum is at most $150008^8/7^4 < 10^{10}$. The weak cutoff rescales these derivatives by 10^{-k} . The capped potential is C^3 with bounded weak fourth derivative, which suffices for these third energy graphs; no fourth radial Hamiltonian graph is invoked.

Define, entirely algebraically,

$$\begin{aligned} N_1 &= A_0, & P &= \{2m(N_1 + 1)\}^{1/2}, \\ B_* &= A_2/(2m) + A_1P/m, & N_2 &= (1001/1000)A_0^2, \\ P_H &= \{2m(N_1N_2 + N_1^2)\}^{1/2}, & U_{rr} &= 2m(N_1 + A_0), \quad N_3 = (1001/1000)A_0^3. \end{aligned}$$

Then $\|H^j u\| \leq N_j$, $j = 1, 2, 3$, throughout the gate and static plateau. To check this directly, $\|H_0^2 \chi\|, \|H_0^3 \chi\| < 1$ by the Gaussian products and the finite weak cutoff. The first graph obeys $\|Hu\| \leq \|H_0 \chi\| + V + F_0 < V + 1 = N_1$ by graph Duhamel and $\int \gamma' = 1$. The identities

$$[H, A] = -(A'' + 2A'\partial_r)/(2m), \quad [H, [H, A]] = \frac{A'''' + 4A'''\partial_r + 4A''\partial_r^2}{4m^2} + A'U'/m$$

and $\int \gamma' = 1$ give

$$\begin{aligned}\|H^2u\| &\leq 1 + 2A_0 + A_0^2 + B_* < N_2, \\ \|H^3u\| &\leq 1 + 3A_0(1 + A_0 + A_0^2/3 + B_*/2) \\ &\quad + 3\{A_2N_1/(2m) + A_1P_H/m\} + (A_4 + 4A_3P + 4A_2U_{rr})/(4m^2) + A_1^2/m < N_3.\end{aligned}$$

For the second line retain the intermediate bound $\|H^2u\| \leq (1 + A_0\gamma)^2 + B_*\gamma$ and integrate $(H^3)' = \gamma'(3AH^2 + 3[H, A]H + [H, [H, A]])$. Smooth approximation on the common graph domain justifies the identities.

To evaluate the smaller reference-current bound use the transported cell construction with $P = B_9$ and C_k through order four. Fix a horizon $a < \pi/2$ (here $a = s_+$; the holding calculation uses $4/3$). With $M_1 = (315/128)/(\eta h)$, $M_2 = (2520/64)/(\eta^2 h^2)$ put

$$\begin{aligned}T_{A1} &= T_1 + M_1T_0, \\ T_{A2} &= T_2 + 2M_1T_1 + M_2T_0, \\ P_t &= mhT_0/2 + \sec(a)T_{A1}, \\ P_{t2} &= m^2h^2T_0/4 + m \tan(a)T_0 + mh \sec(a)T_{A1} + \sec(a)^2T_{A2}, \\ W_1 &= mh/2 + 5V, \\ W_2 &= m(1 + 15/(16\eta)) + 5mh + 10^5V, \\ B_c &= (W_2T_0 + 2W_1P_t + VP_{t2})/(2m) + (V + F_0)VT_0, \\ P_\Phi &= mh/2 + \sec(a)C_1, \\ H_\Phi &= V + \tan(a)/2 + \tan(a)hC_1/2 + c \sec(a)^2 + VT_0 + F_0, \\ B_f &= F_0H_\Phi + F_2/(2m) + F_1P_\Phi/m, \\ B_m &= \{VT_{A2}/(2m) + F_0c\} \sec(a)^2.\end{aligned}$$

Acting with the harmonic core on its residual gives

$$\|HR_{\text{core}}\| \leq Vcb^{-2} + (-b'/b)(hC_3 + C_2)b^{-2}/(4m) + C_4b^{-4}/(4m^2).$$

The preceding B_c, B_f, B_m bound respectively the differentiated outer mismatch, weak mismatch, and their action on the core residual. Thus

$$\begin{aligned}D_9(s) &= D_0 + c \tan s + \epsilon s, \\ Q_9(s) &= c + F_0D_0 + (V + F_0)D_9(g) + Vc \tan s + \frac{hC_3 + C_2}{8m}(\sec^2 s - 1) \\ &\quad + \frac{C_4}{4m^2}(\tan s + \tan^3 s/3) + (B_c + B_f + B_m)s.\end{aligned}\tag{86}$$

These bound $\|u - \Phi\|$ and $\|H(u - \Phi)\|$. The gate derivative source is at most $(V + F_0)D_9(g)$; it is not omitted. Set $P_E = \{2mD_9(Q_9 + F_0D_9)\}^{1/2}$, $P_{E2} = 2m\{Q_9 + (V + F_0)D_9\}$ at $s = a$. The form inequality and half-line Sobolev give $\|E\|_\infty \leq (2D_9P_E)^{1/2}$, $\|E'\|_\infty \leq (2P_EP_{E2})^{1/2}$. With $\rho_* = 6/(7L)$ use

$$\begin{aligned}P_\infty &= (\rho_* \sec a)^{1/2}, \\ P'_\infty &= mhP_\infty/2 + \sec(a)^{3/2}\{(2d_1d_2)^{1/2} + M_1\sqrt{\rho_*}\}, \\ j_* &= (h/2) \tan(a)\rho_* + \{P_\infty\sqrt{2P_EP_{E2}} + P'_\infty\sqrt{2D_9P_E} + \sqrt{2D_9P_E}\sqrt{2P_EP_{E2}}\}/m.\end{aligned}$$

This follows by expanding all three error terms in the bilinear current, including E^*E' . For $a = s_+$, $D_9 < 0.006877$, $Q_9 < 2.364 \times 10^8$, and $j_* < 9.194 \times 10^6$.

Restore $M = 10$, $v = 1$, $T_0 = 0.03$, $\sigma = 0.001$ in SI units and $\hbar = 6.62607015 \times 10^{-34}/(2\pi)$. For the next two current identities, use physical radius $r_{\text{phys}} = 10^{-4}r$ and its normalized conditional

wave, of density $\rho = \sum_i |u_i|^2$; sector sums are implicit in products and r in their integrals denotes r_{phys} . The radial physical mass is $m_{\text{phys}} = 2^{52} \hbar T_0 / (10^{-4})^2$. For $G = \phi(S - vt - S_c)u(S, r_{\text{phys}})$ the radial energy density and clock correction are $e = \Re(u^* H_{\text{phys}} u)$ and $k = -e/(Mv)$. If K is the conditional CDF, write $B_k = \int_0^r k$, $b_k = \int k$. Substituting into the exact rank numerator gives

$$\mathcal{F}(G) = -\phi^2 k j / v - \rho \{2\phi\phi'(B_k - Kb_k) + \phi^2(\partial_S B_k - K\partial_S b_k)\}.$$

Energy continuity gives

$$\partial_S B_k - K\partial_S b_k = \frac{j_E}{Mv^2} - \frac{\dot{\gamma}}{Mv^2} \left\{ \int_0^r (W_{\text{phys}} - F_{\text{phys}}) \rho - K \langle W_{\text{phys}} - F_{\text{phys}} \rangle \right\},$$

where $j_E = (\hbar/(2m_{\text{phys}})) \Im(u^* \partial_{r_{\text{phys}}} (H_{\text{phys}} u) + (H_{\text{phys}} u)^* \partial_{r_{\text{phys}}} u)$. The intrinsic term is bounded by $\hbar a j_* N_1 / (Mv^2 T_0)$. Returning to detector units, for the energy term use the H^2 current $J_2 = m^{-1} \Im(u^* (Hu)' + (Hu)^* u')$, $\|J_2\|_1 \leq (P_H + N_1 P)/m$. For the conservative duration $T_s = 0.0377$ let $\varepsilon = \hbar T_s / (2Mv^2 T_0^2)$ and bound its density by

$$\rho_e = \left[P_\infty + \sqrt{2\tilde{D}\sqrt{2m\tilde{D}(\tilde{Q} + 5 \times 10^{-18}\tilde{D})}} \right]^2, \quad \tilde{D} = D_9 + \varepsilon N_2, \quad \tilde{Q} = Q_9 + \varepsilon N_3.$$

This enlarged density also bounds the unchanged reference. The energy contribution is at most $\varepsilon \rho_e (P_H + N_1 P)/m$. Finally $|B_k - Kb_k| \leq 2\|H_{\text{phys}} u\|/(Mv)$, $\int 2|\phi\phi'| \leq 2p_1$, and $\int \dot{\gamma} dt = 1$ for each clock offset. With $T = 0.04$, the reference functional is bounded by

$$I_G \leq \frac{\hbar a j_* N_1}{Mv^2 T_0} + \frac{\varepsilon \rho_e (P_H + N_1 P)}{m} + \frac{4p_1 T \hbar N_1}{Mv T_0} + \frac{2\hbar A_0}{Mv^2 T_0} + I_q < 2.382267143194879 \times 10^{-16}. \quad (87)$$

The quiet correction I_q is explicitly controlled below. This is a functional of the comparison wave, not an assertion that G solves the autonomous Schrödinger equation.

C.4 The finite-trap quiet boundary

The Gaussian is not stationary for the finite trap. Put $\alpha = (2m)^{-1}$, $w = (\chi_q - 1)U_f$, $u = w\chi$. Then

$$\begin{aligned} H_F \chi &= u, & H_F^2 \chi &= -\alpha u'' + F u, \\ H_F^3 \chi &= \alpha^2 u'''' - \alpha F'' u - 2\alpha F' u' - 2\alpha F u'' + F^2 u. \end{aligned}$$

All terms are supported at $r \geq 2000$. Here is an explicit polynomial recipe for their norms. Set $y = r/L$, $q_f = 1/(2mL^2)$, $U_0(y) = y^2 + 3$, $U_1(y) = 2y/L$, $U_2(y) = 2/L^2$, and $(c_0, c_1, c_2, c_3, c_4) = (1, 1/2, 32/25, 1000, 20000)$. These bounds follow by rescaling the cutoff derivatives $5, 128, 10^6, 2 \times 10^8$ by its width ten. Define

$$\begin{aligned} W_j(y) &= \sum_{k=0}^{\min(2,j)} \binom{j}{k} c_{j-k} U_k(y), \\ (R_0, R_1, R_2, R_3, R_4) &= \left(1, \frac{1+y}{L}, \frac{3+y^2}{L^2}, \frac{3+6y+y^3}{L^3}, \frac{15+10y^2+y^4}{L^4} \right), \\ Z_j(y) &= \sum_{k=0}^j \binom{j}{k} W_{j-k}(y) R_k(y), \quad U_j^* = q_f Z_j(1) 10^{12}/7^{100}. \end{aligned}$$

Each Z_j has nonnegative coefficients and degree at most six. The Gaussian recurrence gives $\|\mathbf{1}_{y \geq 20} y^k \chi\| \leq 10^{12}/7^{100}$ for $k \leq 6$, so $\|u^{(j)}\| \leq U_j^*$. Use

$$\begin{aligned} Q_1 &= 2600/(2mL^2 7^{100}), \\ Q_2 &= \alpha U_2^* + F_0 U_0^*, \\ Q_3 &= \alpha^2 U_4^* + \alpha F_2 U_0^* + 2\alpha F_1 U_1^* + 2\alpha F_0 U_2^* + F_0^2 U_0^*. \end{aligned}$$

The sharper first bound is the direct recurrence for $(y^2 + 3)\chi$ on $y \geq 20$. For $u(s) = e^{-isH_F}\chi$, $-1/8 \leq s \leq 0$, the boundary errors after multiplication by the real clock packet are

$$\Delta_k = p_k Q_{1/8} + \sum_{j=1}^k \binom{k}{j} p_{k-j} Q_j / (vT_0)^j, \quad 0 \leq k \leq 3. \quad (88)$$

In particular $(\Delta_0, \Delta_1, \Delta_2, \Delta_3) < (1.116 \times 10^{-102}, 2.947 \times 10^{-99}, 1.195 \times 10^{-95}, 6.072 \times 10^{-92})$ in the corresponding clock derivative units. The constants p_k are given next. Substituting these in the rank functional pays the negligible I_q ; 10^{-40} is a convenient outward ceiling for that term in (87).

C.5 Clock coefficients and the five-error decomposition

The smoothed cosine-fourth packet has the following derivative ceilings, with $q_s = 1.000001$ and $p_0 = 1$:

$$(p_1, p_2, p_3, p_4) = q_s \left(\sqrt{16/7} \frac{\pi}{2\sigma}, \quad \sqrt{512/35} \left(\frac{\pi}{2\sigma} \right)^2, \quad \sqrt{1024/7} \left(\frac{\pi}{2\sigma} \right)^3, \quad \sqrt{69632/35} \left(\frac{\pi}{2\sigma} \right)^4 \right).$$

They follow by expanding the powers of sine and cosine and integrating on the packet support. Convolution with the nonnegative mollifier contracts every derivative norm; normalization costs at most q_s . Let $d = 0.0001$ seconds and

$$b_1 = (5/2)/d, \quad b_2 = (2520/64)/d^2, \quad b_3 = (7560/16 + 5040/64)/d^3, \quad b_4 = 1360800/d^4.$$

These are bounds for the first four physical-time derivatives of B_9 ; the fourth bound is the absolute coefficient sum. Define

$$\begin{aligned} c_1 &= N_1/(vT_0), \\ c_2 &= N_2/(v^2T_0^2) + b_1 A_0/(v^2T_0), \\ c_3 &= N_3/(v^3T_0^3) + b_1(3A_0N_1 + B_*)/(v^3T_0^2) + b_2 A_0/(v^3T_0), \\ G_1 &= (p_1^2 + c_1^2)^{1/2}, \quad G_2 = p_2 + 2p_1c_1 + c_2, \\ G_3 &= p_3 + 3p_2c_1 + 3p_1c_2 + c_3, \quad Q_G = q_s(4\pi^2/\sigma^2)\sqrt{3/35} + c_1^2. \end{aligned}$$

Differentiating the conditional equation gives these bounds for $\|G_S\|, \|G_{SS}\|, \|G_{SSS}\|$ and its logarithmic-envelope coefficient. In the third derivative use $2H_S H + H H_S = 3H_S H + [H, H_S]$; the commutator costs B_* . The first derivative improves to a sum of squares because the real packet derivative and the unitary conditional derivative have zero real cross term.

The denominator-free rank stability estimate of Section 24, for an error with norms D, e_1, e_2 , is

$$\mathcal{R}[D, e_1, e_2] = \frac{\hbar}{M} \int (4G_1 e_1 + 2e_1^2 + 14DQ_G + 9DG_2 + e_2) dt. \quad (89)$$

The derivatives here are clock derivatives. Decompose the exact error from the same prepared stock as follows. If E_h is the prepared error, U_a the actual propagator, ψ_c the evolution of the product handoff under the clipped gate, and ψ_p agrees with ψ_c until $T_g = 0.0033$ and evolves statically thereafter, put

$$E_p = U_a E_h, \quad E_c = U_a(\phi\chi) - \psi_c, \quad E_l = \psi_c - \psi_p.$$

Before T_g put $E_r = \psi_c - G$, $E_s = 0$; afterward put

$$E_r = U_W(t - T_g)(\psi_c(T_g) - G(T_g)), \quad E_s = U_W(t - T_g)G(T_g) - G(t).$$

Then $\Psi - G = E_p + E_c + E_l + E_r + E_s$ identically. The carried ramp error is counted once, the new static error starts from zero, and the quiet boundary difference is included in E_r . The free clock and its Galilean transport belong to every propagator.

C.6 Propagation of the prepared error under the common operator

Define the old-wave preparation inputs by

$$\begin{aligned} D_{\text{old}} &:= D_o, & P_{10} &:= P_{S,o}, & P_{20} &:= H_{SS}^o, \\ n_{10} &:= n_{1T}, & n_{20} &:= n_{2T}. \end{aligned}$$

Use the defining expressions in (31), (32), and (37), not the rounded enclosures in Section B.7. In particular, H_{SS}^o is the ordinary, phase-restored Hessian bound, not the gauged bound H_{SS} . The old-wave norm D_{old} is distinct from the enlarged-wave norm D_h in Appendix B. The remote force coefficients are explicitly

$$\begin{aligned} F_c &= \mu a_2 F_*/2 + \mu a_0 a_1 f_*^2 + \mu^2 a_1 a_2 F_*^2/(4M) + 10^{-20}, \\ F_{c2} &= \mu a_3 F_*/2 + \mu(a_1^2 + a_0 a_2) f_*^2 + \mu^2(a_2^2 + a_1 a_3) F_*^2/(4M) + 10^{-18}, \end{aligned}$$

where $(a_0, a_1, a_2, a_3) = (140, 43600, 70141750, 1387431060000)$, $F_* = 0.080802$, $f_* = 0.201$, and $\mu = 2^{52} \hbar T_0 / (10^{-4})^2$. These are the product-rule bounds for the retained preparation phase and its scalar selfpotential; $F_c < 5 \times 10^{-6}$ and $F_{c2} < 0.09$. Set $A_{\text{phys}} = \hbar A_0 / T_0$, $F_a = b_1 A_{\text{phys}}$, $F_{a2} = b_2 A_{\text{phys}}$ and collar widths $g_1 = 0.0005$, $g_2 = 0.00025$. Initialize

$$L_{10} = \sqrt{n_{10} P_{20}} + 2\hbar D_{\text{old}}/g_1, \quad L_{20} = \sqrt{n_{20} P_{20}} + 2\hbar n_{10}/g_2.$$

The six nonnegative variables $(P_1, P_2, n_1, L_1, n_2, L_2)$ are bounded by the solution, starting at these values, of

$$\begin{aligned} \dot{P}_1 &= F_a D_{\text{old}} + F_c n_2, & \dot{P}_2 &= 2F_a P_1 + \hbar F_{a2} D_{\text{old}} + 2F_c L_2 + \hbar F_{c2} n_2, \\ \dot{n}_1 &= P_1/(M g_1), & \dot{L}_1 &= P_2/(M g_1) + (F_c + F_a) n_1, \\ \dot{n}_2 &= L_1/(M g_2), & \dot{L}_2 &= P_2/(M g_2) + (F_c + F_a) n_2. \end{aligned}$$

Here the collars move at speed v , cancelling Galilean transport; the inner transition lies where the outer collar equals one. Clock commutators give the first two equations. Weighted continuity for the error and for its first clock momentum gives the other four. The radial kinetic term commutes with every collar. The two initial local-momentum inequalities follow by integration by parts and Cauchy–Schwarz. Thus this is a closed positive comparison system, including the remote preparation force. With $e_j = P_j/\hbar^j$, use its exact time integrals in (89); bound $\int P_1^2 \leq P_1(T) \int P_1$. For $T = 0.04$ the resulting contribution is $< 1.954625 \times 10^{-17}$.

C.7 Short ramp and static correction

Put $a_j = b_j A_0 / (v^j T_0)$, $R_0 = \hbar G_2 / (2M)$ and $R_1 = \hbar G_3 / (2M)$. On $0 \leq t \leq T_g$, commutation with the bounded clock derivatives of the gate gives

$$\begin{aligned} D_r(t) &= \Delta_0 + R_0 t, \\ e_{1r}(t) &= \Delta_1 + R_1 t + a_1 \Delta_0 t + a_1 R_0 t^2/2, \\ e_{3r}(t) &= G_3 + p_3 + 3a_1 p_2 t + 3a_1^2 p_1 t^2 + a_1^3 t^3 + 3a_2 p_1 t + 3a_1 a_2 t^2 + a_3 t, \\ e_{2r}(t) &= \sqrt{e_{1r}(t) e_{3r}(t)}. \end{aligned}$$

The last line is Fourier interpolation $\|E_{SS}\|^2 \leq \|E_S\| \|E_{SSS}\|$. These are clock derivatives and require only the third radial energy graph. After T_g the ramp error's clock norms are conserved by the static propagator. Endpoint ceilings for duration $T = 0.04$ in (89) give $< 4.073932 \times 10^{-16}$. The time T_g is counted from handoff: the trailing support point $0.1009999999 + T_g$ exceeds the gate end 0.1041.

For the static component the energy gauge $U(S) = \exp[-iH_{\text{phys}}(S - S_0)/(\hbar v)]$ transforms the generator to $vp + (p - H_{\text{phys}}/v)^2/(2M)$. Its three commuting corrections are generated by $p^2/(2M)$, $-pH_{\text{phys}}/(Mv)$ and $H_{\text{phys}}^2/(2Mv^2)$. The first two ordinary clock derivatives become powers of $(p - H_{\text{phys}}/v)/\hbar$, which commute with all three corrections. For $T_s = 0.0377$, $\varepsilon = \hbar T_s/(2Mv^2T_0^2)$, the triples (D, e_1, e_2) for these three factors are respectively

$$\begin{aligned} D_f &= \hbar T_s p_2 / (2M), \\ e_{1f} &= \frac{\hbar T_s}{2M} (p_3 + p_2 N_1 / (vT_0)), \\ e_{2f} &= \frac{\hbar T_s}{2M} (p_4 + 2p_3 N_1 / (vT_0) + p_2 N_2 / (v^2 T_0^2)), \\ D_x &= \hbar T_s p_1 N_1 / (MvT_0), \\ e_{1x} &= \frac{\hbar T_s}{Mv} (p_2 N_1 / T_0 + p_1 N_2 / (vT_0^2)), \\ e_{2x} &= \frac{\hbar T_s}{Mv} (p_3 N_1 / T_0 + 2p_2 N_2 / (vT_0^2) + p_1 N_3 / (v^2 T_0^3)), \\ D_E &= \varepsilon N_2, \quad e_{1E} = p_1 \varepsilon N_2 + \varepsilon N_3 / (vT_0), \\ e_{2E} &= p_2 \varepsilon N_2 + 2p_1 \varepsilon N_3 / (vT_0) + \sqrt{2} \varepsilon N_3 / (v^2 T_0^2). \end{aligned}$$

The last term follows from $|e^{-i\varepsilon\lambda^2} - 1| \leq \sqrt{2\varepsilon}|\lambda|$. Thus $\|H^2(e^{-i\varepsilon H^2} - 1)u\| \leq \sqrt{2\varepsilon}\|H^3 u\|$; no fourth radial graph is required. Sum the triples, and use the static versions of G_2 (with its gate term removed) and G_1, Q_G in (89). The static contribution is $< 3.770360 \times 10^{-19}$.

C.8 Both remote clock comparisons

For the clipped evolution let $\mathcal{A}_j = A_{\text{phys}} b_j / v^j$, $j = 1, \dots, 4$, and $\mathcal{A}_0 = A_{\text{phys}}$. The global physical clock moments obey

$$\dot{P}_k \leq \sum_{j=1}^k \binom{k}{j} \hbar^{j-1} \mathcal{A}_j P_{k-j}, \quad P_k(0) = \hbar^k p_k, \quad P_0 = 1.$$

Define their polynomial majorants recursively by equality. This is an explicit finite triangular recursion to degree k , using no radial derivative. Four static left collars of width $g_c = 0.0002$ are supported below 0.10081, below the initial clock support and before activation. Their transitions are nested; where they act all clipped clock forces vanish. Weighted continuity therefore yields the local polynomials

$$l_k(t) = (Mg_c)^{-(4-k)} \mathcal{I}^{4-k} P_4(t), \quad k = 0, 1, 2, \quad \mathcal{I}f(t) = \int_0^t f(s) ds.$$

They bound $\|\chi_0 p^k \psi_c\|$. Set

$$V_c = \mu a_1 F_*/2 + \mu a_0^2 f_*^2/2 + \mu^2 a_1^2 F_*^2/(8M) + 10^{-20}, \quad C_1 = F_c + \mathcal{A}_1, \quad C_2 = F_{c2} + \mathcal{A}_2.$$

The common-minus-clipped source is supported in this collar. Its norm and momentum rates are

$$f_0 = V_c l_0 / \hbar, \quad f_1 = V_c l_1 / \hbar + F_c l_0, \quad f_2 = V_c l_2 / \hbar + 2F_c l_1 + \hbar F_{c2} l_0.$$

Starting from zero, its bounds are the explicit polynomials

$$D_c = \mathcal{I}f_0, \quad P_{1c} = \mathcal{I}(f_1 + C_1 D_c), \quad P_{2c} = \mathcal{I}(f_2 + 2C_1 P_{1c} + \hbar C_2 D_c).$$

This retains the common propagator's force; differentiating a Duhamel integral as though the propagator commuted with p would not suffice. Their exact polynomial time integrals in (89), with $\int P_{1c}^2 \leq P_{1c}(T) \int P_{1c}$, give $< 2.240929 \times 10^{-21}$.

For the clipped-to-static comparison use the moving weight $\chi(t, S) = \min(1, \exp[-10^6(S - 0.1009 - vt)])$. Its derivative satisfies $\chi_t + v\chi_S = 0$ and $|\chi_S| \leq 10^6\chi$. Let $L_k = \|\chi p^k \psi_c\|$, $0 \leq k \leq 3$, and bound them with the positive system

$$\dot{L}_k = (10^6/M)L_{k+1} + \sum_{j=1}^k \binom{k}{j} \hbar^{j-1} \mathcal{A}_j L_{k-j}, \quad L_4 = P_4, \quad L_k(0) = (2/7^{50})\hbar^k p_k.$$

The initial bound follows from $e^{-99.9999} < 2/7^{50}$. For $t \geq T_g$, the weight equals one on the entire remaining gate, so the late error obeys

$$\begin{aligned} D_t &\leq (\mathcal{A}_0/\hbar) \int_0^T L_0, \\ P_{1t} &\leq (\mathcal{A}_0/\hbar) \int_0^T L_1 + \mathcal{A}_1 \int_0^T L_0, \\ P_{2t} &\leq (\mathcal{A}_0/\hbar) \int_0^T L_2 + 2\mathcal{A}_1 \int_0^T L_1 + \hbar\mathcal{A}_2 \int_0^T L_0. \end{aligned}$$

Extending these nonnegative integrals back from T_g to zero is conservative. The static propagator commutes with p . Using the endpoint triples over the duration T in (89) gives $< 2.817683 \times 10^{-29}$. Thus neither remote exact clock tail has been set to zero.

C.9 Final old current, terminal CDF, and reproducible arithmetic

Each of the five component expressions includes its own quadratic $2e_1^2$ term. For their sum use $(\sum_{j=1}^5 e_{1j})^2 \leq 5 \sum_{j=1}^5 e_{1j}^2$. Add four further copies of the five quadratic terms to their sum. This extra allowance is $< 2.117822196256773 \times 10^{-27}$. Together with (87) this gives

$$I_{\text{old}} < 6.655453611235769 \times 10^{-16}. \quad (90)$$

All absolute values precede the live joint-coordinate integral. The components are wave-law integrals; the original-law cap is inserted only in the event conversion.

The scalar terminal comparison, on $T = 0.04$, is independently bounded by

$$D_{\text{end}} = D_{\text{old}} + \frac{\hbar T p_2}{2M} + \frac{T p_1 \hbar N_1}{M v T_0} + \frac{\hbar A_0}{2M v^2 T_0} \frac{5}{2} \left(1 + \frac{2(\sigma + 10^{-10})}{v d} \right) + \frac{T \hbar N_2}{2M v^2 T_0^2} + \Delta_0. \quad (91)$$

The gate coefficient is a supremum over the whole clock packet, not the single-offset integral. It is below 55 and gives $D_{\text{end}} < 3.034377484488931 \times 10^{-11}$. For each radial cut the exact conditional CDF K_Ψ and helper CDF K_u obey

$$\int w(S) |K_\Psi(S, b) - K_u(S, b)| dS \leq \| |\Psi|^2 - |G|^2 \|_1 \leq 2D_{\text{end}}.$$

Indeed subtract the two densities against the centered indicator $\mathbf{1}_{r < b} - K_u(S, b)$, of modulus at most one. Under the exact terminal wave law K_Ψ is uniform conditional on S ; equivariance transports the original cap, giving $2CND_{\text{end}}$ for all $N = 256001$ cuts. This is terminal domination by the exact wave, never a newly imposed cap relative to the helper.

All displayed decimal bounds can be checked by the following finite rational procedure; the formulae above, rather than abbreviated decimals, are the inputs. For $0 \leq t \leq 4/3$ use the 24-term alternating Taylor sums for sine and cosine with the next-term one-sided remainder. Bound square roots by rational bisection, and logarithms by

$$\log x = 2 \sum_{k=0}^{n-1} \frac{z^{2k+1}}{2k+1} + R_n, \quad z = (x-1)/(x+1), \quad |R_n| \leq \frac{2|z|^{2n+1}}{(2n+1)(1-z^2)},$$

after taking reciprocal arguments where useful. Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ with alternating remainders bounds π . Rational arithmetic permits arbitrary refinement. For (80) use the breakpoints $0, s_-/2^{20}, g$ and 64 equally spaced subintervals in every $[s_-/2^j, s_-/2^{j-1}]$, $j = 20, \dots, 1$. Take every nonnegative numerator factor at the right endpoint and the increasing lower gap at the left. No monotonicity of their ratio is assumed. For (83) use 128 equal subintervals and right endpoint rectangles for the increasing product Dq . These give

$$\int_0^{s_-} \frac{G'_+ D^2 + Dq}{G_-} < 0.000517506143180416981,$$

$$\int_{s_-}^{s_+} Dq < 0.000001583255601884495.$$

For a positive system $y' = Ay$, rescale by a positive diagonal matrix S , put $B = S^{-1}AS$, $z_0 = S^{-1}y_0$, and sum $z_k = (TB)^k z_0/k!$. If $\alpha = T\|B\|_\infty < K + 2$, the omitted norm after degree K is at most

$$\|z_K\|_\infty \frac{\alpha}{K+1} \frac{1}{1 - \alpha/(K+2)}.$$

For the time integral sum $Tz_k/(k+1)$ and multiply this tail by $T/(K+2)$. For the homogeneous system append the constant state one, use scales $(10^{-29}, 10^{-47}, 10^{-28}, 10^{-35}, 10^{-33}, 10^{-40}, 1)$ and $K = 80$. For the combined global/local tail system, ordered $(P_0, \dots, P_4, L_0, \dots, L_3)$, use $(1, 10^{-19}, 10^{-38}, 10^{-57}, 10^{-76}, 10^{-42}, 10^{-60}, 10^{-68}, 10^{-72})$ and $K = 100$. In both cases $\alpha < 10$ for $T = 0.04$. This states every matrix entry, initial value, quadrature partition and remainder rule needed to evaluate the bound without a separate coefficient ledger or numerical propagation of the Schrödinger equation.

D Complete prefix estimates for the compensated writer

This appendix closes the clock and pointer derivative estimates used before $t_a = 0.0366$ s. All norms are full spinor Hilbert norms, including both internal sectors. Write $t = 0$ at $t_h = -0.002$ s, and set

$$T = 0.0386 \text{ s}, \quad T_0 = 0.03 \text{ s}, \quad a = vT_0 = 0.03 \text{ m},$$

$$v = 1 \text{ m/s}, \quad M = 10 \text{ kg}, \quad \mu = \mu_Z = 0.01, \quad \kappa = \hbar/M.$$

Clock derivatives are physical S derivatives; pointer derivatives are with respect to the dimensionless Z . All scalar numbers below use SI units for the clock. Removing the same constant Galilean carrier from both exact waves leaves the clock generator $vp_S + p_S^2/(2M)$. The preparation estimates are those of Sections B.5 and B.6; the old-wave bounds are obtained from Sections C.6 and C.5. Thus neither wave nor actual law is replaced at the handoff.

D.1 Finite Gaussian coefficient calculation

Let $w = 0.003$, $b(s) = 24B_9(s/w)$ with its constant extensions, and

$$\gamma(s, Z) = \pi^{-1/4} e^{-(Z-b)^2/2} e^{i\mu b'(Z-b/2)}, \quad y = Z - b.$$

Its first logarithmic derivative is

$$\partial_s \log \gamma = cy + d, \quad c = b' + i\mu b'', \quad d = \frac{i\mu}{2}(bb'' - b'^2).$$

The scalar phase in d is retained. Successive logarithmic derivatives are $cy + d$, $c'y - cb' + d'$, $c''y - 2c'b' - cb'' + d''$, and $c'''y - 3c''b' - 3c'b'' - cb''' + d'''$. The following rational derivative ceilings specify every coefficient used below:

$$(B_0, B_1, B_2) = (24, 19687.5, 105000000),$$

$$(B_3, B_4, B_5) = (4.9 \times 10^{11}, 4.032 \times 10^{17}, 4.7936 \times 10^{20}), \quad |b^{(j)}| \leq B_j.$$

Here $B_1 = 24(315/128)/w$ follows from $B'_9 = 630x^4(1-x)^4$. Differentiating this identity, using $x(1-x) \leq 1/4$ and $|1-2x| \leq 1$, gives $B_2 = 24(2520/64)/w^2$ and $B_3 = 24(7560/16 + 5040/64)/w^3$. For $j = 4, 5$ the sufficient coefficient bound is

$$B_j = \frac{24}{w^j} \sum_{k=5}^9 |\beta_k| \frac{k!}{(k-j)!}, \quad (\beta_5, \dots, \beta_9) = (126, -420, 540, -315, 70).$$

The polynomial and constant extensions are globally C^4 ; their fifth weak derivative is bounded. No endpoint distribution enters these calculations.

Define $C_j = B_j + \mu B_{j+1}$ for $0 \leq j \leq 4$ and $q_0 = C_0 + 1$. The C_j bound the derivatives of $b + i\mu b'$. Define four positive affine polynomials in a formal variable Y :

$$\begin{aligned} d_0 &= \mu(B_0 B_2 + B_1^2)/2, & d_1 &= \mu(B_0 B_3 + B_1 B_2)/2, \\ d_2 &= \mu(B_0 B_4 + B_2^2)/2, & d_3 &= \mu(B_1 B_4 + B_0 B_5 + 2B_2 B_3)/2, \\ L_1 &= d_0 + C_1 Y, & L_2 &= C_1 B_1 + d_1 + C_2 Y, \\ L_3 &= 2C_2 B_1 + C_1 B_2 + d_2 + C_3 Y, \\ L_4 &= 3C_3 B_1 + 3C_2 B_2 + C_1 B_3 + d_3 + C_4 Y. \end{aligned}$$

For a positive polynomial $P(Y) = \sum p_j Y^j$, put

$$\mathcal{N}(P) = \sum_j p_j \sqrt{(2j-1)!!/2^j}, \quad (-1)!! = 1.$$

This uses the exact Gaussian moment $\|y^j \gamma\|^2 = (2j-1)!!/2^j$. Complete, evaluable formulas for the four pointer jets are

$$\begin{aligned} g_0 &= 1, & g_1 &= \mathcal{N}(L_1), & g_2 &= \mathcal{N}(L_1^2 + L_2), \\ g_3 &= \mathcal{N}(L_1^3 + 3L_1 L_2 + L_3), \\ g_4 &= \mathcal{N}(L_1^4 + 6L_1^2 L_2 + 3L_2^2 + 4L_1 L_3 + L_4), \end{aligned} \tag{92}$$

so that $\|\partial_s^j \gamma\| \leq g_j$. In particular,

$$\begin{aligned} g_1 &< 1.529437156625049 \times 10^7, & g_2 &< 2.344303538271762 \times 10^{14}, \\ g_3 &< 3.601493635795900 \times 10^{21}, & g_4 &< 5.545932284532096 \times 10^{28}. \end{aligned}$$

The defining expressions, rather than these shortened displays, are used in the numerical bounds below.

For $A = \partial_Z + Z - \Pi_1(b + i\mu b')$ and $Q(f) = (\|Zf\|^2 + \|p_Z f\|^2)^{1/2}$, the oscillator form identity and $[A, Z] = 1$ give

$$\begin{aligned} Q(f) &\leq \|Af\| + q_0 \|f\|, \\ Q(Zf) &\leq \|A^2 f\| + 2q_0 \|Af\| + (q_0^2 + 1) \|f\|. \end{aligned} \tag{93}$$

They hold fibrewise and then for the Hilbert direct sum over all other coordinates. Differentiating $A\gamma = 0$ gives the weighted jet bounds

$$q_j = q_0 g_j + \sum_{k=1}^j \binom{j}{k} C_k g_{j-k}, \quad 1 \leq j \leq 3, \quad Q(\partial_s^j \gamma) \leq q_j.$$

Finally, for the writer coefficients $F = b/\mu + \mu b''$ and $C_b = b^2/(2\mu) + \mu b b''/2$, define

$$\begin{aligned} F_* &= B_0/\mu + \mu B_2 = 1052400, \\ F_1 &= B_1/\mu + \mu B_3 = 4901968750, \\ F_2 &= B_2/\mu + \mu B_4 = 4032010500000000, \\ K_1 &= B_0 B_1/\mu + \mu(B_1 B_2 + B_0 B_3)/2 = 69183187500, \\ K_2 &= (B_1^2 + B_0 B_2)/\mu + \mu(B_2^2 + 2B_1 B_3 + B_0 B_4)/2 = 48535884509765625. \end{aligned} \tag{94}$$

These bound $|F|, |F'|, |F''|, |C'_b|, |C''_b|$, respectively.

D.2 Exact intertwiner and pulse integrals

Let $G_* = V(S)(\psi_o g_0)$, where ψ_o is the full exact old wave, and V applies the above Weyl displacement in sector one and the identity in sector zero. Since $i\hbar v\gamma_S = H_Z\gamma$, product differentiation gives, up to an irrelevant overall sign in the forced error equation,

$$R/\hbar = \kappa(\gamma_S\psi_{o,S} + \gamma_{SS}\psi_o/2).$$

This identity holds separately in the orthogonal sectors and hence for any normalized qubit input. Its support is exactly the open writer pulse; the stationary displaced plateau contributes no source. Write $\delta = \Psi - G_*$ and $I_j = \int_0^T \|\mathbf{1}_{\text{pulse}} \partial_S^j \psi_o\| dt$. The old characteristic helper has clock centre $t_{\text{lab}} + 0.104$ and half-width 0.0010000001. Its earliest pulse contact is 0.0364499999 s, leaving $\Delta = 0.0001500001$ s. Throughout those late slices the initial offset lies in an endpoint strip of relative width $d = 0.1500002$. With $\sigma = 0.001$ and $m_\phi = 1.000001$, define the rational bounds

$$\beta = m_\phi^2 \frac{64}{35} (11/7)^8 \frac{d^9}{9}, \quad \eta^2 = m_\phi^2 \frac{1024}{35} (11/7)^8 \frac{d^7}{7\sigma^2}.$$

The inequality $\cos(\pi x/(2\sigma)) \leq \pi(\sigma - x)/(2\sigma)$, positive convolution and Jensen's inequality prove that β bounds the strip mass and η its amplitude first-derivative norm. The mollification radius is included in d . On these slices $\|u_S\| \leq N_1/a$, $N_1 = 34359738369$. The old-wave estimates, evaluated through elapsed time 0.0404, give $\|\psi_o - G_o\| < 3.1 \times 10^{-11}$ and $\|\partial_S(\psi_o - G_o)\| < 36000$. Consequently

$$\begin{aligned} I_0 &\leq T(3.1 \times 10^{-11}) + \Delta\sqrt{\beta} =: J_0, \\ I_1 &\leq T(36000) + \Delta(\eta + N_1\sqrt{\beta}/a) =: J_1, \\ I_2 &\leq T(1.315 \times 10^{24}) =: J_2. \end{aligned} \tag{95}$$

The last bound uses the full old Hessian, including its error. The preparation transfer and old right collar imply $\|\delta(0)\| \leq 3 \times 10^{-30} + 1.2 \times 10^{-37}$; the latter term bounds $(V - I)\psi_o g_0$ without discarding the tail. Duhamel therefore proves the uniform ceiling

$$\begin{aligned} D_* &:= 3 \times 10^{-30} + 1.2 \times 10^{-37} + \hat{\kappa}(g_1 J_1/a + g_2 J_0/(2a^2)), \\ \|\delta(t)\| &\leq D_* < 5.055628819933504 \times 10^{-22}. \end{aligned} \tag{96}$$

where $\hat{\kappa} = 1.055 \times 10^{-35} > \kappa$. All subsequent sharp bounds use the defining expression for D_* .

D.3 Closing the simultaneous pointer hierarchy

For this first, deliberately coarse closure take $h_- = 1.054 \times 10^{-34} < \hbar$, and use $\hat{\kappa}$ in positive numerators and h_- in denominators. The old physical force ceilings are $F_o = 8.00000000001 \times 10^{-6}$ and $F_{o,2} = 1$. The handoff bounds proved in the preparation appendix are

Quantity	Outward upper bound
$Q(\delta(0))$	6×10^{-24}
$\ \delta_S(0)\ $	0.073
$Q(\delta_S(0))$	300000
$\ \Psi_{SS}(0)\ $	2.5×10^{27}
$A_h = \ A\Psi(0)\ $	$5.049592328644200595 \times 10^{-24}$
$A_{2,h} = \ A^2\Psi(0)\ $	$1.228242965202911600 \times 10^{-17}$
$U_h = \ \partial_S(A\Psi)(0)\ $	276122.9698216115699

These already include the remote Weyl change. Differentiating the residual once gives the coefficients $1, 3/2, 1/2$. Define its integrated bounds

$$\begin{aligned} R_Q &= \hat{\kappa}(q_1 J_1/a + q_2 J_0/(2a^2)), \\ R_S &= \hat{\kappa}(g_1 J_2/a + 3g_2 J_1/(2a^2) + g_3 J_0/(2a^3)), \\ R_{SQ} &= \hat{\kappa}(q_1 J_2/a + 3q_2 J_1/(2a^2) + q_3 J_0/(2a^3)). \end{aligned}$$

Oscillator rotation and the differentiated error equation give the successive bounds

$$\begin{aligned} Q_d &= 6 \times 10^{-24} + F_* T D_*/T_0 + R_Q, \\ e_S &= 0.073 + F_o T D_*/h_- + T(F_1 Q_d + K_1 D_*)/(v T_0^2) + R_S, \\ P_1 &= 1.146 \times 10^{12} + g_1/a + e_S, \\ A_1 &= A_h + \hat{\kappa} \left[\frac{C_1}{a} (J_1 + g_1 J_0/a + T e_S) + \frac{C_2}{2a^2} (J_0 + T D_*) \right]. \end{aligned}$$

In particular $Q_d < 6.847 \times 10^{-16}$, $e_S < 1.482 \times 10^6$, and $A_1 < 6.192373945249072 \times 10^{-23}$. The last row follows from the exact identity

$$(i\partial_t - H/\hbar - 1/(\mu T_0))A\Psi = -\kappa\Pi_1(c'\Psi_S/a + c''\Psi/(2a^2)), \quad c = b + i\mu b'. \quad (97)$$

For the second power the exact equation is

$$(i\partial_t - H/\hbar - 2/(\mu T_0))A^2\Psi = -2\kappa\Pi_1(c'\partial_S(A\Psi)/a + c''A\Psi/(2a^2)) - \kappa\Pi_1(c')^2\Psi/a^2.$$

The last source is essential and comes from $[A, \partial_S] = \Pi_1 c'/a$.

For candidate constants (X, P, U, B) bounding $(Q(\delta_S), \|\Psi_{SS}\|, \|\partial_S(A\Psi)\|, \|A^2\Psi\|)$, put

$$Q_2 = (q_0^2 + 1)D_* + 2q_0 A_1 + B, \quad P_{1Q} = q_0(1.146 \times 10^{12}) + q_1/a + X.$$

A simultaneous constant supersolution consists of the following four strict inequalities:

$$\begin{aligned} X &> 300000 + F_* T e_S/T_0 + F_o T Q_d/h_- + T(K_1 Q_d + F_1 Q_2)/(v T_0^2) + R_{SQ}, \\ P &> 2.5 \times 10^{27} + T \left[\frac{2F_o P_1 + F_{o,2}}{h_-} + \frac{2(F_1 P_{1Q} + K_1 P_1)}{v T_0^2} + \frac{F_2(q_0 + Q_d) + K_2}{v^2 T_0^3} \right], \\ U &> U_h + T \left[\frac{F_o A_1}{h_-} + \frac{F_1(B + q_0 A_1) + K_1 A_1}{v T_0^2} + \hat{\kappa} \left(\frac{C_1 P}{a} + \frac{3C_2 P_1}{2a^2} + \frac{C_3}{2a^3} \right) \right], \\ B &> A_{2,h} + T\hat{\kappa}(2C_1 U/a + C_2 A_1/a^2 + C_1^2/a^2). \end{aligned} \quad (98)$$

These follow from the twice differentiated Schrödinger equation, (97), and (93). In particular the force on $A\Psi$ costs $B + q_0 A_1$, not merely A_1 . For $(X, P, U, B) = (10^{14}, 10^{40}, 2 \times 10^{11}, 2 \times 10^{-17})$ the respective right sides are strictly below

$$\begin{aligned} \mathcal{R}_X &< 4.011662441023 \times 10^{12}, \quad \mathcal{R}_P < 6.718077927095 \times 10^{39}, \\ \mathcal{R}_U &< 1.452034044217 \times 10^{11}, \quad \mathcal{R}_B < 1.809106526629 \times 10^{-17}. \end{aligned}$$

Every dependence on the other three unknowns has been retained. The positive integral inequalities and a simultaneous first-crossing argument prove the four bounds. The large 10^{40} bound is used only inside this hierarchy; the next subsection supplies the much sharper error Hessian needed for the clock current.

D.4 Old third clock derivative with complete coefficients

Only an ordinary third spatial clock derivative of the old wave is needed; no fourth radial energy graph is invoked. Sections B.3 and B.4 give the following outward bounds for the gauged old preparation error E_g :

$$\begin{aligned} P_2 &= 3.238008492819540878 \times 10^{-49}, \\ P_4 &= 1.075566867715738537 \times 10^{-76}, \\ n_2 &= 5.970369147764257548 \times 10^{-38}, \end{aligned}$$

where $\|p_S^j E_g\| \leq P_j$ and n_2 bounds the second left collar. For a cutoff equal to one through 0.10005, zero from 0.10015, and $g = 0.0001$, the localized interpolation identities give

$$L_1 = \sqrt{n_2 P_2} + 10\hbar_+ n_2 / g, \quad L_2 = \sqrt{n_2}(\sqrt{P_4} + 18\hbar_+ \sqrt{P_2}/g), \quad \hbar_+ = 1.055 \times 10^{-34}.$$

This cutoff covers every preparation-phase derivative, whose support ends at 0.10001, and lies inside the full-one region of the old second collar, which ends at 0.100249999873. For the preparation-rate jets use

$$\begin{aligned} (a_0, a_1, a_2) &= (140, 43600, 70141750), \\ (a_3, a_4) &= (1387431060000, 986084475600000), \\ \mu_r^+ &= 1.425 \times 10^{-12}, \quad F_p = 0.080802, \end{aligned}$$

and put $r_j = \mu_r^+ F_p a_j / 2$ for $j = 1, 2, 3$. Expanding $p_S^3(e^{i\theta} E_g)$, with $\hbar|\partial_S^j \theta| \leq r_j$, proves the handoff bound

$$\begin{aligned} E_{3,h} := h_-^{-3} &\left[\sqrt{P_2 P_4} + 3r_1 L_2 + 3(r_1^2 + \hbar_+ r_2) L_1 \right. \\ &\left. + (r_1^3 + 3\hbar_+ r_1 r_2 + \hbar_+^2 r_3) n_2 \right] + 46933757821.201. \end{aligned} \quad (99)$$

The last term bounds the original packet's third derivative. The helper's radial state is S independent at handoff and its preparation phase is zero there. Thus $\|\psi_{o,SSS}(0)\| \leq E_{3,h} < 2.250395507380083 \times 10^{42}$.

Here are the old-wave input ceilings used on the enlarged horizon $T_o = 0.0404$, computed with the formulas of Sections C.6 and C.5:

Input	Outward upper bound
$D_o = \sup \ \psi_o - G_o\ $	$3.034654348976846863 \times 10^{-11}$
$e_{1,o} = \sup \ \partial_S(\psi_o - G_o)\ $	35444.22634847288176
$e_{2,o} = \sup \ \partial_S^2(\psi_o - G_o)\ $	$1.047307159416726327 \times 10^{21}$
$G_1 = \sup \ G_{o,S}\ $	1145324612300.000003
$G_2 = \sup \ G_{o,SS}\ $	$1.313080270080687692 \times 10^{24}$
$F_{c,1}$	$4.388993584524007892 \times 10^{-6}$
$F_{c,2}$	0.08054047024909008647

For clarity the enlargement keeps the same ramp interval 0.0033 and uses static duration $T_o - 0.0033 = 0.0371 < 0.0377$. The helper age remains below $4/3 < \pi/2$. Its packet ceilings are

$$p_1 = 1.000001\sqrt{16/7}\pi/(2\sigma), \quad p_2 = 1.000001\sqrt{512/35}(\pi/(2\sigma))^2,$$

so that, with $N_2 = 1.001(2^{35} + 1)^2$,

$$G_1 \leq \sqrt{p_1^2 + (N_1/a)^2}, \quad G_2 \leq p_2 + 2p_1 N_1/a + N_2/a^2 + 25000N_1/a.$$

The three error entries are the sums of the transported preparation, short ramp, static, common-to-clipped and clipped-to-static component bounds evaluated at these durations; no new propagation assumption is introduced by extending the time argument.

The remote third force, including the preparation phase-square term, has the complete upper expression

$$F_{c,3} = \mu_r^+ a_4 F_p / 2 + \mu_r^+ (3a_1 a_2 + a_0 a_3) (0.201)^2 + \frac{(\mu_r^+)^2}{4M} (3a_2 a_3 + a_1 a_4) F_p^2 + 2.393692459677192 \times 10^{-27}. \quad (100)$$

The last summand is the third quiet-force derivative obtained from the finite product-rule calculation in Section C.4. Thus $F_{c,3} < 68.48117358490407$. Put

$$W_0 = \hbar_+ (2^{35} + 1) / T_0, \quad F_{a,1} = 25000 W_0, \\ F_{a,2} = \frac{2520 W_0}{64(0.0001)^2}, \quad F_{a,3} = \frac{(7560/16 + 5040/64) W_0}{(0.0001)^3}.$$

The remote common-minus-clipped force is supported on $S \leq 0.10001$. The helper and all its S derivatives vanish there for the entire prefix, since its left support starts at 0.1009999999 and moves right. The third differentiated unitary equation therefore gives

$$\frac{d}{dt} \|\psi_{o,SSS}\| \leq R_3 := \hbar^{-1} [3F_{c,1} e_{2,o} + 3F_{c,2} e_{1,o} + F_{c,3} D_o + 3F_{a,1} (G_2 + e_{2,o}) + 3F_{a,2} (G_1 + e_{1,o}) + F_{a,3}].$$

In particular the large helper Hessian is not charged on the remote force support. Integrating the linear envelope once more yields

$$J_3 := T E_{3,h} + \frac{1}{2} T^2 R_3 < 9.746858547719537661 \times 10^{46}, \quad I_3 \leq J_3. \quad (101)$$

D.5 Localized first derivative and refined mixed moment

Take a fixed decreasing Lipschitz cutoff $\chi = 1$ on $S \leq 0.126$, $\chi = 0$ on $S \geq 0.136$, with $|\chi'| \leq 1/g_L$, $g_L = 0.01$. Every old clock-force derivative is supported in its full-one region ($S \leq 0.106$), while the residual starts at $S = 0.14145$. The positive constant drift favors departure from this left region. The forced continuity identity, with no source on the support of χ , gives

$$n_L(t) := \|\chi \delta(t)\| \leq n_{L,h} + \frac{\kappa}{g_L} \int_0^t \|\delta_S(u)\| du, \quad n_{L,h} = 7.901146211666425714 \times 10^{-36}. \quad (102)$$

Since $AG_* = 0$, improve the oscillator bound to $\tilde{Q} = A_1 + q_0 D_*$. Let $\tilde{R}_S = (\kappa/\hat{\kappa}) R_S$ and similarly $\tilde{R}_{SQ} = (\kappa/\hat{\kappa}) R_{SQ}$. For the candidate $e = 0.1$, define

$$n_* = n_{L,h} + \kappa T e / g_L, \\ \mathcal{E} = 0.073 + F_o T n_* / \hbar + T (F_1 \tilde{Q} + K_1 D_*) / (v T_0^2) + \tilde{R}_S. \quad (103)$$

The positive two-variable comparison closes because $\mathcal{E} < 0.07327295767972046 < e$. Consequently $\|\delta_S\| \leq 0.1$ and $n_L \leq n_* < 1.197180 \times 10^{-35}$ throughout the prefix. The improvement retains the old force but multiplies it by its actual localized error norm.

On that force support $A = \partial_Z + Z$ and $AG_* = 0$. Applying the oscillator identity on each S slice, without differentiating a sharp indicator, gives $Q(\mathbf{1}_{\text{old force}} \delta) \leq \sqrt{A_1^2 + n_*^2}$. Set $B = 2 \times 10^{-17}$ in Q_2 above. The differentiated oscillator rotation equation now proves the much sharper uniform bound

$$X_* := 300000 + F_* T e / T_0 + F_o T \sqrt{A_1^2 + n_*^2} / \hbar + T (K_1 \tilde{Q} + F_1 Q_2) / (v T_0^2) + \tilde{R}_{SQ}, \quad Q(\delta_S) \leq X_* < 616735. \quad (104)$$

This is the mixed moment required in the clock estimate; the coarse 10^{14} bound from (98) is not substituted here.

D.6 Second clock derivative and the two absolute currents

Two residual derivatives give exactly

$$R_{SS}/\hbar = \kappa \left(\gamma_S \psi_{o,SSS} + \frac{5}{2} \gamma_{SS} \psi_{o,SS} + 2 \gamma_{SSS} \psi_{o,S} + \frac{1}{2} \gamma_{SSSS} \psi_o \right).$$

Its integrated norm is bounded by the explicit expression

$$R_{SS}^* = \kappa (g_1 J_3/a + 5g_2 J_2/(2a^2) + 2g_3 J_1/a^3 + g_4 J_0/(2a^4)).$$

The refined new-minus-old handoff Hessian from Section B.6, plus both derivatives of the remote Weyl change, gives the following expression. Here $n_{2,R} = 5.970369147764257548 \times 10^{-38}$ is the *right* second-collar ceiling from Section B.5. It equals the numerical left-collar ceiling n_2 because reflecting the moving-cutoff construction gives the same positive momentum inequality: the right boundary translates at v plus the drift ceiling and the left boundary at v minus that ceiling. The right collar is one throughout the writer support; its handoff full-one edge is below 0.103750000127, while $S_{\text{on}} = 0.14145$. Thus the required right-tail amplitude is bounded independently of the left-tail localization used in (99). Explicitly,

$$\begin{aligned} H_h = & 1.828509471006939478 \times 10^{19} + \frac{2(2.534074040624599816 \times 10^{-57})}{\hbar^2} \\ & + \frac{2g_1(1.390399439216243060 \times 10^{-43})}{a\hbar} + \frac{g_2 n_{2,R}}{a^2}. \end{aligned} \quad (105)$$

For $H(t) = \|\delta_{SS}(t)\|$, integration by parts with χ gives

$$\|\mathbf{1}_{\text{old force}} \delta_S\| \leq \sqrt{n_* H(t)} + 2D_*/g_L.$$

The cutoff cost is retained. Twice differentiating the error equation therefore yields the differential comparison $H' \leq 2\alpha\sqrt{H} + f(t)$, where $\alpha = F_o\sqrt{n_*}/\hbar$, $f \geq 0$, and $H(0) + \int_0^T f(t) dt \leq \mathcal{R}$ with

$$\begin{aligned} \mathcal{R} = & H_h + R_{SS}^* + \frac{4F_o T D_*}{\hbar g_L} + \frac{F_{o,2} T n_*}{\hbar} \\ & + \frac{2T(F_1 X_* + K_1 e)}{v T_0^2} + \frac{T(F_2 \tilde{Q} + K_2 D_*)}{v^2 T_0^3}. \end{aligned} \quad (106)$$

Indeed, writing $F(t) = H_h + \int_0^t f(u) du$, the function $[\alpha t + \sqrt{F(t)}]^2$ has derivative at least $2\alpha[\alpha t + \sqrt{F(t)}] + f(t)$ and dominates the initial value. Scalar differential comparison therefore proves

$$H(t) \leq H_* := (F_o T \sqrt{n_*}/\hbar + \sqrt{\mathcal{R}})^2 < 1.117214708915152634 \times 10^{21}. \quad (107)$$

Regularization at zero justifies this comparison when a norm vanishes. Smooth-domain approximation and these uniform finite estimates justify the oscillator and weak clock commutations without an oscillator cutoff.

On $S < S_{\text{on}}$, $G_* = \psi_o g_0$ and $A = \partial_Z + Z$. The oscillator form and graph identities imply, after restriction to that region,

$$\|\delta_Z\| \leq e_Z := \sqrt{A_1^2 + D_*^2}, \quad \|\delta_{ZZ}\| \leq e_{ZZ} := \sqrt{B^2 + 4A_1^2 + 3D_*^2}.$$

For example $\|p_Z^2 f\|^2 \leq \|(p_Z^2 + Z^2)f\|^2 + 2\|f\|^2$ and $\|(\mathcal{B}^\dagger \mathcal{B} + 1)f\|^2 = \|\mathcal{B}^2 f\|^2 + 4\|\mathcal{B}f\|^2 + \|f\|^2$ with $\mathcal{B} = \partial_Z + Z$ prove the second bound. The conditional-rank stability functional then gives

$$I_Z \leq \frac{T}{\mu T_0} \left(2\sqrt{2}e_Z + 2e_Z^2 + \frac{23\sqrt{3}}{2}D_* + e_{ZZ} \right) < 2.574814383253091344 \times 10^{-15}. \quad (108)$$

Here the comparison's radial direction is independent of Z ; $\|g'_0\| = 1/\sqrt{2}$, $\|g''_0\| = \sqrt{3}/2$ and $\|g_0'^2/g_0\| = \sqrt{3}/2$ supply the displayed coefficients. For the clock functional use the old characteristic helper instead. Expanding its positive stability majorant with the additional error δ gives

$$\begin{aligned}\Delta I_S &\leq \kappa T \left[4(G_1 + e_{1,o})e + 2e^2 + 14D_*Q_G + 9D_*G_2 + H_* \right], \\ Q_G &= 1.000001 \frac{4\pi^2}{\sigma^2} \sqrt{3/35} + (N_1/a)^2, \\ \Delta I_S &< 4.547786946512146880 \times 10^{-16}.\end{aligned}\tag{109}$$

This is an increment of an absolute-current upper bound, not a subtraction of signed currents. All hidden-coordinate absolute values precede integration. Both estimates apply on the common whole-prefix good-clock event; its complement is charged separately in the event composition. The current components must be added before optimizing the single rank collar.

Reproducible outward arithmetic. Equations (92)–(109), together with the outward input tables, form a finite coefficient specification. Every displayed decimal input is an exact rational ceiling. One fully rational evaluation uses a bracket for π of width 2×10^{-69} about $3.141592653589793238462643383279502884197169399375105820974944592307817$ and $\hbar = 6.62607015 \times 10^{-34}/(2\pi)$. For a nonnegative rational $x = n/d$, replace each square root by

$$\text{sqrt}_+(x) = \frac{1 + \left\lfloor \sqrt{\lfloor 10^{200}n/d \rfloor} \right\rfloor}{10^{100}}.$$

All these coefficient expressions have positive inputs; use upper numerators and lower denominators. This procedure gives $X_* < 616734.119066647397144$, $H_* < 1.117214708915152634 \times 10^{21}$, and the strict current bounds (108)–(109). It specifies the calculations inside the manuscript and requires no external coefficient file.

E Copying and retention for the exact loaded wave

All estimates in this appendix concern the same normalized enlarged wave Ψ and its original actual law. Write $C = 270000/67499$, $m = 2^{52}$, $h = 1/128$, $M = 10$ kg, $v = 1$ m/s, $T_0 = 0.03$ s, $\ell_0 = 10^{-4}$ m, and $\mu = \mu_Z = 0.01$. Radial and pointer coordinates are dimensionless; S and the elapsed time t below have physical units. The origin $t = 0$ is the handoff at laboratory time -0.002 s. The transferred entrance estimates are those proved in Section B.5; no law or wave is assigned anew there.

E.1 Gaussian derivatives and the complete clock residual

Put $b(s) = 24B_9(s/w)$, $w = 0.003$, with the constant extensions specified in the model. Direct differentiation gives the following rational ceilings $b_j \geq \|b^{(j)}\|_\infty$:

$$(b_0, b_1, b_2, b_3, b_4) = \left(24, \frac{39375}{2}, 105000000, 490000000000, 403200000000000000 \right). \tag{110}$$

Indeed $B'_9 = 630x^4(1-x)^4$; the first three derivative bounds follow from $x(1-x) \leq 1/4$ and $|1-2x| \leq 1$, and the coefficient sum of $B_9^{(4)}$ is 1360800. These bounds apply on both constant extensions because B_9 has four matching endpoint derivatives.

For the exact sector-one pointer put $y = Z - b$ and

$$g_b = \pi^{-1/4} e^{-y^2/2} e^{i\mu b'(Z-b/2)}, \quad c = b' + i\mu b'', \quad d = \frac{i\mu}{2}(bb'' - b'^2).$$

Then $g'_b = (cy + d)g_b$ and

$$g''_b = ((cy + d)^2 + c'y - cb' + d')g_b, \quad d' = \frac{i\mu}{2}(bb''' - b'b'').$$

In particular the scalar phase is retained. Define

$$\begin{aligned} c_0 &= b_1 + \mu b_2, & c_1 &= b_2 + \mu b_3, \\ d_0 &= \mu(b_0 b_2 + b_1^2)/2, & d_1 &= \mu(b_0 b_3 + b_1 b_2)/2, \\ g_1 &= d_0 + c_0/\sqrt{2}, & g_2 &= d_0^2 + c_0 b_1 + d_1 + (2d_0 c_0 + c_1)/\sqrt{2} + \sqrt{3} c_0^2/2. \end{aligned} \quad (111)$$

The Gaussian moments $\|y^j g_b\|^2 = (2j - 1)!!/2^j$ prove $\|g_b^{(j)}\| \leq g_j$ for $j = 1, 2$. The sector-zero pointer is constant, so the same ceilings apply to both sectors, uniformly in the normalized qubit input.

The characteristic comparison through readout is

$$\begin{aligned} G(t, S, r, Z) &= \phi(S - 0.102 - vt) \sum_{i=0}^1 \alpha_i u_i(\sigma, r) g_i(s, Z), \\ \sigma &= \frac{S/v - 0.104 \text{ s}}{T_0}, & s &= \frac{S - 0.14145 \text{ m}}{vT_0}. \end{aligned} \quad (112)$$

Here u_i is the exact smooth-activation radial family, continued backwards by the quiet Hamiltonian at negative ages. On the entire compact support of ϕ , the preparation phase potential has ended. Substitution into the boosted Schrödinger equation leaves precisely the residual $\pm \hbar^2 G_{SS}/(2M)$. The sign is immaterial for the norm estimates, but both pointer and activation derivatives in G_{SS} must be included.

Let $N_0 = 1$, $N_1 = V + 1$, $N_2 = 1001(V + 1)^2/1000$ and $N_3 = 1001(V + 1)^3/1000$, where $V = 2^{35}$; these are the energy-graph bounds proved in Section C.3. Set

$$\begin{aligned} F_0 &= \frac{2010^2}{2m 100^4} + \frac{3}{2m 100^2}, & F_1 &= F_0/2 + \frac{2010}{m 100^4}, & W_1 &= 2^{46}, \\ K_j &= \sqrt{2m(N_j N_{j+1} + F_0 N_j^2)}, & j &= 0, 1, 2. \end{aligned}$$

The form inequality $H_\gamma \geq p_r^2/(2m) - F_0$, applied to $H_\gamma^j u_i$, proves $\|\partial_r H_\gamma^j u_i\| \leq K_j$. Only the third Hamiltonian graph is needed. Since

$$\partial_\sigma^2 u_i = -H_\gamma^2 u_i - i\gamma'_\sigma(W - F)u_i, \quad 0 \leq \gamma'_\sigma \leq 750,$$

the product-rule bounds, including the gate derivative, are

$$\begin{aligned} U_1 &= N_1 + g_1, \\ U_2 &= N_2 + 750(V + F_0) + 2N_1 g_1 + g_2, \\ L_0 &= K_0, \\ L_1 &= K_1 + g_1 K_0, \\ L_2 &= K_2 + 750\{W_1 + F_1 + (V + F_0)K_0\} + 2g_1 K_1 + g_2 K_0. \end{aligned}$$

For the normalized mollified clock packet, convolution contraction and its normalization factor at most 1.000001 give

$$p_1 = 1.000001 \sqrt{16/7} \frac{\pi}{0.002}, \quad p_2 = 1.000001 \sqrt{512/35} \left(\frac{\pi}{0.002} \right)^2$$

as ceilings on $\|\phi'\|$ and $\|\phi''\|$. Therefore

$$G_2 = p_2 + \frac{2p_1 U_1}{vT_0} + \frac{U_2}{(vT_0)^2}, \quad G_{2r} = p_2 L_0 + \frac{2p_1 L_1}{vT_0} + \frac{L_2}{(vT_0)^2} \quad (113)$$

bound $\|G_{SS}\|$ and $\|\partial_r G_{SS}\|$.

Write $E = \Psi - G$. Its entrance norm e_h is D_h from Section B.5 plus the backward-quiet discrepancy. For that discrepancy, if $e = (e^{-i\sigma H_F} - 1)\chi$ and $|\sigma| \leq 1/8$, then

$$\|e\| \leq \|H_F \chi\|/8, \quad \|H_F e\| \leq \|H_F^2 \chi\|/8, \quad \|\partial_r e\|^2 \leq 2m\|e\|(\|H_F e\| + F_0\|e\|).$$

The quiet tail estimates in Section C.4 make its physical momentum correction less than 10^{-100} kg, m/s. Let $P_{r,h}$ be the old ordinary radial error momentum, the loaded-minus-old momentum from Section B.5, and this quiet correction added together.

For completeness the global radial force used in propagating E is explicit. Put $m_{\text{phys}} = m\hbar T_0/\ell_0^2$, $\omega = 1/(mT_0)$, $R = 0.201$ m, $a = 140$ s $^{-1}$, $a' = 43600$ s $^{-2}$ and $F_{\text{max}} = 2R^2$. Define

$$\begin{aligned} Q_0 &= m_{\text{phys}}\omega^2 R^2/2 + 3\hbar\omega/2, \\ Q_r &= m_{\text{phys}}\omega^2 R + 5000Q_0, & W_r &= \frac{\hbar(m\hbar/2 + 5V)}{T_0\ell_0}, \\ V_{p,r} &= m_{\text{phys}}(a' + a^2)R + \frac{m_{\text{phys}}^2(a')^2 F_{\text{max}}R}{2M}, & V_r &= V_{p,r} + Q_r + W_r. \end{aligned}$$

These follow by differentiating the displayed preparation, weak-trap and capped detector potentials; all remote preparation terms remain. The writer has zero radial derivative. Unitary Duhamel, followed by the ordinary radial momentum commutator, consequently gives

$$\begin{aligned} e(t) &\leq e_h + r_0 t, & r_0 &= \hbar G_2/(2M), \\ P_r(t) &\leq P_{r,h} + r_r t + V_r(e_h t + r_0 t^2/2), & r_r &= \hbar^2 G_{2r}/(2M\ell_0). \end{aligned} \quad (114)$$

The odd extension at the Dirichlet origin has no boundary term. Approximation on the common smooth core and these uniform bounds justify the differentiated evolution without a fourth radial graph. At $T_c = 0.0404$ s, denote the right sides by e_* and $P_{r,*}$ and put $p_* = \ell_0 P_{r,*}/(\hbar m h)$. Evaluation gives

$$e_* < 3.034680 \times 10^{-11}, \quad P_{r,*} < 2.208802 \times 10^{-20} \text{ kg, m/s}, \quad p_* < 0.000595293.$$

E.2 A closed clock and annihilator estimate through the hold

The exact boosted generator contains $p_s + \epsilon p_s^2/2$, $\epsilon = \hbar/(Mv^2 T_0)$. Define $\mathcal{A} = \partial_Z + Z - \Pi_1(b + i\mu b')$. Since the radial potentials commute with this operator, its complete commutator is

$$(i\partial_r - H - 1/\mu)\mathcal{A}\Psi = -\epsilon\Pi_1[(b' + i\mu b'')\partial_s + (b'' + i\mu b''')/2]\Psi. \quad (115)$$

For example $[\mathcal{A}, H_{\text{ptr}}] = (\mathcal{A} + \Pi_1 c)/\mu - \Pi_1 F$ and $[\mathcal{A}, p_s] = -i\Pi_1 c'$, where $c = b + i\mu b'$; $c/\mu - F - ic' = 0$ cancels the prescribed driving terms. The clock Laplacian supplies both terms on the right of (115).

Let $P(t) = \|-i\hbar\partial_S\Psi(t)\|$ be the boosted clock momentum, $n(t)$ the moving left-collar norm of width $g = 0.0005$ m, and $R_Z(t) = (\|Z\Psi\|^2 + \|p_Z\Psi\|^2)^{1/2}$. The entrance estimates allow the conservative values

$$P(0) \leq 2 \times 10^{-27}, \quad n(0) \leq 2 \times 10^{-26}, \quad R_Z(0) \leq 2,$$

in SI units where applicable; the much sharper actual entrance $A_h = \|\mathcal{A}\Psi(0)\|$ is retained. Set

$$F_* = b_0/\mu + \mu b_2, \quad F'_* = b_1/\mu + \mu b_3, \quad C'_* = b_0 b_1/\mu + \mu(b_1 b_2 + b_0 b_3)/2.$$

The Heisenberg equations for (Z, p_Z) are a rotation with bounded forcing $\Pi_1 F$; its variation-of-constants formula gives $R_Z(t) \leq 2 + F_* t/T_0$, without exponentiating the oscillator frequency. The translating collar and clock commutator give

$$n' \leq \frac{P}{Mg}, \quad P' \leq F_p n + F_a + \frac{\hbar}{vT_0^2}\{F'_* R_Z + C'_*\}, \quad F_p = 8 \times 10^{-6}, \quad F_a = 10^{-17}. \quad (116)$$

Here F_p and F_a have units of force. Explicit ceilings from the original potentials, with $a'' = 70141750 \text{ s}^{-3}$, are

$$\begin{aligned} F_p^{\text{calc}} &= \frac{m_{\text{phys}} a'' F_{\text{max}}}{2v} + \frac{m_{\text{phys}} a a' R^2}{v} + \frac{m_{\text{phys}}^2 a' a'' F_{\text{max}}^2}{4Mv^3} < F_p, \\ Q_S &= 2m_{\text{phys}} \omega^2 a R^2 + 3\hbar \omega a + 5000 Q_0, \\ F_a^{\text{calc}} &= Q_S + 25000(\hbar V/T_0 + Q_0) < F_a. \end{aligned}$$

The C'_* term retains the clock force of the scalar compensation. All inequalities apply to the full wave, including remote clock tails. The oscillator estimates extend from finite-energy approximants by the form-domain bounds just obtained.

For $q = 0.0002 \text{ kg, m/s}$ and $k = 0.04 \text{ s}^{-1}$, $Y = P + qn$ satisfies $Y' \leq kY + f_0 + f_1 t$, where

$$Y_0 = 2 \times 10^{-27} + q(2 \times 10^{-26}), \quad f_0 = F_a + \frac{\hbar(F'_* 2 + C'_*)}{vT_0^2}, \quad f_1 = \frac{\hbar F'_* F_*}{vT_0^3}.$$

Indeed $q/(Mg) = F_p/q = k$. With $T = 0.092 \text{ s}$ and $E_T = (1 - kT)^{-1} \geq e^{kT}$, positive integration proves

$$\begin{aligned} P(t) &\leq E_T(Y_0 + f_0 T + f_1 T^2/2), \\ \int_0^T P(t) dt &\leq \mathcal{P} := E_T(Y_0 T + f_0 T^2/2 + f_1 T^3/6), \\ A_* &:= A_h + \frac{c_0 \mathcal{P}}{MvT_0} + \frac{\hbar c_1 T}{2Mv^2 T_0^2} \geq \sup_{0 \leq t \leq T} \|\mathcal{A}\Psi(t)\|. \end{aligned} \tag{117}$$

The last line is unitary Duhamel applied to (115). The resulting ceilings are

$$\mathcal{P} < 2.667201 \times 10^{-18} \text{ kg, m}, \quad A_* < 9.510237 \times 10^{-12}.$$

Equivariance and Cauchy–Schwarz imply $\mathbb{E}_{|\Psi|^2} \int_0^T |\dot{S} - v| dt \leq \mathcal{P}/M$. Thus actual-law domination and Markov’s inequality bound the failure of the entire-path deviation $\int |\dot{S} - v| \leq 10^{-5} \text{ m}$ by $C\mathcal{P}/(10^{-5}M) < 1.067 \times 10^{-13}$. The handoff core $|S_h - 0.102| \leq 0.000839 \text{ m}$ has failure at most

$$e_{\text{core}} = C(\sqrt{\beta_{\text{core}}} + D_h)^2, \quad \beta_{\text{core}} = (1.000001)^2 \frac{128}{35} \left(\frac{\pi}{2}\right)^8 \frac{(0.1610001)^9}{9}. \tag{118}$$

To see this, normalize the unsmoothed $\cos^4$ packet on its 0.001 m half-width. Its squared density has coefficient $64/35$ after scaling to unit half-width; the two edges give the factor $128/35$. On either edge use $\sin x \leq x$, integrate the eighth power, and enlarge the edge width by the smoothing radius 10^{-10} m . Positive convolution and Jensen preserve this two-sided bound, with the displayed normalization factor. The D_h triangle then uses the exact handoff wave, rather than replacing its marginal.

On the intersection of these two good events, the first possible crossing of S_{on} is after 0.036601 s , and the last possible completion of the pulse is before 0.038389 s . More directly, the pathwise bound $|S_t - (0.104 + vt_{\text{lab}})| \leq 0.000849 \text{ m}$ implies $S_t \geq 0.14154 \text{ m}$ for every $t_{\text{lab}} \in [0.0384, 0.09]$. This proves completion throughout the whole hold, including the exclusion of later returns. The same two exceptional events are charged only once below.

E.3 The actual earlier label and a positive capture event

Let $H_r(r) \in [0, 1]$ be the soft radial classifier, agreeing with the sharp label $\ell(r)$ outside its bands B and having $|H'_r| \leq 10/h$. Put $I_0 = [-8, 8]$, $I_1 = [16, 32]$ and

$$M_r(r, Z) = (1 - H_r(r))\mathbf{1}_{I_0^c}(Z) + H_r(r)\mathbf{1}_{I_1^c}(Z).$$

For each absolutely continuous exact trajectory, with witness and readout at laboratory times $t_a = 0.0366$, $t_c = 0.0384$,

$$\mathbf{1}_{\{Z_c \notin I_{\ell(r_a)}\}} \leq \mathbf{1}_B(r_a) + \text{Var}_{[t_a, t_c]} H_r(r_t) + M_r(r_c, Z_c). \quad (119)$$

Outside B , $H_r(r_a)$ is the binary earlier label; replacing this coefficient by $H_r(r_c)$ changes the mismatch by at most its total variation. This proves the inequality without assigning a sampled internal sector.

Throughout this copy interval the entire comparison packet has conditional age in $[71/60, 4/3]$. Use the B_9 -taper lens expressions D_9, Q_9 of (86), evaluated at $4/3$, and denote these by D, Q ; set

$$q_r = \frac{\sqrt{2mD(Q + F_0D)}}{mh}. \quad (120)$$

They give $D < 0.007640094$, $Q < 262597634.037$ and $q_r < 0.003820673$. The contraction scale is at most $\cos(\sigma - 1/300)$. Since $\sigma - 1/300 \geq 59/50$ and $\cos(59/50) < 2/5$, the compact comparison and its radial derivative vanish on the bands $[h/5, 3h/10]$ and their translates. Hence $\|\mathbf{1}_B G\| \leq D$ and $\|\mathbf{1}_B \partial_r G\| \leq mhq_r$. Expanding the exact current of $G + E$, taking its modulus before integrating any hidden variable, gives

$$\int_B |J_r[\Psi]| \leq h\{Dq_r + (D + e_*)p_* + e_*q_r\}. \quad (121)$$

For example the four terms before division by m are $D(mhq_r)$, $D\|\partial_r E\|$, $e_*(mhq_r)$ and $e_*\|\partial_r E\|$.

On the completed-clock region, let P_g project in each internal sector onto the normalized real Gaussian centered at 0 or 24. The oscillator ladder spectrum proves the fibre inequality

$$\mathcal{A}^* \mathcal{A} \geq 2(I - P_g). \quad (122)$$

Both operators commute with completed-clock localization. Thus the localized excited component has norm at most $A_*/\sqrt{2}$. The ground-state wrong-inner probability is $q_I = \text{erfc}(8)$, and contraction and the norm triangle give a pointer cost at most $C(\sqrt{q_I} + A_*/\sqrt{2})^2$. To combine it with the radial mismatch, use the pointwise inequality

$$M_r(r, Z) \leq |H_r(r) - i| + \mathbf{1}_{I_i^c}(Z), \quad i = 0, 1,$$

and sum against the exact orthogonal component densities. The square-root multiplier of the first term kills the compact sector- i lens comparison, so its total contribution is at most $C(D + e_*)^2$. Together with the initial band and (121), this proves the copy cost, apart from the already identified exceptional events,

$$C \left[2(D + e_*)^2 + \frac{3}{5} \{Dq_r + (D + e_*)p_* + e_*q_r\} \right] + C(\sqrt{q_I} + A_*/\sqrt{2})^2. \quad (123)$$

Here $10(t_c - t_a)/T_0 = 3/5$. Each endpoint and variation refers to the same exact trajectory and its own earlier radial label.

E.4 Absolute current over the entire holding interval

At each full configuration, including S, r, Z , the exact identity is

$$J_Z[\Psi] = \frac{1}{\mu} \Im(\Psi^\dagger \mathcal{A} \Psi) + b' \Psi^\dagger \Pi_1 \Psi. \quad (124)$$

The second term has not been discarded: it vanishes on every visited point of a good clock path because the pulse is completed there. For $R_0 = [-10, 10]$, $R_1 = [14, 34]$, a path starting in I_i and leaving R_i crosses one of the four width-two bands $[-10, -8]$, $[8, 10]$, $[14, 16]$, $[32, 34]$. Linear

cutoffs of slope $1/2$ on these disjoint bands require unit variation for such an exit. Restricted equivariance, actual-law domination and then Cauchy–Schwarz therefore give

$$\begin{aligned}\mathbb{P}(\text{a hold exit, correct capture, good clock}) &\leq \frac{C}{2\mu} \int_{1.28}^3 \int |\Psi| |\mathcal{A}\Psi| \, dq \, d\tau \\ &\leq \frac{C}{2\mu} \frac{43}{25} A_* < 3.271570 \times 10^{-9}.\end{aligned}\quad (125)$$

All hidden-coordinate moduli are taken before integration. The variation counts recrossings and every time of the interval; an endpoint estimate is not being substituted for a path estimate. The global annihilator norm bounds the restricted current, so no unfinished part of the comparison wave has been removed.

E.5 The loaded radial exit and the evaluated sum

For the radial barrier increasing from zero at 0.099 m to one at 0.100 m, the slope is 1000 m^{-1} . Let

$$d_j^2 = \frac{(2j+1)!!}{2^j 100^{2j}}, \quad B_j = (d_j^2 + 2hd_j d_{j+1})^{1/2}, \quad T_G = \sqrt{22/7^{49}}, \quad \beta = \frac{B_2}{2m} + \frac{hB_1}{2} + VT_G + F_0 B_0.$$

Here d_j are the half-line Gaussian derivative norms, equivalently the norms of its unitary odd extension. For an integrable absolutely continuous f the grid sum obeys

$$\sup_{0 \leq x < h} \left| h \sum_{k \in \mathbb{Z}} f(x + kh) - \int_{\mathbb{R}} f \right| \leq h \int_{\mathbb{R}} |f'|.$$

This follows by comparing the value at the chosen point of each cell with its integral and then summing the fundamental theorem of calculus bounds. With $f = |\chi_{100}^{(j)}|^2$, $\int |f'| \leq 2d_j d_{j+1}$ proves the folded bound B_j^2 . The periodic reference has mass $h/2$ per cell and circle norm one. Its kinetic energy starts at zero and increases by at most V under the nondecreasing gate, proving $\|\partial_r u_{\text{ref},i}^\gamma\|_{L^2(0,2)} \leq mh/2$. Consequently the norms of $\sqrt{2}\chi_{100}^{(j)} u_{\text{ref},i}^\gamma$ and $\sqrt{2}\chi'_{100} \partial_r u_{\text{ref},i}^\gamma$ are bounded by B_j and $B_1 mh/2$, respectively. For the auxiliary $\sqrt{2}\chi_{100} u_{\text{ref},i}^\gamma$, the product-rule residual consists respectively of the second Gaussian derivative, the cross derivative, the outer cutoff tail and the weak trap, and their sum is exactly β above. For the decreasing Gaussian density beyond 990, periodic cell masses give a folded tail at most $\text{Tail}(990) + h\rho(990)$. Writing $z = 9.9$, integration by parts gives

$$\int_z^\infty y^2 e^{-y^2} \, dy \leq \left(\frac{z}{2} + \frac{1}{4z} \right) e^{-z^2}.$$

The radial density is $4y^2 e^{-y^2} / \sqrt{\pi}$ in this scaled coordinate. Thus $\text{Tail}(990) < 21/7^{49}$ and $\rho(990) < 4/7^{49}$ in detector coordinates, by $e^2 > 7$ and $\sqrt{\pi} > 1$. Since $h < 1/4$, their folded sum is below T_G^2 . Unitary integration for age at most $4/3$ yields the full-wave bound

$$\|\mathbf{1}_{r_{\text{phys}} \geq 0.099} \Psi\| \leq T_r := \frac{4}{3}\beta + T_G + e_*.$$

Negative quiet ages obey the same larger bound by the quiet estimate used in (114); pointer tensor factors have norm one.

The same loaded clock estimate gives a uniform left-collar bound $n_* = n_h + \mathcal{P}/(Mg)$. Thus the exact radial momentum starts below $P_{r,0} = \hbar\sqrt{3/2}/0.01 + P_{r,h}$ and grows at most at rate $F_r = W_r + Q_r + V_{p,r} n_*$. There is no writer radial force. The initial outer tail plus the absolute soft-barrier current prove

$$e_{\text{exit}} = C \left[(\sqrt{21/7^{49}} + D_h)^2 + \frac{1000T_r}{m_{\text{phys}}} (P_{r,0}T_c + F_r T_c^2/2) \right] < 9.127218 \times 10^{-8}. \quad (126)$$

This bounds initial outside configurations as well as every later exit through readout under the loaded wave.

For a directly evaluable decomposition, define

$$\begin{aligned}
e_{\text{base}} &= C\{2D^2 + (3/5)Dq_r\}, \\
e_{\text{corr}} &= C\{2[(D + e_*)^2 - D^2] + (3/5)[(D + e_*)p_* + e_*q_r]\}, \\
e_{\text{inner}} &= C\left(\sqrt{1/(14 \cdot 7^{32})} + A_*/\sqrt{2}\right)^2, & e_{\text{hold}} &= 43CA_*/(50\mu), \\
e_{\text{dev}} &= C\mathcal{P}/(10^{-5}M).
\end{aligned}$$

The inner-tail replacement follows from $\text{erfc}(8) < e^{-64}/(8\sqrt{\pi}) < 1/(14 \cdot 7^{32})$. All constants in these expressions have now been specified by rational operations, square roots, π , and the lens functions D_9, Q_9 in (86). Outward evaluation gives

Term	Upper bound
e_{base}	0.000537032962395581630703
e_{corr}	0.000010915589215098519870
e_{inner}	$1.81324790661460 \times 10^{-22}$
e_{hold}	$3.271569843081749 \times 10^{-9}$
e_{core}	0.000004378860028173163338
e_{dev}	$1.06689598045912 \times 10^{-13}$
e_{exit}	$9.127217662854526 \times 10^{-8}$

Their sum is strictly below 0.000552421956. The event proof is (119), the positive endpoint bound, (125), and the union with the three common exceptional events (clock core, clock deviation, radial exit). When combined with the earlier-label theorem, those same events are counted once. None of the estimates supplies an independent internal-sector variable, a new actual-law marginal, or an additional physical source beyond the stated effective Hamiltonian.

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